

CH 412/512: Physical Chemistry: Statistical Mechanics and Kinetics

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1 Recapitulation of Thermodynamics:

1.1 Basics of Thermodynamics

In your previous classes, you have studied thermodynamics using a more common approach. You learnt the zeroth, first, second, and the third laws, and use them to solve thermodynamic problem. In this course we will take a more abstract postulate based approach. However, before we go there, we will recap some basics of thermodynamics:

1.1.1 System, Surrounding, and Universe

System: The part of the universe that we are focusing on, such as a water in a glass or a protein molecule in cell.

Surroundings: Everything outside the system that can interact with it, such as the glass that contains the water and the air around it.

Universe: The system plus its surroundings.

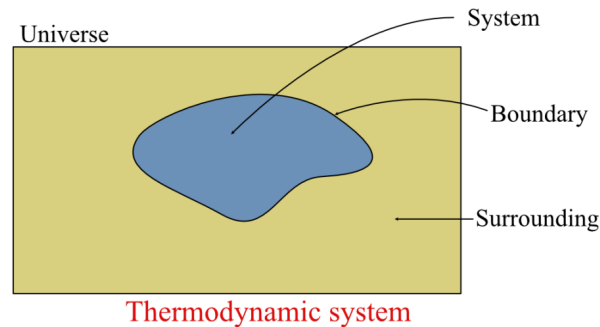


Figure 1: System, Surrounding, and Universe

1.1.2 Open, Closed, and Isolated Systems

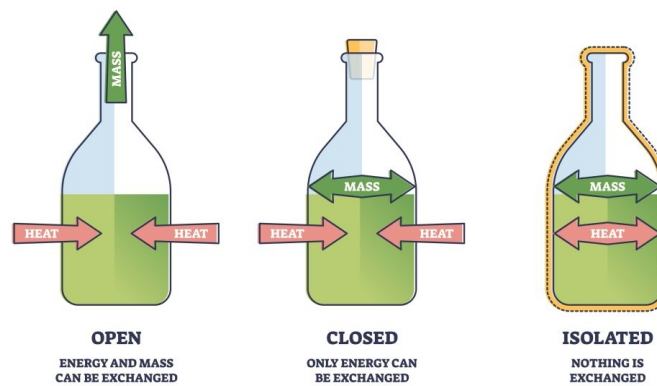


Figure 2: Open, Closed, and Isolated Systems

1.1.3 Laws of thermodynamics

0th Law: If two systems are each in thermal equilibrium with a third system, then they are in thermal equilibrium with each other.

1st Law: Energy cannot be created or destroyed, but only converted from one form to another. The total energy of an isolated system is always constant.

2nd Law: The total entropy of universe (or an isolated system) can only increase but can never decrease; natural processes tend toward a state of greater disorder.

3rd Law: As the temperature of a perfect crystal approaches absolute zero (0 Kelvin or -273.15°C), its entropy approaches a constant minimum value, typically zero.

1.1.4 Intensive and Extensive Variables

Extensive Variables are quantities that directly depend on the system size i.e. Number of particles present in the system. For example, volume V , total internal energy U , and entropy S are all Extensive variables. They all individually are proportional to N , i.e. number of moles or the number of particles present in the system. Intensive variables are exact opposite. They do not directly depend on the system size i.e. N . For example, temperature (T) and pressure (P). Doubling the number of particles will not necessarily double the temperature. Intensive variables are often ratio of two extensive variables, this makes sense as the N dependence cancels out when we take the ratio.

1.2 The Postulate Approach

Now we will study thermodynamics using the postulate approach to formalize the mathematical framework. You **do not** need to memorize the exact statements of the postulates.

- **Postulate 1:** Thermodynamic systems have equilibrium states, that are characterized completely by the extensive variables of the system, e.g., internal energy U , the volume V , and the number of particles or mole numbers $N_1, N_2, \dots, N_r$ of the different chemical components.
- **Postulate 2:** There exist single-valued, continuous, and differentiable function of the extensive variables called **entropy**: $S = S(U, V, N_1, N_2, \dots, N_r)$. In equilibrium state the other extensive variables (U , V , etc.,) take on those values that maximizes entropy (S) of the system, subject to its internal constraints.
- **Postulate 3:** The **entropy** of a composite system is **additive** over the constituent subsystems. The entropy is **continuous and differentiable** and is a **monotonically increasing function of the energy**.
- **Postulate 4:** The entropy of any system is non-negative and vanishes at zero temperature. (This is equivalent to the 3rd law).

1.3 Explanation of the Postulates

1.3.1 Explanation of Postulate 1

The equilibrium states are the most stable observable states of a system. Most of thermodynamics and statistical mechanics is concerned with characterizing the equilibrium states of a system. In thermodynamics we ask questions like: (1) if we keep two metal blocks of temperature T_1 and T_2 adjacent to each other, in equilibrium what will be the temperature of the whole system? (2) If we inject a anti-cancer drug to a patient, in equilibrium, what fraction of the drug will be bound to the disease causing protein? (3) If we design a new metal organic framework (MOF) for absorbing CO_2 from the atmosphere, in equilibrium how much CO_2 can it absorb per kg?

1.3.2 Explanation of Postulate 2

Entropy S in a function of internal energy U , the volume V , and the mole numbers $N_1, N_2, \dots, N_r$ of the chemical components:

$$S = S(U, V, N_1, N_2, \dots, N_r) \quad (1.1)$$

For the time being let's focus on single component systems for simplicity. Let's say there is box of volume V filled with N moles of Ar gas (I am purposefully choosing an inert gas so that no new components are generated through any chemical process). This system has internal energy U and entropy S . Therefore, S is a function of U , V , and N :

$$S = S(U, V, N) \quad (1.2)$$

What do we mean by "function" here? Let's look at a simple example. Suppose I ask how long it takes to go from Eugene to Portland by driving. The answer will depend on the distance and your speed of driving. If the distance is l and you drive at a constant speed v (for simplicity), the time it would take to reach Portland is $\tau = l/v$. Therefore τ is a function of l and v : $\tau = \tau(l, v)$. The τ , however, is not a function of the temperature of Portland (T_P). So $\tau \neq \tau(T_P)$.

Now, how do we maximize the variable S ? By setting the first derivative to zero. Therefore, when $dS = 0$, we reached equilibrium. This can be written as:

$$dS = \left(\frac{\partial S}{\partial U} \right)_{V,N} dU + \left(\frac{\partial S}{\partial V} \right)_{U,N} dV + \left(\frac{\partial S}{\partial N} \right)_{U,V} dN \quad (1.3)$$

We can set the RHS to zero, solve, and find conditions for equilibrium.

1.3.3 A Note on the Differential of Function

Once we know which variables are functions of which other variables, we can try to measure the change of one of those variables with respect to the others. Let's say f is a function of variables x and y .

$$f = f(x, y) \quad (1.4)$$

Now a small change in f can be expressed in terms of small changes in x and y in the following way

$$df = \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy \quad (1.5)$$

For the problem of measuring time of going from Eugene to Portland, we can write:

$$\begin{aligned} \tau &= l/v \\ \implies l &= \tau v \end{aligned} \quad (1.6)$$

Now if we take differential of the distance

$$dl = \left(\frac{\partial l}{\partial \tau} \right)_v d\tau + \left(\frac{\partial l}{\partial v} \right)_\tau dv \quad (1.7)$$

Translating this idea for the entropy function we can write:

$$dS = \left(\frac{\partial S}{\partial U} \right)_{V,N} dU + \left(\frac{\partial S}{\partial V} \right)_{U,N} dV + \left(\frac{\partial S}{\partial N} \right)_{U,V} dN \quad (1.8)$$

Exercise: Say, I drive at a constant speed of 65 miles per hour, and I reach Portland from Eugene in two hours. However, if I drive 67 miles per hour for 2.1 hours, I reach Vancouver, WA, instead. So, how far is Vancouver, WA, from Eugene? Solving using Eq. 1.5 and by multiplying time and velocity. Do these estimates agree?

1.3.4 Explanation of Postulate 3

Let's say there are two systems with entropy S_1 and S_2 . Their internal energies, volumes, and number of particles are U_1 and U_2 , V_1 and V_2 , and N_1 and N_2 respectively. According to this postulate the entropy of the combined system will be $S_{total} = S_1 + S_2$. Note that this total entropy will now be a function of the total internal energy, total volume, and total number of moles.

$$S_1(U_1, V_1, N_1) + S_2(U_2, V_2, N_2) = S_{total}(U_1 + U_2, V_1 + V_2, N_1 + N_2)$$

This is because the volume, number of moles, and internal energy are also additive. Now consider that system 1 and 2 are identical i.e. $U_1 = U_2$, $V_1 = V_2$, and $N_1 = N_2$. We call the common values U , V , and N . So the above equation becomes

$$S(U, V, N) + S(U, V, N) = S_{total}(2U, 2V, 2N) \implies S(2U, 2V, 2N) = 2S(U, V, N)$$

So if we double the variables entropy also doubles. This makes entropy also an extensive variable which means its value depends on the system size.

1.4 Relationship Between Extensive and Intensive Variables

Now if you look carefully, all the partial derivatives that contribute to dS are some kind of intensive variable. But what are they? To address that we need to do some rearrangement. If

$$S = S(U, V, N)$$

we can write,

$$U = U(S, V, N)$$

In the same way we can write $l = \tau v$ from $\tau = l/v$. Now from the latter equation we can write

$$dU = \left(\frac{\partial U}{\partial S} \right)_{V,N} dS + \left(\frac{\partial U}{\partial V} \right)_{S,N} dV + \left(\frac{\partial U}{\partial N} \right)_{S,V} dN \quad (1.9)$$

This equation has three intensive variables: $\left(\frac{\partial U}{\partial S} \right)_{V,N}$, $\left(\frac{\partial U}{\partial V} \right)_{S,N}$, and $\left(\frac{\partial U}{\partial N} \right)_{S,V}$. They are intensive variables because they are ratios of extensive variables; therefore, system size dependence cancels out. Let us give them some names.

$$\begin{aligned} \left(\frac{\partial U}{\partial S} \right)_{V,N} &= T \quad (\text{Temperature}) \\ \left(\frac{\partial U}{\partial V} \right)_{S,N} &= -P \quad (P = \text{Pressure}) \\ \left(\frac{\partial U}{\partial N} \right)_{S,V} &= \mu \quad (\text{Chemical Potential}) \end{aligned} \quad (1.10)$$

By now you all know empirically what temperature and pressure is. But now we get a more formal definition. Temperature is the measure of how internal energy changes as a function of entropy, when the volume and number of particles are kept constant. Pressure is the measure of how internal energy changes as a function of volume, when the entropy and number of particles are kept constant. Similarly chemical potential is the measure of how internal energy changes as a function of number of particles, when the entropy and volume are kept constant.

But what is with the sign of the pressure? Because it can be shown that if we increase volume without changing entropy and the number of particles, the internal energy goes down. So $\left(\frac{\partial U}{\partial V} \right)_{S,N} < 0$. But the pressure we measure in daily life is a positive quantity. To account for that we chose $-P$ as the value of this derivative.

But this does not solve our problem, We do not know what are the derivatives of entropy. We only found the derivatives of the internal energy.

Let us now find the expressions for $\left(\frac{\partial S}{\partial U} \right)_{V,N}$, $\left(\frac{\partial S}{\partial V} \right)_{U,N}$, and $\left(\frac{\partial S}{\partial N} \right)_{U,V}$ from Eq. 1.9 and 1.15.

$$\begin{aligned} dU &= \left(\frac{\partial U}{\partial S} \right)_{V,N} dS + \left(\frac{\partial U}{\partial V} \right)_{S,N} dV + \left(\frac{\partial U}{\partial N} \right)_{S,V} dN \\ \implies dU &= TdS - PdV + \mu dN \\ \implies TdS &= dU + PdV - \mu dN \\ \implies dS &= \frac{1}{T}dU + \frac{P}{T}dV - \frac{\mu}{T}dN \end{aligned} \quad (1.11)$$

Comparing with Eq. 1.8:

$$\begin{aligned}
\left(\frac{\partial S}{\partial U}\right)_{V,N} &= \frac{1}{T} \\
\left(\frac{\partial S}{\partial V}\right)_{U,N} &= \frac{P}{T} \\
\left(\frac{\partial S}{\partial N}\right)_{U,V} &= -\frac{\mu}{T}
\end{aligned}
\tag{1.12}$$

1.5 Conditions of Equilibrium

Postulate 2 says equilibrium is reached when entropy is maximized. This is equivalent to the second law, which says the entropy of the universe always increases in a spontaneous process. In calculus, we can find the maximum (or extremum to be precise) by setting the derivative of the quantity to zero. As we already have an expression for dS :

$$dS = \frac{1}{T}dU + \frac{P}{T}dV - \frac{\mu}{T}dN$$

We can just set the right-hand side to zero and find out the conditions of equilibrium. Problem solved, right? **No!** There are two problems.

1. Although we can measure T , P , V , N , etc., there is no way to experimentally measure dU or dS .
2. The second law (and postulate 2) implies that in an equilibrium state, the **entropy of the universe is maximised**, not the entropy of the system itself. This is why, for example, if we keep water at 273K, it freezes spontaneously, although the process of freezing reduces the entropy of the water. In fact, the loss of entropy in water is compensated by the gain of entropy of the surroundings to ensure that the entropy of the universe increases. Measuring any property of the universe is virtually impossible, as most processes of practical relevance involve some form of heat or material exchange with the surroundings, which includes literally anything in the world (universe!) apart from your system.

Our objective is therefore to **express change in entropy in terms of (i) measurable quantities of (ii) only the system**.

1.6 Getting Rid of Non-measurable Quantities: Legendre Transform

One approach to replacing non-measurable extensive variables like internal energy and entropy with measurable intensive parameters is to use the Legendre transform. If we have a function $f = f(x, y)$, the legendre transform of f with respect to the variable x is defined as

$$\mathcal{L} = f - \left(\frac{\partial f}{\partial x}\right)_y x$$

Now the legendre transform $\mathcal{L}$ is no longer a function of x but only of $(\frac{\partial f}{\partial x})_y$ and y :

$$\mathcal{L} = \mathcal{L}\left(y, \left(\frac{\partial f}{\partial x}\right)_y\right)$$

So if we want to avoid calculating x , we can work with the quantity $\mathcal{L}$ instead of f .

Let us prove this property of the Legendre transform. Let's say $z = \left(\frac{\partial f}{\partial x} \right)_y$. Therefore,

$$\begin{aligned}
\mathcal{L} &= f - zx \\
\Rightarrow d\mathcal{L} &= df - zdx - xdz \\
\Rightarrow d\mathcal{L} &= \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy - zdx - xdz \\
\Rightarrow d\mathcal{L} &= zdx + \left(\frac{\partial f}{\partial y} \right)_x dy - zdx - xdz \\
\Rightarrow d\mathcal{L} &= \left(\frac{\partial f}{\partial y} \right)_x dy - xdz \\
\Rightarrow \mathcal{L} &= \mathcal{L}(y, z)
\end{aligned} \tag{1.13}$$

Now use this technique to remove non-measurable extensive variables from our thermodynamic expressions. We use the internal energy expression for simplicity (so that we don't have to deal with lots of division by T).

$$U = U(S, V, N) = TdS - PdV + \mu dN$$

In this example entropy is the quantity we cannot measure. So let us create a legendre transform of U with respect to S .

$$A = U - \left(\frac{\partial U}{\partial S} \right)_{V, N} S = U - TS$$

$$dA = dU - TdS - SdT = -SdT - PdV + \mu dN$$

$$\Rightarrow A = A(T, V, N)$$

Where all the variables T, V, and N are measurable. This term A is called Helmholtz Free Energy. Now if we can identify an extremum principle for this quantity A, we will have a practical way of characterizing the equilibrium state.

1.7 Maximizing Entropy of the Universe is Equivalent to Minimizing Free Energy of the System

The second problem we had was that measuring any property of the universe is impossible. We need an approach that requires only properties of the system. Let's consider a constant volume situation of a closed system (no particle enters or exits the system). Therefore $dV = 0$ and $dN = 0$.

$$dU = TdS - PdV + \mu dN$$

$$\Rightarrow dU = TdS \quad \text{when} \quad dV = 0, \quad dN = 0$$

The total internal energy of the universe is constant. So

$$dU_{\text{univ}} = dU_{\text{sys}} + dU_{\text{surr}} = 0$$

$$\Rightarrow dU_{\text{sys}} = -dU_{\text{surr}} = -T_{\text{surr}} dS_{\text{surr}}$$

$$\Rightarrow dS_{\text{surr}} = -\frac{dU_{\text{sys}}}{T_{\text{surr}}}$$

Now, if we consider a constant temperature process $T_{\text{sys}} = T_{\text{surr}} = T$.

So for a spontaneous process at constant T , V , and N ,

$$\begin{aligned}
 dS_{\text{univ}} &\geq 0 \\
 \implies dS_{\text{sys}} + dS_{\text{surr}} &\geq 0 \\
 \implies dS_{\text{sys}} - \frac{dU_{\text{sys}}}{T} &\geq 0 \\
 \implies dU_{\text{sys}} - TdS_{\text{sys}} &\leq 0 \\
 \implies dA_{\text{sys}} &\leq 0
 \end{aligned} \tag{1.14}$$

as $dA = d(U - TS) = dU - TdS - SdT = dU - TdS$ when T is constant.

Therefore, when the entropy of the universe increases, the Free Energy of the system decreases. So, instead of maximizing entropy S of the universe, we can minimize A to find the equilibrium state at constant T , V , and N conditions.

This is actually a very general result. Note that if you create any Legendre transform of internal energy in terms of S , V , or N , minimizing that Lagrange multiplier implies maximization of the entropy of the universe, under the condition that the corresponding intensive variable is held constant for the system.

Most chemical and biological processes occur in constant temperature and pressure conditions. So if we can create a Legendre transform, which is a function of T , P , and N , minimizing that function will also imply maximization of the entropy of the universe, thereby reaching equilibrium at the constant temperature and pressure conditions.

To generate a function of T , P , and N , we need to get rid of V dependence of $A(T, V, N)$. We can do so by taking a Legendre transformation again to generate a function G :

$$G = A - \left(\frac{\partial A}{\partial V} \right)_{T,N} V$$

From the relationship $dA = -SdT - PdV + \mu dN$, we know $\left(\frac{\partial A}{\partial V} \right)_{T,N} = -P$. So our new Free energy function is

$$G = G(T, P, N) = A + PV = U - TS + PV$$

This is called the Gibbs Free Energy.

Following the example of $A(T, V, N)$, we can explicitly show that at constant T , P , and N , in a spontaneous process, Gibbs Free Energy always decreases. Therefore, in a constant $T - P$ process, we can identify the equilibrium state by minimizing the Gibbs Free Energy (G).

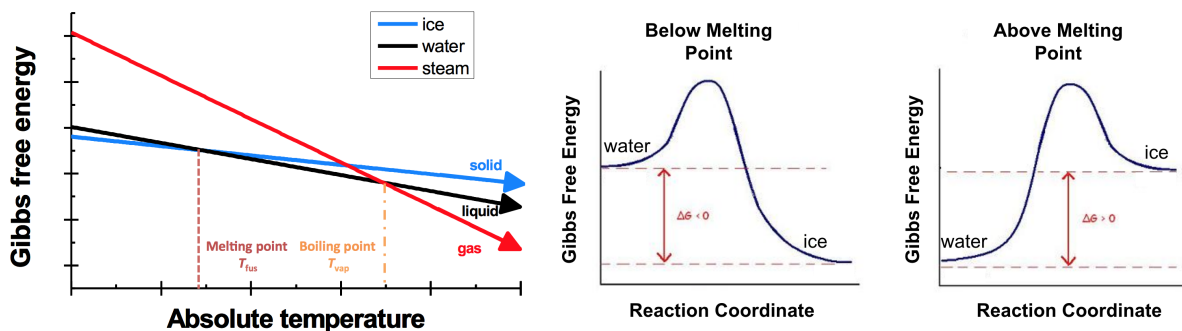


Figure 3: Change of Gibbs Free Energy as a function of temperature.

For example, if we plot the Gibbs free energy vs. temperature for H_2O , below the melting point, the G is lower for ice, but above the melting point, G is lower for water. If we could plot the Gibbs Free Energy against the reaction coordinate of the phase transition, we could see that, below the melting point, ice has a lower free energy, and the trend reverses above the melting point. So, below the melting point, the equilibrium state is ice, while above the melting point, the equilibrium state is water.

Unfortunately, we cannot provide an explicit quantitative example here, as we did not specify how to calculate the free energy and entropy of any realistic system. We will do so in the subsequent chapter.

1.8 Summary

- There exist two extensive properties, entropy and internal energy, that are functions of the other extensive variables of the system: $U = U(S, V, N)$ and $S = S(U, V, N)$.
- Partial derivatives of U and S with respect to their argument provide the experimentally measurable intensive variables.

$$\begin{aligned}
 \left(\frac{\partial U}{\partial S} \right)_{V,N} &= T \quad (\text{Temperature}) \\
 \left(\frac{\partial U}{\partial V} \right)_{S,N} &= -P \quad (P = \text{Pressure}) \\
 \left(\frac{\partial U}{\partial N} \right)_{S,V} &= \mu \quad (\text{Chemical Potential})
 \end{aligned} \tag{1.15}$$

- Equilibrium states can be obtained by maximizing the entropy of the universe. This is equivalent to minimizing the system's Helmholtz free energy: $A = A(T, V, N) = U - TS$ in a constant $T - V$ process and the Gibbs Free energy: $G(T, P, N) = U - TS + PV$ in constant $T - P$ conditions. As measuring the entropy of the universe is impossible, we describe equilibrium thermodynamic processes in terms of the appropriate free energy.

2 Probability and Its Relation to Entropy

Probability and statistics are at the center of statistical mechanics and thermodynamics. Molecules often have more than one state and structure. The relative probability between these structures is one of the most important quantities for predicting reactions or other molecular properties. (Example: crystal polymorph, reaction yield etc. Markovnikov product in HBr addition to alkene). The basic principle of probability is

$$P(\text{desired outcome}) = \frac{\text{number of desired events}}{\text{number of all possible outcome}}$$

2.1 Microstates and Macrostates

To introduce the concepts of microstates and macrostates, we will work with the example of flipping a fair coin. A coin flip has two outcomes, head (H) or tail (T), with equal probability. So $P(H) = 1/2$ and $P(T) = 1/2$. But what happens when we want to predict the outcomes of many coin flips? For example, what is the probability of getting exactly 1 H in 4 coin flips?

$$\begin{aligned} &TTTT(0) \\ &HTTT(1), THTT(1), TTHT(1), TTTH(1) \\ &HHTT(2), HTHT(2), HTTH(2), THHT(2), THTH(2), TTTH(2) \\ &TTHH(3), HTTH(3), HHTH(3), HHHT(3) \\ &HHHH(4) \end{aligned} \tag{2.1}$$

Out of the total 16 outcomes, 4 have only one head. So the probability of getting one head is $4/16 = 0.25$. This outcome of observing 1 one head in 4 coin flips is called a **macrostate**, whereas each specific outcome ($HTTT$, $THTT$, etc.) is called **microstate**. One macrostate can contain one or more microstates. Here, we can see five macrostates (0H, 1H, 2H, 3H, and 4H). They contain 1, 4, 6, 4, and 1 microstates, respectively. The number of microstates in a given macrostate is called **degeneracy**. The macrostate (2H) has the highest number (6) of microstates. Therefore, this is the most likely macrostate of the system.

These observations can be summarized in **two postulates**.

1. **Equal a priori probability:** Each microstate has equal probability in the absence of any constraints on the system.
2. **The observed macrostate is the one with the highest number of microstates.**

When we predict bulk properties, we deal with Avogadro's number of molecules. In such an extensive system, the most probable macrostate becomes the only observable state of the system as the likelihood of observing other macrostates becomes very low. This can be shown by increasing the number of coin flips. However, you will soon realize that it can become repetitive and complex to continue counting the number of heads and tails as the number of coin flips increases. This is the same problem we face in molecular science. So let us look at how we can address them.

2.2 Theory, Simulation, and Experiment

Here we will digress a little bit and discuss three different approaches toward science: **(i) Theory, (ii) Simulation, and (iii) Experiment**.

In **Experiment**, we try to write all possible combinations of heads and tails for a certain number of coin flips and measure the number of heads and tails. We could also physically flip a coin hundreds of thousands of times to estimate the probabilities of each macrostate.

In **Simulation**, we write a computer program to randomly generate the letters 'H' and 'T' with a 50-50 probability and run this program N number of times. Then we repeat this exercise several hundred times and compute the average probabilities of getting a certain number of 'H's and 'T's.

In **Theory**, we derive a mathematical expression for the quantity we want to compute (e.g., the probability of getting exactly N_1 heads in N coin flips) using empirical understanding about the system (e.g., there is a 50-50 chance of getting a head or a tail).

All three approaches are valid ways of doing science; however, one approach is more applicable than the other, depending on the problem we want to solve. For example, we may be assigned the task of finding out the green coverage (the fraction of land area covered with greenery) in your city. Depending on the location and configuration of the green areas, as well as your existing knowledge, either theory, simulation, or experiment may be the most suitable approach.

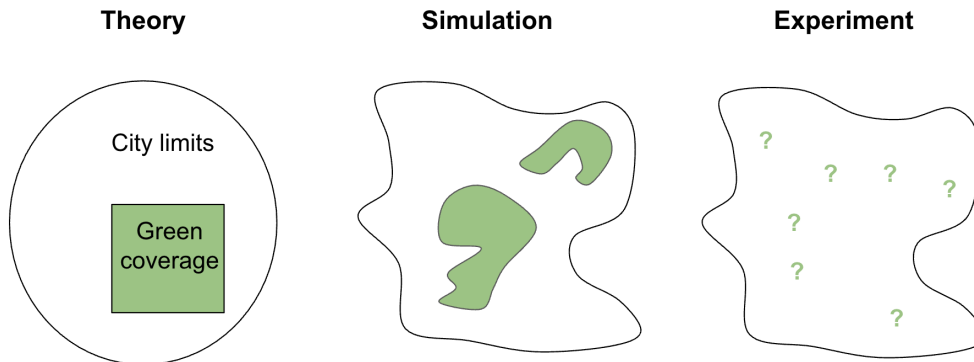


Figure 4: Difference among Theory, Simulation, and Experiment approaches towards science explained with an example of calculating the green coverage of a city.

Like the leftmost picture, if your city is a perfect circle and the green area is located in one location as a perfect square, the **Theory** is the best approach. We can write an expression like the following for the fraction of green coverage p_g :

$$p_g = \frac{l_g^2}{\pi r_c^2},$$

where l_g is the length of a side of the green square and r_c is the radius of the city. But it is clear that this is too simplistic a situation. In real life, neither the city limits nor the green areas are perfect squares or circles; therefore, it is not possible to write an exact expression of green coverage. However, you may have access to the map of the city, which shows the green areas (middle picture). In that case, we can perform a **Simulation**, where we randomly generate points in the 2D space inside the city limits and check what fractions of the points fall within the green areas. With an increasing number of points, this will provide a fairly accurate estimate of the fraction of green coverage. In the worst-case scenario depicted in the right picture, we may have no idea where the green areas are. There, our only option is to perform **Experiment**, where we must physically survey the land, identify the green areas, and calculate their square footage.

In the rest of the course, we will primarily deal with model problems that can be solved using the **Theory** approach, but in the real world, we often have to do **Simulation** or **Experiment**. However, we need to learn the **Theory** approach first, as it lays the groundwork for designing effective simulation and analysis tools to make meaningful predictions of the real world.

2.3 Calculating the Number of Microstates in a Macrostate

With the discussion of theory, simulation, and experiment out of the way, we can now look at our problem at hand: calculating the number of microstates in a macrostate. For the example of a coin flip, the **Theory** is the best approach of all, as the probability of getting heads and tails in an individual coin flip is known beforehand.

If we flip N coins, the number of ways we can get n heads is

$$W = \frac{N!}{n!(N-n)!}$$

It is important to understand this equation. If we have N different objects, we can arrange them in $N!$ different ways; this is known as permutations. The factorial (!) symbol means: $N! = 1 \times 2 \times 3 \times \dots \times (N-1) \times N$. However, when we flip a coin N times, and get n heads, then each result of heads looks the same: if you think of the result $HTHH$, it looks the same if we interchange any of the heads with any other head. Therefore, the number of ways

of arranging the N flips is $N!$ divided by $n!$ (to take into account that all the heads look the same), and obviously divided by $(N - n)!$ to take into account that all the tails are the same also. Notice how this fits in with flipping the coins 4 times. In this case, to get 2 heads (and therefore 2 tails) $W \approx 4!/(2! \times 2!) = 6$ as expected.

Now, to calculate the probability of each outcome, we need to compute the number of all accessible microstates. We know that each coin flip has two possible outcomes (H or T) and each coin flip is independent of the others. So if we flip a coin N times, we have 2^N possible outcomes. For example, with four coin flips, the total number of possible outcomes is $2^4 = 16$. Taking them together, we can write a general expression for the probability of observing n heads in N coin flips:

$$p(n|N) = \frac{N!}{2^N n!(N - n)!}$$

When we plot this function below, we see that the probability of getting 50% heads, i.e., the most likely microstate, gets more and more 'peaked' with increasing the number of coin flips, and the likelihood of observing other macrostates goes down drastically. Therefore, **for a large system, the observable macrostate is the one with the largest number of microstates.**

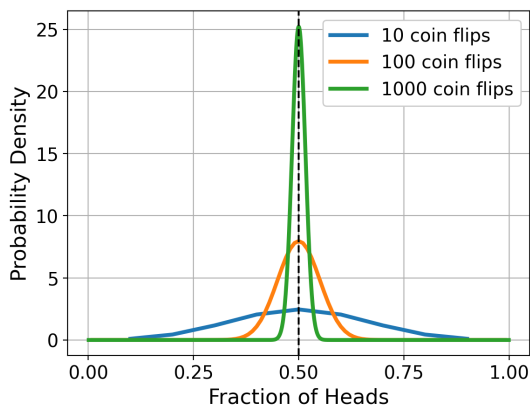


Figure 5: Obtaining 50% heads is the most likely event as we increase the number of coin flips.

2.4 Situations with More Than Two Outcomes

So far, we have considered only the coin flip example, which has a binary outcome, i.e., either head or tail. In molecular systems, we typically have more than one conformation for a given molecule, so we need to generalize our expressions to accommodate multi-state scenarios. One example of this is the conformers of *n*-butane molecule, which can remain in *Syn*, *Gauche*, *Eclipsed*, or *Anti* state. Please check this webpage for an interactive experience of the different conformers. https://sites.science.oregonstate.edu/~gablek/CH334/Chapter2/bare_rotation.htm

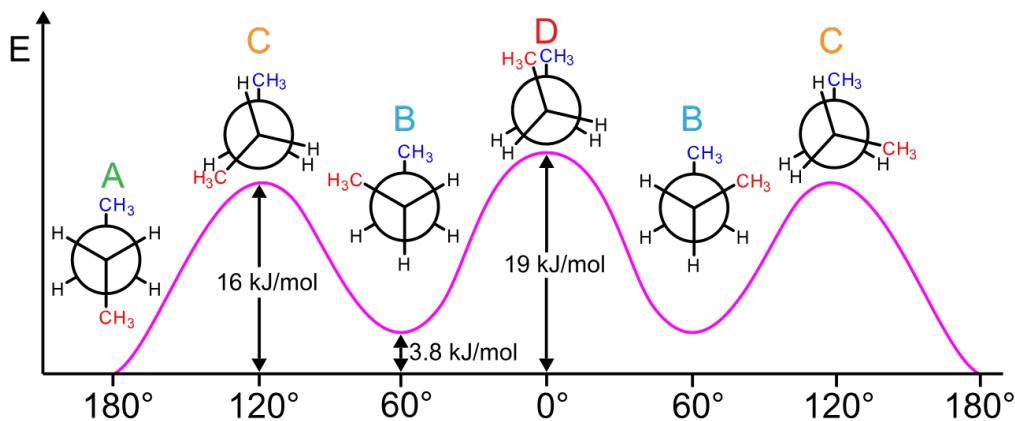


Figure 6: Newman projections of butane conformations & their relative energy differences (not total energies). Conformations form when butane rotates about one of its single covalent bonds. The torsional/dihedral angle is shown on the x-axis.

As there is an energy difference between the different conformers, their relative probabilities are not exactly equal. Besides, there are other intermediate states at intermediate values of the torsional angle. So this system is more complex than what we can deal with using **Theory**; we need **Simulation** for that. However, this example shows the multi-state nature of molecular systems.

To gain a better understanding, we examine a simpler model of N particles, each of which can occupy t distinct energy levels. These energy levels are similar to Bohr's atomic orbits. The particles are distinguishable, i.e., we can explicitly identify particle 5 in energy level 3, and so on. Below is a figure of a 7-particle system with 5 energy levels.

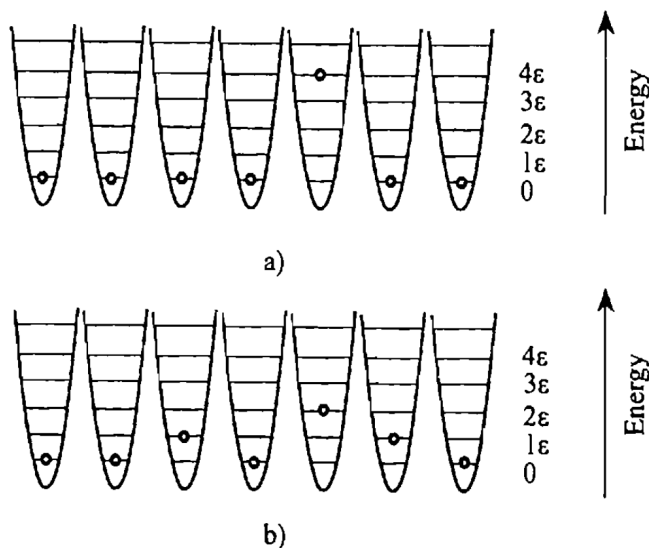


Figure 7: Figure Courtesy: Wark, J. S., and A. M. Glazer. Statistical Mechanics: A Survival Guide. Oxford University Press, 2001.

The energy levels of the particles are $E_1 = 0, E_2 = 1\epsilon, E_3 = 2\epsilon, E_4 = 3\epsilon, E_5 = 4\epsilon$ and there are n_1, n_2, n_3, n_4, n_5 particles in the five levels.

$$N = \sum_{i=1}^5 n_i \quad \text{and} \quad U = \sum_{i=1}^5 n_i E_i$$

where N is the total number of particles and U is the total internal energy of the system. In the case of coin flipping, we did not need to write it this way, as there are only two possible outcomes. Therefore, if there are n heads, there can only be $N - n$ tails. Here, there is more variability. Also, we have the additional constraint of the total internal energy. The number of ways in which we can arrange these particles in five energy levels is:

$$W = \frac{N!}{n_1!n_2!n_3!n_4!n_5!} = \frac{N!}{\prod_{i=1}^5 n_i!}$$

and the total number of possible microstates is 5^N for N independent particles in 5 energy states.

Let us now determine how likely the configurations (a) and (b) are from the picture above.

(a) 1 particle in level 5 and 6 particles in level 1:

$$W = \frac{7!}{1!6!} = 7$$

(b) 1 particle in level 3, 2 particles in level 2, and 4 particles in level 1:

$$W = \frac{7!}{1!2!4!} = 105$$

Therefore, the macrostate (b) has more microstates and is more likely to be observed.

2.5 Concept of Entropy

2.5.1 Boltzmann Entropy

We learned that the macrostate with the maximum number of microstates is the most likely one. We also saw that by increasing the number of observations (e.g., coin flip), the macrostate becomes so much more likely than the others that it can almost become the only observed macrostate (e.g., if you flip 10,000 coins, you are much more likely to observe close to 50% heads than by flipping just 4 coins). Now we need to translate this idea to thermodynamics. In the module "recapitulation of thermodynamics", we noted there are certain equilibrium states that are attained when entropy is maximized. Those **equilibrium states are the most likely macrostates**. For example, when we bring two blocks of metal with different temperatures next to each other, the equilibrium state will be reached when both blocks attain the same temperature. That state is the most likely macrostate, as when two adjacent blocks have the same temperature, their constituent molecules will have the highest number of microstates associated with them. As equilibrium is attained when entropy is maximized, entropy increases monotonically with the increase in the number of macrostates. Boltzmann has formalized this relationship as:

$$S = k_B \ln W,$$

where W is the number of microstates and k_B is known as Boltzmann's constant, which is a universal constant of value $1.38 \times 10^{-23} JK^{-1}$.

Why the natural logarithm is used instead of a proportional relationship?

- Consider a thermodynamic system having two subsystems, A and B, with multiplicities W_A and W_B , respectively. The multiplicity of the total system will be the product $W_{total} = W_A W_B$. Compatibility with thermodynamics requires that entropies be extensive, meaning that the system entropy is the sum of subsystem entropies: $S_{total} = S_A + S_B$. The logarithm function satisfies this requirement. If $S_A = k_B \ln W_A$ and $S_B = k_B \ln W_B$, then $S_{total} = k_B \ln W_{total} = k_B \ln W_A W_B = k_B \ln W_A + k_B \ln W_B = S_A + S_B$. This argument illustrates why S should be a logarithmic function of W .
- The number of microstates can be really huge in a bulk system, and the corresponding logarithm is much more tractable than the W itself. Say if you want to calculate the number of microstates for getting exactly 5 heads in 10 coin flips: $W_{5/10} = \frac{10!}{5!5!} = 252$, which is not so big. But for 100 coin flips: $W_{50/100} = \frac{100!}{50!50!} \approx 10^{29}$. Now, let's imagine what will happen if we flip Avogadro's number of coins, a number that is closer to the number of molecules in a bulk system. Logarithm comes to our rescue.
- Logarithm is also a monotonously increasing function, so if the number of microstates increases, entropy also increases.
- The number of microstates is often calculated using factorials of large numbers. There is a useful relationship about the natural logarithm of factorials, which helps in computing these large numbers. This is called the **Sterling's approximation** which states:

$$\ln N! \approx N \ln N - N + \ln \sqrt{2\pi N} \quad \text{for large } N.$$

Let's check Sterling's theorem on our earlier example of coin flip:

$$\ln W_{5/10} = \ln \left(\frac{10!}{5!5!} \right) = \ln 10! - \ln 5! - \ln 5! \approx 5.554 \quad (\text{Using Stirling's approximation})$$

$$W_{5/10} = e^{5.554} \approx 258 \quad (\text{true value } 252, 2.3\% \text{ error})$$

$$\ln W_{50/100} = \ln \left(\frac{100!}{50!50!} \right) = \ln 100! - \ln 50! - \ln 50! \approx 66.78 \quad (\text{Using Stirling's approximation})$$

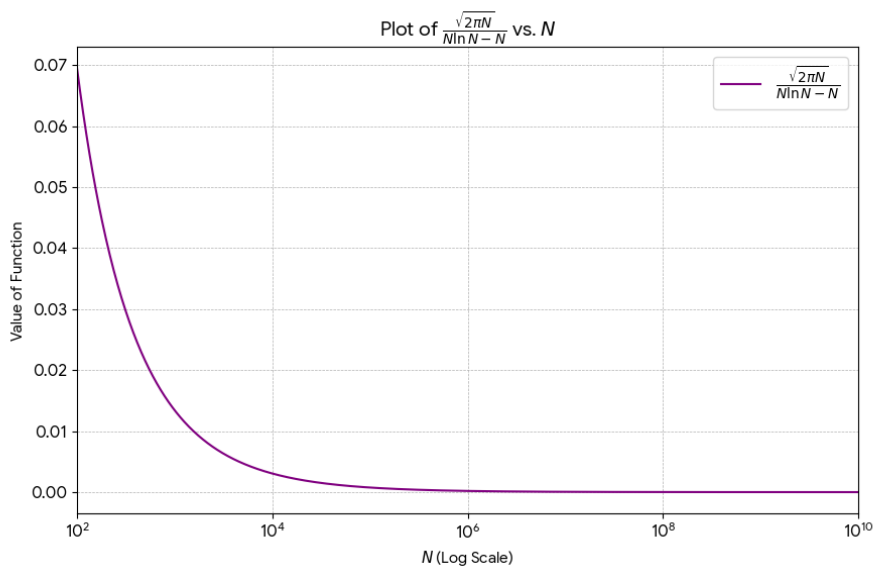
$$W_{50/100} = e^{66.78} \approx 1.005 \times 10^{29} \quad (\text{true value } 1.009 \times 10^{29}, 0.4\% \text{ error})$$

Also note that increasing the number of observations by a factor of 10 increased the value of $\ln W$ by a factor of 13, rather than 10^{27} or so, keeping the increase manageable.

For very large values of N , however, the approximation can be simplified as:

$$\ln N! \approx N \ln N - N$$

This is because the $\sqrt{2\pi N}$ term is proportional to $N^{1/2}$, whereas other terms are proportional to N or $N \ln N$. So the third term becomes negligible at very high values of N , particularly for Avogadro's number. Below, we plot the ratio of the $\sqrt{2\pi N}$ term with the rest of the expression and see how it goes to zero at larger values of N .



2.5.2 Shannon Entropy from Information Theory

Mathematician Claude Shannon introduced a different expression for entropy in terms of the probabilities of different outcomes.

$$S = - \sum_i p_i \ln p_i$$

where i is the index of an outcome and p_i is the corresponding probability.

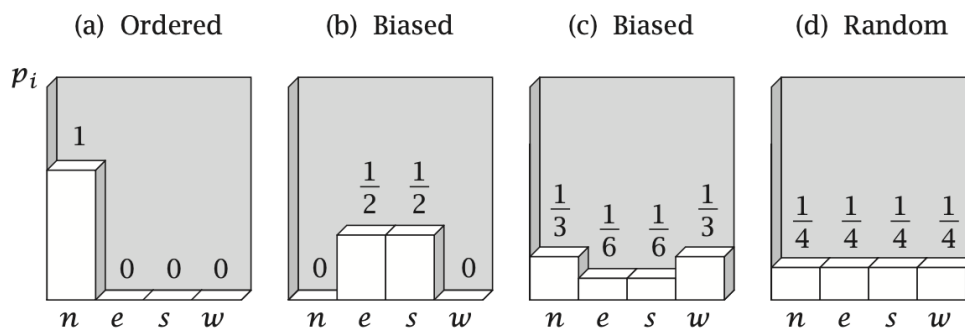
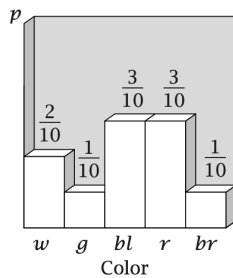


Figure 5.1 Spin a hundred pencils. Here are four (of a large number) of possible distributions of outcomes. (a) All pencils could point north (n). This is the most *ordered* distribution, $S = -1 \ln 1 = 0$. (b) Half the pencils could point east (e) and half could point south (s). This distribution has more entropy than (a), $S = -2[(1/2) \ln(1/2)] = 0.69$. (c) One-third of the pencils could point n , and one-third w , one-sixth e , and one-sixth s . This distribution has even more entropy, $S = -2[(1/3) \ln(1/3) + (1/6) \ln(1/6)] = 1.33$. (d) One-quarter of the pencils could point in each of the four possible directions. This is the distribution with highest entropy, $S = -4[(1/4) \ln(1/4)] = 1.39$.

Figure 8: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.



EXAMPLE 5.2 Colors of socks. Suppose that, on a given day, you sample the distribution of the colors of the socks that 30 people are wearing. Figure 5.2 shows a possible outcome. The entropy of this distribution is

$$S = -0.2 \ln 0.2 - 0.6 \ln 0.3 - 0.2 \ln 0.1 = 1.50.$$

The entropy function just reports the relative flatness of a distribution function. The limiting cases are the most ordered, $S = 0$ (everybody wears the same color socks) and the most disordered, $S = \ln t = \ln 5 = 1.61$ (all five sock colors are equally likely).

Figure 9: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.

2.5.3 Equivalence of Shannon and Boltzmann Entropy

It can be explicitly shown that the Shannon and the Boltzmann entropy formulas are equivalent (as long as we include the k_B factor, which is not there in the original Shannon entropy expression). But we will not go through all the complex math (If you are interested, check Dill and Bromberg, page 82). We will show the equivalence for some specific cases.

- In a system, where all microstates are equally probable, the probability of observing each microstate is $p_i = 1/W$, where W is the total number of microstates. Here, the Boltzmann entropy is $S = k_B \ln W$, and the Shannon entropy is

$$S = -k_B \sum_{i=1}^W p_i \ln p_i = -k_B \sum_{i=1}^W \frac{1}{W} \ln \frac{1}{W} = -k_B W \cdot \frac{1}{W} \ln \frac{1}{W} = k_B \ln W$$

- In the example of flipping 4 coins, the Boltzmann entropy of each macrostate is as follows:

$$S(0H) = k_B \ln 1 = 0; S(1H) = k_B \ln 4; S(2H) = k_B \ln 6; S(3H) = k_B \ln 4; \text{ and } S(4H) = k_B \ln 1 = 0.$$

The Shannon Entropy for each Macrostate is:

$$S(0H) = -k_B(p_H \ln p_H + p_T \ln p_T) = -k_B \ln 1 = 0$$

$$S(1H) = -k_B(p_H \ln p_H + p_T \ln p_T) = -k_B(0.25 \ln 0.25 + 0.75 \ln 0.75) = 0.56k_B$$

$$S(2H) = -k_B(p_H \ln p_H + p_T \ln p_T) = -k_B(0.5 \ln 0.5 + 0.5 \ln 0.5) = k_B \ln 2 = 0.69k_B$$

$$S(3H) = -k_B(p_H \ln p_H + p_T \ln p_T) = -k_B(0.75 \ln 0.75 + 0.25 \ln 0.25) = 0.56k_B$$

$$S(4H) = -k_B(p_H \ln p_H + p_T \ln p_T) = -k_B \ln 1 = 0$$

These values are the same as the Boltzmann Entropy barring a constant factor.

Note: The Shannon entropy calculation does not require you to know how many microstates are there in a given macrostate. When it is too cumbersome to obtain the number of microstates, Shannon entropy is the way to go.

2.6 Summary

- The principle of probability states

$$P(\text{desired outcome}) = \frac{\text{number of desired events}}{\text{number of all possible outcome}}$$

- The postulate of **equal a priori probability** states that each microstate has equal probability when the system is in equilibrium without any external constraint. The second postulate states that the **most probable macrostate is the one with the highest number of associated microstates** and may be the only macrostate observed in a bulk system.
- In a binary system, for example, in the case of a coin flip, the number of ways we can get n heads out of N coin flips is

$$W = \frac{N!}{n!(N-n)!}$$

If we have N different objects, we can arrange them in $N!$ different ways; this is known as permutations. The factorial (!) symbol means: $N! = 1 \times 2 \times 3 \times \dots \times (N-1) \times N$.

- The total number of outcomes for N independent binary events (e.g., coin flip) is 2^N . Therefore, the probability of observing n heads in N coin flips is:

$$p(n|N) = \frac{N!}{2^N n!(N-n)!}$$

- The expressions for the binary system can be generalized to systems with more than two possible outcomes per event or particle. For a system with N particles, each of which can occupy t distinct energy levels, the number of microstates for the macrostate $n_1, n_2, n_3, \dots, n_t$ is

$$W = \frac{N!}{n_1!n_2!n_3!\dots n_{t-1}!n_t!} = \frac{N!}{\prod_{i=1}^t n_i!}$$

and the total number of microstates across all macrostates is t^N .

- Entropy can be computed using either the Boltzmann or the Shannon formula.

$$\boxed{S = k_B \ln W} \quad \text{or} \quad \boxed{S = - \sum_{i=1}^N p_i \ln p_i}$$

where, W is the number of microstates and the p_i is the probability of observing i -th outcome. These two expressions are equivalent to a factor of k_B .

3 Boltzmann Probability and Partition Function

3.1 Boltzmann Probability

Now we are going to learn about another formula from Boltzmann. He demonstrated that the population of a state, i.e., the number of particles in a state, is proportional to the exponential of the negative energy of that state. So if there is a state with energy E and the population of that state is n_E ,

$$n_E \propto e^{-\frac{E}{k_B T}}. \quad (3.1)$$

We often write it as $n_E \propto e^{-\beta E}$ where $\beta = 1/k_B T$. We will use the β symbol more often to avoid writing a fraction and $k_B T$ every time. Note that it is possible to prove this relationship explicitly; we will skip that for the time being. You can watch the derivation of this idea for the special case of ideal gas in this link https://www.youtube.com/watch?v=_6AFgt2gkr0&t=2s

Now suppose there are two states in a system, A and B. These can be the folded and unfolded states of a protein or the reactant and product states of a unimolecular chemical reaction. Their energies are E_A and E_B respectively. If there are n_A particles (such as proteins or molecules) in state A and n_B particles in state B

$$n_A \propto e^{-\beta E_A} \quad \text{and} \quad n_B \propto e^{-\beta E_B}.$$

Therefore the probability of being in state A (p_A) and state B (p_B) are given by:

$$p_A = \frac{n_A}{n_A + n_B} = \frac{e^{-\beta E_A}}{e^{-\beta E_A} + e^{-\beta E_B}}$$

$$p_B = \frac{n_B}{n_A + n_B} = \frac{e^{-\beta E_B}}{e^{-\beta E_A} + e^{-\beta E_B}}$$

What is the relative probability of being in state A over state B?

$$\frac{p_A}{p_B} = \frac{e^{-\beta E_A}}{e^{-\beta E_B}} = e^{-\beta(E_A - E_B)}$$

If $E_A > E_B$, $p_A/p_B < 1$. That means the probability of being in a higher energy state is lower than being in a lower energy state. This is intuitive. This is the reason air pressure is lower at high altitude, because the potential energy increases with height $E = mgh$.

3.2 Partition Function

Now imagine there are many more than just two states. In actual chemical systems, this is more common because a molecule can have many different energy states. For example, a protein not only exists in folded or unfolded states but also in many intermediate states. Butane molecule can have many different Gauche and staggered conformations. Let us say there are N states with energies $E_1, E_2, E_3, \dots, E_N$. The probability of being in the k -th state is:

$$p_k = \frac{n_k}{n_1 + n_2 + n_3 + \dots + n_N} = \frac{e^{-\beta E_k}}{e^{-\beta E_1} + e^{-\beta E_2} + e^{-\beta E_3} + \dots + e^{-\beta E_N}} = \frac{e^{-\beta E_k}}{\sum_{i=1}^N e^{-\beta E_i}}$$

Exercise: Check that this probability is normalized, i.e., the sum of the probabilities of all states adds up to 1.

Turns out that the summation in the denominator has some interesting properties. We call this summation, the **partition function** (Q).

$$Q = \sum_{i=1}^N e^{-\beta E_i}$$

It might also be the case that more than one states have the same energy. This situation is known as **degeneracy**. Let's say there are five states with energies E_1 , E_2 , E_2 , E_3 , and E_3 . States 2 and 3 are degenerate, and states 4 and 5 are degenerate. So, the partition function of this systems is

$$Q = e^{-\beta E_1} + e^{-\beta E_2} + e^{-\beta E_2} + e^{-\beta E_3} + e^{-\beta E_3} = e^{-\beta E_1} + 2e^{-\beta E_2} + 2e^{-\beta E_3}$$

We often see that a system has several macrostates with unique energies; however, the microstates within a specific macrostate share the same energy. In a system with several macrostates and microstates, we can write the partition function in two different ways:

$$Q = \sum_{i=1}^N e^{-\beta E_i} \quad N = \# \text{ of microstates}$$

$$Q = \sum_{j=1}^M g_j e^{-\beta E_j} \quad M = \# \text{ of macrostates and } g_j \text{ is the degeneracy of the } j\text{-th macrostate}$$

You might wonder why bother with the degeneracy expression of the partition function if we can apply the expression for the microstates directly? Isn't it much simpler? Let's see the advantage of the second expression in a hypothetical example. Suppose we live in a world where every time you get heads in a coin flip, you gain an amount of energy, E , but no energy is gained if you get tails. (sounds crazy, but bear with me!). In this fictitious world, let us perform four coin flips and try to predict the probability of getting two heads.

TTTT (Energy = 0)

HTTT, THTT, TTHT, TTTH (Energy = E)

HHTT, HTHT, HTTH, THHT, THTH, TTHH (Energy = $2E$)

THHH, HTHH, HHTH, HHHT, (Energy = $3E$)

HHHH (Energy = $4E$)

The partition function of this system in terms of microstates is

$$Q = 1 + e^{-\beta E} + e^{-\beta E} + e^{-\beta E} + e^{-\beta E} + e^{-\beta E} + e^{-2\beta E} + e^{-2\beta E} + e^{-2\beta E} + e^{-2\beta E} + e^{-2\beta E} + e^{-2\beta E} + e^{-3\beta E} + e^{-3\beta E} + e^{-3\beta E} + e^{-3\beta E} + e^{-4\beta E}$$

The partition function of this system in terms of macrostates and degeneracy is

$$Q = \sum_{n=0}^4 g_n e^{-\beta n E} = \sum_{n=0}^4 \frac{4!}{n!(4-n)!} e^{-\beta n E} = 1 + 4e^{-\beta E} + 6e^{-2\beta E} + 4e^{-3\beta E} + e^{-4\beta E}$$

They are equivalent, but one is far easier to calculate for a large number of coin flips than the other. When the number of coin flips is much higher, we can use combinatorial formulas from the last module to compute the degeneracies. Note that we used $e^{-\beta \cdot 0} = e^0 = 1$. Now the probability of getting two heads is

$$p(2H) = \frac{6e^{-2\beta E}}{1 + 4e^{-\beta E} + 6e^{-2\beta E} + 4e^{-3\beta E} + e^{-4\beta E}}$$

Exercise: If $\beta = 1$ and $E = 1$, what is the most probable number of heads to be observed in four coin flips?

Interactive Tutorial

3.2.1 Violation of equal a priori probability postulate?

Here arises one contradiction. What happened to the equal a priori probability hypothesis? Aren't all microstates supposed to have the same probability? Actually, that postulate is correct when there are no constraints on the system, i.e., the temperature and energy can take any possible value. However, if we add constraints to the system, particularly in terms of constant temperature and constant average energy (the absolute value of the energy can fluctuate over time, such as in a closed system) the probability distribution is no longer uniform across states. The mathematical derivation of this is provided in Dill and Bromberg, pages 86-89, but we will skip that here. This situation is more realistic, as most chemical and biological processes occur in a constant temperature environment with some reasonable constraints on the total energy accessible to the system.

3.3 Computing thermodynamic quantities from the partition function

We mentioned in the last subsection that the partition function has some interesting properties, but we did not elaborate on them further. We will do so now. From the partition function, we can calculate (almost) all thermodynamic properties that we need to know about a system. Let's derive the expression of these properties one at a time.

3.3.1 Internal Energy (U)

The internal energy of a system is the average energy of all the particles in all energy states of the system. If, for a single particle system, there are N states with energies $E_1, E_2, E_3, \dots, E_N$, and the probabilities of being in these states are $p_1, p_2, p_3, \dots, p_N$, the average energy is

$$\langle E \rangle = p_1 E_1 + p_2 E_2 + p_3 E_3 + \dots + p_N E_N = \sum_{i=1}^N p_i E_i$$

Now we know that

$$p_i = \frac{1}{\sum_{i=1}^N e^{-\beta E_i}} e^{-\beta E_i} = \frac{1}{Q} e^{-\beta E_i}$$

Therefore, internal energy

$$U = \langle E \rangle = \sum_{i=1}^N p_i E_i = \frac{1}{Q} \sum_{i=1}^N E_i e^{-\beta E_i} \quad (3.2)$$

On the other hand, if we take a partial derivative of the partition function with respect to β ,

$$\frac{\partial Q}{\partial \beta} = \frac{\partial}{\partial \beta} \sum_{i=1}^N e^{-\beta E_i} = \sum_{i=1}^N -E_i e^{-\beta E_i} = - \sum_{i=1}^N E_i e^{-\beta E_i} \quad (3.3)$$

Combining Equations 3.2 and 3.3, we can write

$$U = -\frac{1}{Q} \frac{\partial Q}{\partial \beta} = -\frac{\partial}{\partial \beta} \ln Q$$

So if we know the partition function of a system, we need to take the partial derivative with respect to β to get the internal energy. **This is how Statistical Mechanics forms a bridge between microscopic and macroscopic observables.** Partition function is a microscopic property as it is a sum over the energies of all individual microstates, which we can only obtain from the information about individual atoms and molecules. Internal energy, on the other hand, is a bulk thermodynamic property. Statistical mechanics provides a method for calculating bulk thermodynamic properties from microscopic information.

Let us now find a relationship between the U and Q in terms of the temperature T instead of β . According to the chain rule of derivatives

$$U = -\frac{\partial \ln Q}{\partial \beta} = -\frac{\partial \ln Q}{\partial T} \cdot \frac{\partial T}{\partial \beta} = -\frac{\partial \ln Q}{\partial T} \cdot \frac{1}{\frac{\partial \beta}{\partial T}}$$

Now as $\beta = 1/k_B T$,

$$\frac{\partial \beta}{\partial T} = \frac{\partial}{\partial T} \frac{1}{k_B T} = -\frac{1}{k_B T^2} \implies \frac{1}{\frac{\partial \beta}{\partial T}} = -k_B T^2$$

Therefore:

$$U = k_B T^2 \frac{\partial \ln Q}{\partial T}$$

3.3.2 Entropy (S)

Using the Shannon entropy formula

$$\begin{aligned}
S &= -k_B \sum_{i=1}^N p_i \ln p_i \\
&= -k_B \sum_{i=1}^N p_i \ln \left(\frac{e^{-\beta E_i}}{Q} \right) \\
&= -k_B \sum_{i=1}^N p_i \ln(e^{-\beta E_i}) + k_B \sum_{i=1}^N p_i \ln Q \\
&= k_B \beta \sum_{i=1}^N p_i E_i + k_B \langle \ln Q \rangle \\
&= k_B \cdot \frac{1}{k_B T} \cdot \langle E \rangle + k_B \ln Q \\
&= \frac{\langle E \rangle}{T} + k_B \ln Q
\end{aligned} \tag{3.4}$$

3.3.3 Helmholtz Free Energy (A)

The Helmholtz Free Energy (A) can be computed from the partition function in the following manner:

$$A = U - TS = U - T \left(\frac{U}{T} + k_B \ln Q \right) = -k_B T \ln Q$$

We will skip talking about other types of free energies and chemical potentials until a later stage. As the systems we have discussed so far do not have any pressure-volume relationships, Gibbs' Free Energy and chemical potentials will be relevant only when we discuss ideal gases.

3.4 Fluctuations and Response Functions

3.4.1 Constant volume heat capacity (C_V)

Now we will discuss one property that you may have learned in Thermodynamics, but we have not discussed so far. The constant volume heat capacity is defined as the amount of energy absorbed by a system to increase its temperature by one unit in a constant volume process. In statistical mechanics, these types of variables are known as **response functions**. Response functions describe how one thermodynamic function responds to a change in another thermodynamic function under controlled conditions.

As there is no P-V work involved (constant volume) and all the heat absorbed is only going to increase the internal energy of the systems:

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

We have previously derived an expression for the average energy in terms of the partition function Q . But in a constant temperature process, the energy is not necessarily constant, and it fluctuates. To estimate the extent of energy fluctuation, we must calculate the standard deviation or variance of the energy. The variance of energy (or any quantity) is defined by

$$\sigma_E^2 = \frac{1}{M} \sum_{i=1}^M (E_i - \langle E \rangle)^2$$

where M is the number of observations. The standard deviation is

$$\sigma_E = \sqrt{\frac{\sum_{i=1}^M (E_i - \langle E \rangle)^2}{M}}$$

The expression of variance can be rearranged as:

$$\begin{aligned}
\sigma_E^2 &= \frac{1}{M} \sum_{i=1}^M (E_i - \langle E \rangle)^2 \\
&= \frac{1}{M} \sum_{i=1}^M (E_i^2 - 2E_i \langle E \rangle + \langle E \rangle^2) \\
&= \frac{1}{M} \sum_{i=1}^M E_i^2 - \frac{2\langle E \rangle}{M} \sum_{i=1}^M E_i + \frac{\langle E \rangle^2}{M} \sum_{i=1}^M 1 \\
&= \langle E^2 \rangle - 2\langle E \rangle^2 + \langle E \rangle^2 \\
&= \langle E^2 \rangle - \langle E \rangle^2
\end{aligned} \tag{3.5}$$

We already showed that in terms of the partition function $\langle E \rangle = k_B T^2 \frac{\partial \ln Q}{\partial T}$

$$\begin{aligned}
\langle E^2 \rangle &= \sum_{i=1}^N p_i E_i^2 \\
&= \frac{1}{Q} \sum_{i=1}^N e^{-\beta E_i} E_i^2
\end{aligned} \tag{3.6}$$

Now

$$\begin{aligned}
C_V &= \left(\frac{\partial U}{\partial T} \right)_V \\
&= \left(\frac{\partial U}{\partial \beta} \right) \left(\frac{\partial \beta}{\partial T} \right) \\
&= \frac{\partial}{\partial \beta} \left(- \frac{\partial \ln Q}{\partial \beta} \right) \left(- \frac{1}{k_B T^2} \right) \\
&= \frac{1}{k_B T^2} \frac{\partial^2 \ln Q}{\partial \beta^2}
\end{aligned} \tag{3.7}$$

Now

$$\begin{aligned}
\frac{\partial^2 \ln Q}{\partial \beta^2} &= \frac{\partial}{\partial \beta} \left(\frac{1}{Q} \frac{\partial Q}{\partial \beta} \right) \\
&= \frac{1}{Q} \frac{\partial^2 Q}{\partial \beta^2} - \frac{1}{Q^2} \left(\frac{\partial Q}{\partial \beta} \right)^2 \\
&= \frac{1}{Q} \frac{\partial^2 Q}{\partial \beta^2} - \langle E \rangle^2 \\
&= \frac{1}{Q} \frac{\partial^2}{\partial \beta^2} \sum_{i=1}^N e^{-\beta E_i} - \langle E \rangle^2 \\
&= \frac{1}{Q} \frac{\partial}{\partial \beta} \left(- \sum_{i=1}^N E_i e^{-\beta E_i} \right) - \langle E \rangle^2 \\
&= \frac{1}{Q} \sum_{i=1}^N E_i^2 e^{-\beta E_i} - \langle E \rangle^2 \\
&= \langle E^2 \rangle - \langle E \rangle^2 = \sigma_E^2
\end{aligned} \tag{3.8}$$

Therefore

$$C_V = \frac{\sigma_E^2}{k_B T^2} = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}$$

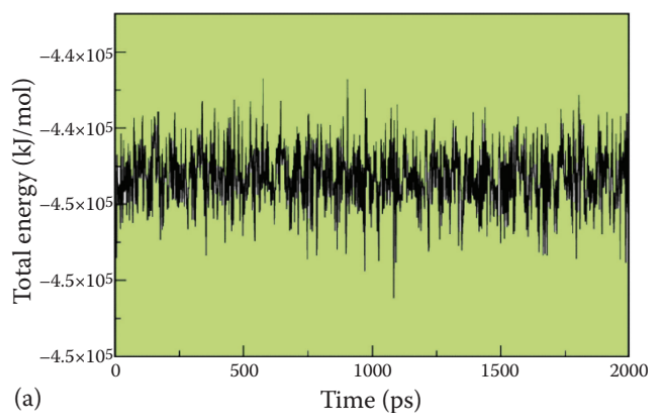


Figure 10: Energy Fluctuations in a box of water at constant temperature. Figure Courtesy: Biman Bagchi. Statistical Mechanics for Chemistry and Materials Science. CRC Press, 2018.

It means that the heat capacity is directly proportional to the fluctuation of the internal energy. Such relationships between response functions and fluctuations are quite general and can be shown for other properties, such as constant-pressure heat capacity or isothermal compressibility. For example, specific heat is given by the mean square fluctuation of energy or enthalpy, compressibility by the mean square fluctuation of number density, dielectric constant by the fluctuation of the total electric dipole moment, to name a few. In fact, each response function has an associated exact expression in terms of fluctuations. This is also a connection between macroscopic observable (response functions such as heat capacity) and microscopic properties (energy fluctuations at the atomic level).

3.4.2 Practical Example:

Heating a pot of water versus a piece of metal, both at room temperature.

High heat capacity (water): When you heat the pot of water, the temperature of the water increases slowly. This is because water has a high heat capacity and can absorb a lot of energy before its temperature changes significantly. Microscopically, the incoming energy is distributed in many ways—vibrating the water molecules, exciting their rotations, and stretching intermolecular hydrogen bonds. The energy is “soaked up” by these many microscopic motions, so the overall average kinetic energy (and thus the temperature) rises slowly. This ability to absorb and distribute energy across many modes corresponds to larger internal energy fluctuations.

Low heat capacity (metal): When you heat the piece of metal, its temperature rises much more quickly. Metal has a lower heat capacity, so it takes less energy to increase its temperature. The incoming energy is more focused on increasing the vibrations of the metal’s lattice atoms. There are fewer “degrees of freedom” to absorb the energy, so the temperature (average kinetic energy) changes more dramatically for the same amount of heat. This means the microscopic energy fluctuations are smaller relative to the macroscopic change in temperature. Conclusion: A material that can distribute energy among many modes (like water) has a higher heat capacity and larger internal energy fluctuations than a material with fewer energy-absorbing modes (like metal).

3.5 Examples of Partition Function

3.5.1 A two-level system

Consider a system that has N distinguishable particles with two energy levels for each particle: a ground state with energy zero and an excited state with energy $\epsilon = \epsilon_0 > 0$. This model is useful for describing many different problems: the behaviors of atoms or molecules that are excited by electromagnetic radiation, conduction of electrons in semiconductors, or the behavior of spins in magnetic fields. We want to find the average particle energy $\langle \epsilon \rangle$, the heat capacity C_V , the entropy, and the free energy per particle from the partition function. The partition function for a two-level system is:

$$q = 1 + e^{-\beta\epsilon_0}$$

The relative populations of the two states are given by the Boltzmann distribution:

$$p_1 = \frac{1}{q} = \frac{1}{1 + e^{-\beta\epsilon_0}} \quad \text{and} \quad p_2 = \frac{e^{-\beta\epsilon_0}}{q} = \frac{e^{-\beta\epsilon_0}}{1 + e^{-\beta\epsilon_0}}$$

The average energy of one particle is

$$\langle \epsilon \rangle = p_1 \cdot 0 + p_2 \cdot \epsilon_0 = \frac{\epsilon_0 e^{-\beta\epsilon_0}}{1 + e^{-\beta\epsilon_0}}$$

Now let us compute this same quantity from the partition function and see whether we get the same result.

$$\langle \epsilon \rangle = -\frac{\partial}{\partial \beta} \ln q = -\frac{\partial}{\partial \beta} \ln(1 + e^{-\beta\epsilon_0}) = -\frac{\frac{\partial}{\partial \beta} e^{-\beta\epsilon_0}}{1 + e^{-\beta\epsilon_0}} = \frac{\epsilon_0 e^{-\beta\epsilon_0}}{1 + e^{-\beta\epsilon_0}}$$

These two values agree with each other.

If we plot the energy of the two-state system as a function of temperature, at low temperatures, most molecules are in the ground state, so the system has low energy. As the temperature increases, the energy of the system increases and approaches the value $\epsilon_0/2$ per particle because energy levels 0 and ϵ_0 become equally populated.

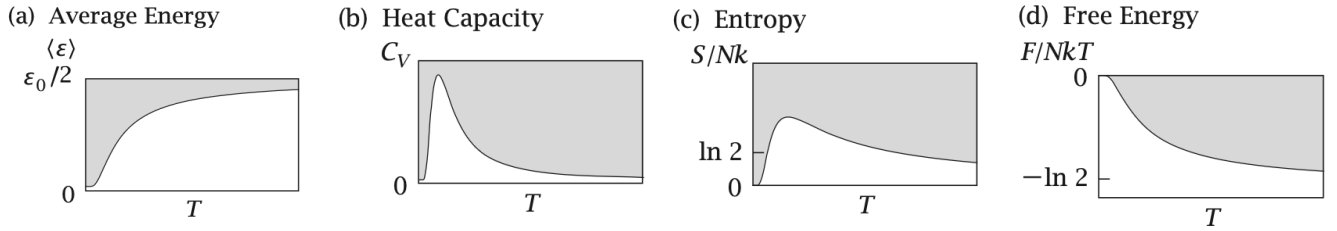


Figure 11: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.

Next, the heat capacity C_V can be calculated as follows:

To compute the heat capacity, use the definition $C_V = (\partial U / \partial T)$ from thermodynamics. Using Equation (10.36) to convert the total energy to the average energy per particle, $U = N\langle \epsilon \rangle$, you have

$$C_V = N \left(\frac{\partial \langle \epsilon \rangle}{\partial T} \right)_{V,N} = N \left(\frac{\partial \langle \epsilon \rangle}{\partial \beta} \right) \left(\frac{d\beta}{dT} \right) = -\frac{N}{kT^2} \left(\frac{\partial \langle \epsilon \rangle}{\partial \beta} \right), \quad (10.48)$$

where the right-hand expressions convert from T to β to make the next step of the differentiation simpler. Take a derivative of the form $d(u/v) = (vu' - uv')/v^2$, where $u = \epsilon_0 e^{-\beta\epsilon_0}$ and $v = 1 + e^{-\beta\epsilon_0}$, to get

$$\begin{aligned} \left(\frac{\partial \langle \epsilon \rangle}{\partial \beta} \right) &= \frac{(1 + e^{-\beta\epsilon_0})(-\epsilon_0^2 e^{-\beta\epsilon_0}) - \epsilon_0 e^{-\beta\epsilon_0}(-\epsilon_0 e^{-\beta\epsilon_0})}{(1 + e^{-\beta\epsilon_0})^2} \\ &= \frac{-\epsilon_0^2 e^{-\beta\epsilon_0}}{(1 + e^{-\beta\epsilon_0})^2}. \end{aligned} \quad (10.49)$$

Substitute Equation (10.49) into the right side of Equation (10.48) to find the heat capacity C_V in terms of the energy level spacing ϵ_0 :

$$C_V = \frac{N\epsilon_0^2}{kT^2} \frac{e^{-\beta\epsilon_0}}{(1 + e^{-\beta\epsilon_0})^2}. \quad (10.50)$$

Figure 12: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.

Heat capacity peaks are characteristic of bond-breaking and melting processes. At low temperatures, the thermal energy kT from the bath is too small to excite the system to its higher energy level. At intermediate temperatures, the system can absorb heat from the bath, and particles are excited into the higher-energy state. At the highest

temperatures, the system takes up no further energy from the bath because it has already taken up the maximum energy it can contain.

The entropy can be calculated as:

$$\begin{aligned}
 S &= \frac{\langle E \rangle}{T} + k_B \ln Q \\
 &= \frac{N\langle \epsilon \rangle}{T} + Nk_B \ln q \\
 &= \frac{N\epsilon_0 e^{-\beta\epsilon_0}}{T(1 + e^{-\beta\epsilon_0})} + Nk_B \ln(1 + e^{-\beta\epsilon_0})
 \end{aligned}$$

The free energy is:

$$\begin{aligned}
 A &= -k_B T \ln Q \\
 &= -Nk_B T \ln q \\
 &= -Nk_B T \ln(1 + e^{-\beta\epsilon_0})
 \end{aligned}$$

You do not need to memorize any of these expressions. This is just to show that when you encounter a new problem, it is possible to calculate different thermodynamic properties of the systems from its partition function.

3.5.2 Polymer Folding and Unfolding

Let's look at a more realistic application of the partition function in the case of polymer folding or polymer collapse transition. The polymer collapse transition is also the simplest model for protein folding. Let us look at a simple six-bead polymer. The compact or folded state of the polymer is most stable where the number of intrachain contacts are maximized. This is because most monomeric units are hydrophobic and they prefer to interact with themselves over the surrounding water molecule. This is a gross generalization but it will help us learn a few things. However, it is hard to predict the exact energy of each folded, unfolded and misfolded conformation. So we make a simple approximation that the stabilization energy is proportional to the number of intrachain contacts. All possible conformations of this six-bead polymer is shown in the figure below.

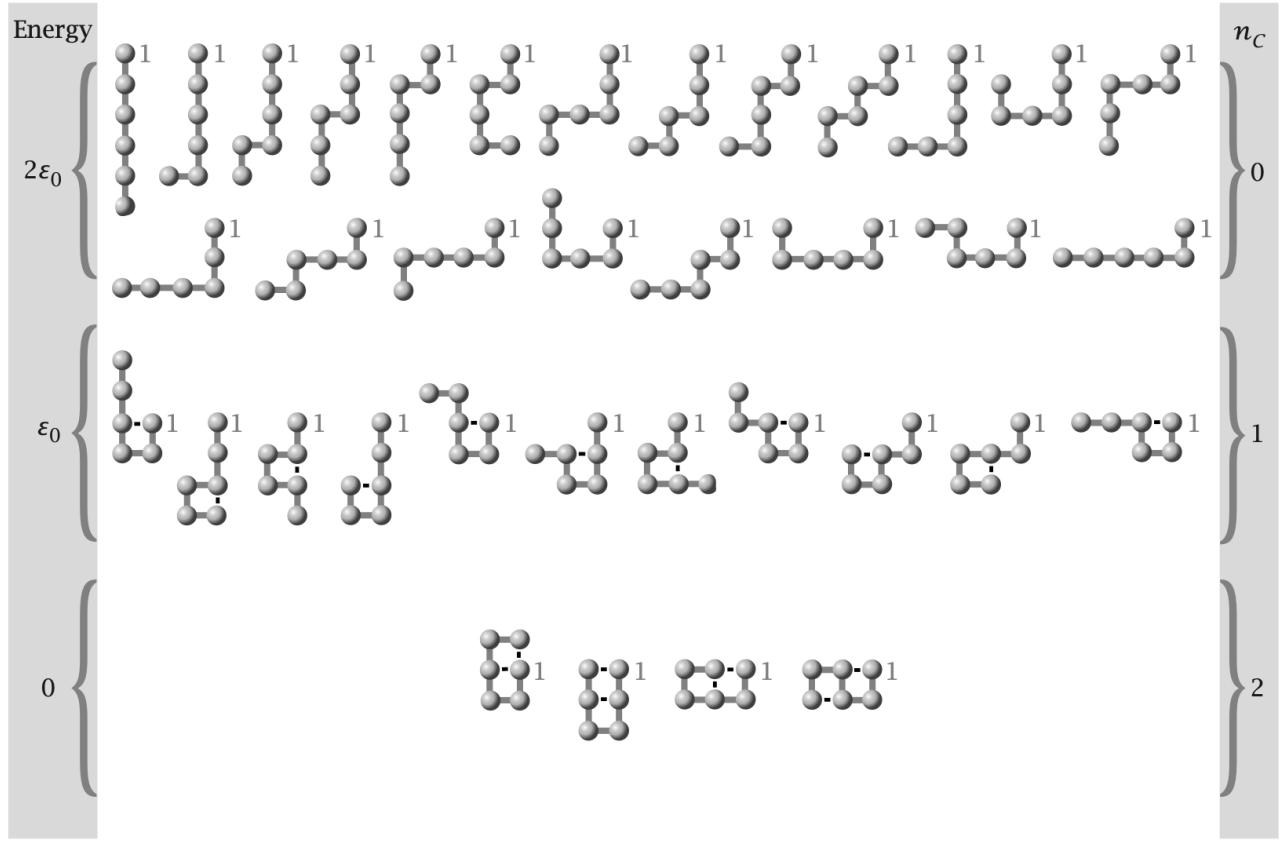


Figure 10.7 For the six-bead polymer chain, there are $W(0) = 4$ microstates having $n_c = 2$ contacts (contacts are shown as short dashes), $W(\epsilon_0) = 11$ microstates having 1 contact, and $W(2\epsilon_0) = 21$ microstates having 0 contacts.

Figure 13: Figure Courtesy: Ken Dill and Sarina Bromberg. *Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience*. Garland Science, 2010.

We assume that the most compact states with 2 intrachain contacts n_c (excluding the bonds) have energy zero. Loss of each contact leads to gaining ϵ_0 amount of energy. So fully extended conformations where there is no intrachain contact, have energy $2\epsilon_0$ as shown in the figure. What would be the partition function of this polymer? We use the macrostate degeneracy expression of partition function for simplicity.

$$Q = 4 + 11e^{-\beta\epsilon_0} + 21e^{-2\beta\epsilon_0}$$

Now if we compute the probability of each macrostate:

$$p(0, n_c = 2) = \frac{4}{4 + 11e^{-\beta\epsilon_0} + 21e^{-2\beta\epsilon_0}}$$

$$p(1, n_c = 1) = \frac{11e^{-\beta\epsilon_0}}{4 + 11e^{-\beta\epsilon_0} + 21e^{-2\beta\epsilon_0}}$$

$$p(2, n_c = 0) = \frac{21e^{-2\beta\epsilon_0}}{4 + 11e^{-\beta\epsilon_0} + 21e^{-2\beta\epsilon_0}}$$

Plotting these probabilities as a function of temperature we have the following observation:

Figure 10.8 Six-bead-chain polymer collapse of Example 10.4. At low temperatures T , the ground state is most stable ($p(0)$ is largest). At high temperatures, the open state is most stable ($p(2)$ is largest). Near the middle of the thermal transition, the intermediate state is most populated ($p(1)$ is largest).

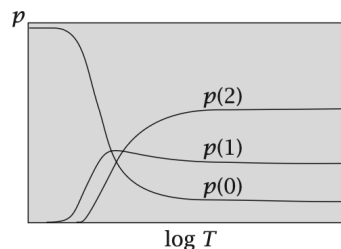


Figure 14: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.

3.5.2.1 Melting point

In protein folding, RNA folding, and DNA helix formation, we use a term melting point (different from the melting of solid to liquid). In case of biomacromolecules, melting point is the temperature at which the probability of being folded and unfolded is exactly equal. If we consider the $n_c = 1$ states as some sort of unfolded states (technically they do not have the most compact structure required to be categorized as folded), can you calculate the melting point of this polymer? At the melting point (T_m),

$$\begin{aligned}
 p(0, n_c = 2) &= p(1, n_c = 1) + p(2, n_c = 0) \\
 \Rightarrow \frac{4}{4 + 11e^{-\beta\epsilon_0} + 21e^{-2\beta\epsilon_0}} &= \frac{11e^{-\beta\epsilon_0}}{4 + 11e^{-\beta\epsilon_0} + 21e^{-2\beta\epsilon_0}} + \frac{21e^{-2\beta\epsilon_0}}{4 + 11e^{-\beta\epsilon_0} + 21e^{-2\beta\epsilon_0}} \\
 \Rightarrow 4 &= 11e^{-\beta\epsilon_0} + 21e^{-2\beta\epsilon_0} \\
 \Rightarrow 21x^2 + 11x - 4 &= 0 \quad \text{where } x = e^{-\beta\epsilon_0}
 \end{aligned}$$

This is a quadratic equation with two solutions: $x_1 = -0.770$ and $x_2 = 0.247$. From physical intuition we know that $e^{-\beta\epsilon_0} > 0$, so we can only consider the positive solution. Therefore:

$$e^{-\beta\epsilon_0} = 0.247 \Rightarrow \frac{\epsilon_0}{k_B T_m} = -\ln(0.247) \Rightarrow T_m = -\frac{\epsilon_0}{k_B \ln(0.247)}$$

So if we can figure out the value of the stabilization energy for each contact (ϵ_0) using either experiment or theory such as quantum mechanics, we can predict the melting point of this polymer.

Exercise: Compute the free energy, entropy, and the heat capacity of the polymer.

Interactive Tutorial

3.5.3 Free Energy Difference: the determinant of stability

Now, if I ask you the question, "How stable is the folded state of the polymer?", how would you answer? This question arises very often in chemistry and biochemistry. The stability of a drug-target bound structure relative to the unbound state determines how strongly the drug binds and its ability to cure diseases. The stability of ice crystals with respect to water determines whether water will freeze. We have learned to compute the free energy of a whole system, but not to compare the free energy change associated with a transformation like protein folding or water freezing. For that, we need to define partition functions for subsets of the system. For example, to compute the free energy change associated with the folding of the six-bead polymer, we need to construct two different partition functions, one for the folded macrostate with four microstate, and another one for the unfolded and misfolded state with $11+21 = 32$ microstates. This approach can be tractable for small model systems but becomes impossible to work with when we study realistic molecular systems. So we need to obtain a straightforward expression for calculating the free energy difference without having to compute partition functions for every arbitrarily defined state. Let's say there are two different groups of microstates (such as folded and unfolded, bound or unbound, etc.) between which we want to calculate the free energy difference. Each group may contain one or more macrostates.

The first group A , contains microstates $i = 1$ to $i = k$, while the second group B , contains microstates $i = k + 1$ to $i = N$. (There are a total of N microstates).

$$Q_A = \sum_{i=1}^k e^{-\beta E_i} \quad \text{and} \quad Q_B = \sum_{i=k+1}^N e^{-\beta E_i}$$

Note

$$Q = \sum_{i=1}^N e^{-\beta E_i} = \sum_{i=1}^k e^{-\beta E_i} + \sum_{i=k+1}^N e^{-\beta E_i} = Q_A + Q_B$$

and

$$p_A = \frac{\sum_{i=1}^k e^{-\beta E_i}}{\sum_{i=1}^N e^{-\beta E_i}} = \frac{Q_A}{Q} \quad \text{and similarly} \quad p_B = \frac{Q_B}{Q}$$

We know the expression for free energy is $A = -k_B T \ln Q$. So the free energy difference between groups A and B is:

$$\begin{aligned} \Delta A &= A_A - A_B \\ &= -k_B T (\ln Q_A - \ln Q_B) \\ &= -k_B T (\ln Q_A - \ln Q - \ln Q_B + \ln Q) \\ &= -k_B T \left(\ln \frac{Q_A}{Q} - \ln \frac{Q_B}{Q} \right) \\ &= -k_B T \left(\ln p_A - \ln p_B \right) \\ &= -k_B T \ln \frac{p_A}{p_B} \end{aligned}$$

So, the free energy difference between two groups of states is proportional to the ratio of their probabilities. This is significantly more useful than the partition function expression, as relative probabilities can be measured from experiments and simulations. Say, for example, the issue of protein folding. One can use single-molecule fluorescence experiments to measure the distance between two probes located at the two ends of a protein. When the distance between them is significant, we can consider the protein to be unfolded (or misfolded), and when the distance is small, we can consider the protein to be folded (see the six-bead example). Below is a plot of a single molecule trace of the end-to-end distance of a protein with time stamps for each transition event between folded and unfolded.

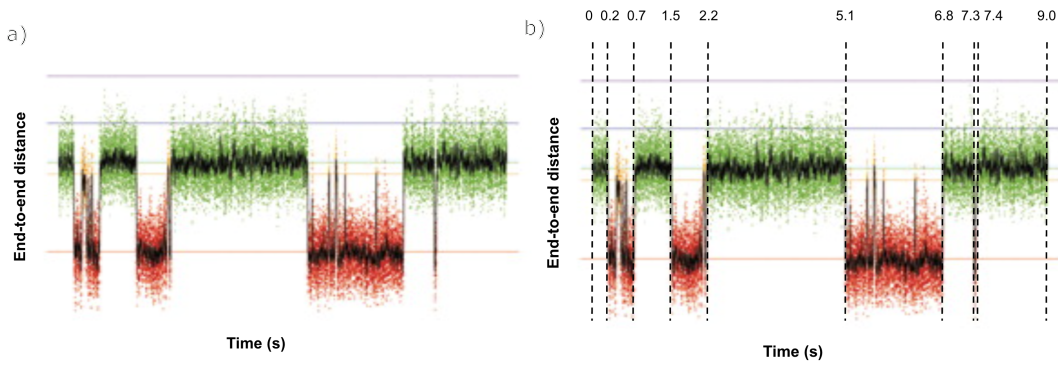


Figure 15: Single molecule fluorescence spectroscopy results of protein molecule showing folding-unfolding transitions as a function of time.

The red-colored region has a lower end-to-end distance, corresponding to the folded conformation. The total time trace is 9s long. Out of that, it remains folded for $(0.7 - 0.2) + (2.2 - 1.5) + (6.8 - 5.1) + (7.4 - 7.3)s = 3.0s$. So

$p(\text{folded}) = 3/9 = 1/3$, and $p(\text{unfolded}) = 6/9 = 2/3$. So the free energy difference between the folded and the unfolded state is

$$\Delta A = -k_B T \ln \frac{p(\text{folded})}{p(\text{unfolded})} = -k_B T \ln \frac{1/3}{2/3} = -k_B T \ln(1/2) = k_B T \ln 2$$

This is a positive number, which means the folded state is less stable than the unfolded state. Here, we refer to the fact that a reduction in the system's free energy means an increase in the universe's entropy, indicating a thermodynamically favorable process. Here, transitioning from an unfolded to a folded state leads to an increase in free energy, indicating that it is destabilizing for the system to remain folded.

Exercise: Calculate the folding free energy for a six-bead polymer. Assume $\epsilon_0 = 1k_B T$. At what temperature range is the folded state more stable?

3.6 Single particle (q) and multi-particle (Q_N) partition function for distinguishable and indistinguishable particles

When we discuss the probabilities of independent events, we can calculate the probability of two independent events occurring simultaneously by multiplying the probabilities of the individual events. For example, if we flip a coin and roll a six-sided dice, the probability obtaining a head and the number 6 is $1/2 \times 1/6 = 1/12$. Similarly, we can obtain the partition function of a system consisting of N distinguishable particles (Q_N) by just multiplying the partition function of each particles q_i :

$$Q_N = \prod_{i=1}^N q_i = q_1 q_2 q_3 \dots q_N$$

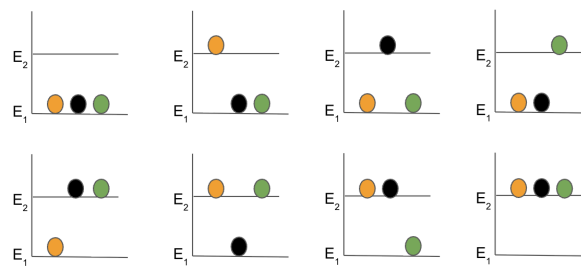
Consequently, if the energy levels of all the particles are identical, then their single-particle partition functions q will also be identical. In that case, we can write:

$$Q_N = q^N$$

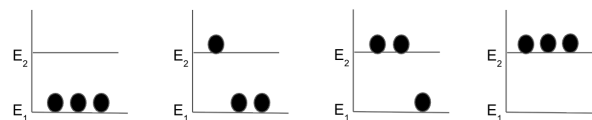
This expression is particularly useful in chemical systems, as in a bulk material or solution, where all molecules typically experience an identical environment and therefore have identical single-particle partition functions. If we can identify the energy levels associated with a single molecule or atom, we can then calculate the partition function of the bulk material and predict bulk properties, such as the heat capacity of a block of metal.

The word **distinguishable** is essential here, as if these particles were indistinguishable, i.e., they can be interchanged without changing the microstate of the system, this expression will not be valid. Below is a depiction of three particles in two energy levels, both for the distinguishable and indistinguishable cases.

Distinguishable Particles



Indistinguishable Particles



The single particle partition function is

$$q = e^{-\beta E_1} + e^{-\beta E_2}$$

If the particles are distinguishable, there are $2^3 = 8$ possible rearrangements. The total energy of the first configuration is $3E_1$ as all three particles are in energy state E_1 . The total energy of the second configuration is $2E_1 + E_2$ as two particles are in energy state E_1 , and one is in E_2 , and so on. If we now explicitly calculate the partition function of the three-particle system in the distinguishable scenario:

$$\begin{aligned}
Q_3 &= e^{-\beta(3E_1)} + e^{-\beta(2E_1+E_2)} + e^{-\beta(2E_1+E_2)} + e^{-\beta(2E_1+E_2)} + e^{-\beta(E_1+2E_2)} + e^{-\beta(E_1+2E_2)} + e^{-\beta(E_1+2E_2)} + e^{-\beta(3E_2)} \\
&= (e^{-\beta E_1})^3 + 3(e^{-\beta E_1})^2(e^{-\beta E_2}) + 3(e^{-\beta E_1})(e^{-\beta E_2})^2 + (e^{-\beta E_2})^3 \\
&= (e^{-\beta E_1} + e^{-\beta E_2})^3 \\
&= q^3
\end{aligned}$$

This result is in agreement with the formula we stated earlier.

However, if the particles are **indistinguishable**, there are only four states and:

$$Q_3 = e^{-\beta(3E_1)} + e^{-\beta(2E_1+E_2)} + e^{-\beta(E_1+2E_2)} + e^{-\beta(3E_2)} \neq q^3$$

If we use q^3 , we over-count the state in which the particles are in different energy levels. In general, there is no simple expression for the N -particle partition function for indistinguishable particles. This is a problem, as when molecules can move around, for example, in a gas or liquid phase, they are indistinguishable. We therefore need to find an approximate method to calculate the multiparticle partition function from the single-particle one, for indistinguishable particles.

We notice that overcounting occurs only for states where particles are distributed across more than one energy level. In the case of molecular systems, there is usually an infinite number of energy levels accessible to the system (because we cannot explicitly fix the amount of energy in a system). So the number of particles is always significantly lower than the number of energy levels. In that situation, the probability that two particles are on the same level is extremely low. This is especially true if the energy levels are separated by a very small amount of energy, often $\ll k_B T$. We can therefore ignore the contribution from the states in which one energy level is occupied by more than one particle.

Distinguishable Particles

E_4				B			B		B	AB			A		A	A
E_3			B			B		AB	A			A		A		B
E_2		B			AB	A	A				A			B	B	
E_1	AB	A	A	A							B	B	B			

Indistinguishable Particles

E_4				A			A		A	AA
E_3			A			A		AA	A	
E_2		A			AA	A	A			
E_1	AA	A	A	A						

Let us explain this argument with an example. Say there are two particles in four energy levels. The number of ways we can arrange two particles in four levels is $4^2 = 16$. Out of these, there are 4 arrangements in which two particles are in the same state. So the probability of both particles being in the same state is $1/4$. If there are n energy levels, this probability becomes $\frac{n}{n^2} = \frac{1}{n}$. So, the larger the number of energy levels, the lower the likelihood that two particles will be on the same level.

Now, let us see how the partition function differs between the distinguishable and indistinguishable cases. If they are distinguishable, we name them A and B ; otherwise, we name both particles as A . First, we ignore the shaded states where more than one particle is in a single energy level. The two-particle partition function for the distinguishable

case is

$$Q_2^{(\text{dis})} = e^{-\beta(E_1^A + E_2^B)} + e^{-\beta(E_1^A + E_3^B)} + e^{-\beta(E_1^A + E_4^B)} + e^{-\beta(E_2^A + E_3^B)} + e^{-\beta(E_2^A + E_4^B)} + e^{-\beta(E_3^A + E_4^B)} + e^{-\beta(E_1^B + E_2^A)} + e^{-\beta(E_1^B + E_3^A)} + e^{-\beta(E_1^B + E_4^A)} + e^{-\beta(E_2^B + E_3^A)} + e^{-\beta(E_2^B + E_4^A)} + e^{-\beta(E_3^B + E_4^A)}$$

The superscript of the energies is not a power but the index of the particle. The second line is identical to the first line, except that the particles are swapped. So

$$Q_2^{(\text{dis})} = 2(e^{-\beta(E_1 + E_2)} + e^{-\beta(E_1 + E_3)} + e^{-\beta(E_1 + E_4)} + e^{-\beta(E_2 + E_3)} + e^{-\beta(E_2 + E_4)} + e^{-\beta(E_3 + E_4)})$$

If the particles are indistinguishable, such as in a swap, it would not count as a different state. So partition functions for the:

$$Q_2^{(\text{indis})} = e^{-\beta(E_1 + E_2)} + e^{-\beta(E_1 + E_3)} + e^{-\beta(E_1 + E_4)} + e^{-\beta(E_2 + E_3)} + e^{-\beta(E_2 + E_4)} + e^{-\beta(E_3 + E_4)}$$

We see that:

$$Q_2^{(\text{indis})} = \frac{1}{2!} Q_2^{(\text{dis})} \approx \frac{q^2}{2!}$$

as the N particle partition function is $Q_N = q^N$ for distinguishable particles. The $2!$ term in the denominator takes care of the overcounting of identical states for the two indistinguishable particles. If there are N **indistinguishable particles**,

$$Q_N = \frac{q^N}{N!}$$

as N particles can be rearranged in $N!$ ways. This expression will be useful in the study of an ideal gas where molecules are indistinguishable.

3.7 Summary

- The probability of a system being in energy state E is

$$p \propto e^{-\beta E}$$

and therefore the relative probability of being in state A over state B is

$$\frac{p_A}{p_B} = e^{-\beta(E_A - E_B)}$$

- The partition function Q is given by

$$Q = \sum_{i=1}^N e^{-\beta E_i} = \sum_{j=1}^M g_j e^{-\beta E_j}$$

where i and j are indices of microstates and macrostates, respectively, and g_j is the degeneracy of the j -th macrostate. Also the probability of being in state i is

$$p_i = \frac{e^{-\beta E_i}}{Q}$$

- Almost all thermodynamic quantities can be calculated from the partition function, such as

$$\text{Internal Energy: } U = \langle E \rangle = -\frac{\partial}{\partial \beta} \ln Q$$

$$\text{Entropy: } S = \frac{U}{T} + k_B \ln Q$$

$$\text{Free Energy: } A = -k_B T \ln Q$$

- Response functions, such as the constant volume heat capacity, are proportional to the fluctuations of the corresponding extensive variable, such as the internal energy.

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}$$

- The N -particle partition function Q_N is related to the single particle partition function q as

$$\text{Distinguishable particles: } Q_N = q^N \quad \text{Indistinguishable particles: } Q_N = \frac{q^N}{N!}$$

- If A and B are two macrostates or groups of macrostates, the free energy difference between them can be computed from their relative probability. The lower the free energy, the more stable the state.

$$\Delta A = -k_B T \ln \frac{p_A}{p_B}$$

4 Molecular Partition Function

Now we are equipped with the fundamental understanding of the partition functions and basic principles of statistical mechanics. We will now study some relatively simple and theoretically solvable models of molecules and materials. For that, we need to identify the energy levels of such systems. The atoms in molecules of materials can have primarily 4 types of energy:

1. **Vibrational Energy:** e.g., bond vibrations, vibration of atoms in crystal lattice, etc. **Model:** Simple Harmonic Oscillator.
2. **Translational Energy:** Movement of non-bonded atoms or molecules in 3D space. **Model:** Ideal Gas.
3. **Rotational Energy:** Rotation of the whole molecule or specific bonds and torsions (say the butane example). **Model:** Rigid Rotor.
4. **Electronic Energy:** The electronic energy levels and orbitals. **Model:** Atomic and Molecular Orbital Theory and its advanced variants.

For the purpose of this course, we will assume that these four different contributions to the molecular energy are independent of each other. This has a significant advantage. Due to the multiplicability of partition functions of independent systems, we can just compute the partition functions for each different types of energy states and multiply them to get the total molecular partition function:

$$Q_{\text{mol}} = Q_{\text{trans}} Q_{\text{rot}} Q_{\text{vib}} Q_{\text{elec}}$$

However, please remember that this assumption of independence is not always valid in practical systems, and different energetic modes such as rotation and vibration are often coupled and interact with each other. Such situations are usually not tractable using analytical theory and can be studied using computer simulations.

4.1 Quantum Mechanics

When discussing the energies of large macroscopic objects, such as a baseball or a stone, we employ classical mechanics. In classical mechanics, the total energy of a system is given by the sum of kinetic and potential energies. The kinetic energy is usually $E_{\text{kin}} = \frac{1}{2}mv^2$, where m and v are the mass and velocity respectively. The potential energy varies from system to system. For example, in the simple harmonic oscillator, such as a mass attached to a spring, the potential energy is $E_{\text{pot}} = \frac{1}{2}kx^2$, where k is the spring constant and x is the displacement from the equilibrium position. Therefore, the total energy of a harmonic oscillator is $E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$. Note that this is a continuous function, i.e., the energy can take any possible value between zero and infinity.

At the microscopic scale, atoms and molecules do not behave this way. Their dynamics and energetics are governed by quantum mechanics. One significant consequence of quantum mechanics is that the energy of the system is quantized at discrete values. The allowable values of the energies can be obtained by solving the Schrödinger Equation. We will leave further discussion on quantum mechanics for your next class in the PChem sequence: Chem 413. For the time being, it will suffice to know that molecular systems have discrete energy levels, which, unlike classical energy expressions, are generally independent of the exact position and velocity of the particle. The expressions for the energy levels can be derived from quantum mechanics.

The remainder of this module will be organized as follows. We will examine each of the partition functions, vibrational, rotational, and translational, individually. In each subsection, we will first state the quantum mechanical energy levels of the corresponding model system without proof. Then, we will compute the partition function, from which we will subsequently derive thermodynamic quantities, such as internal energy and heat capacity. Finally, we will combine all these different partition functions to derive a general partition function for a model diatomic molecule.

4.2 Vibrational Motion: Simple Harmonic Oscillator

4.2.1 Energy levels

The energy levels of a one-dimensional simple harmonic oscillator are given by:

$$E_n = \left(n + \frac{1}{2}\right)h\nu \quad \text{where } n = 0, 1, 2, \dots, \infty$$

Here ν is the frequency of oscillation, i.e., the number of full cycles of oscillations completed in a unit time. The h is known as Planck's Constant. It is a universal constant with a value of $6.62607015 \times 10^{-34}$ Joule-second. The n is called a quantum number, which can have discrete integer values. This factor ensures the energy is discrete and determines the energy level. So the energy levels of an 1D harmonic oscillators are: $\frac{1}{2}h\nu, \frac{3}{2}h\nu, \frac{5}{2}h\nu, \dots$. Now, you may understand why we discussed discrete energy states extensively in the previous modules. Going forward, we have to deal with energy levels like this instead of continuous total energies.

However, there is one aspect of the energy levels of a harmonic oscillator that differs from the energy levels we studied previously. In our earlier examples, like the polymer folding, the total number of energy levels was finite. Here, the energy levels can go up to infinity. To compute partition functions, therefore, we need to learn how to compute infinite sums.

4.2.2 Partition function

The partition function of harmonic oscillator is:

$$\begin{aligned} Q &= \sum_{n=0}^{\infty} e^{-\beta E_n} \\ &= \sum_{n=0}^{\infty} e^{-\beta(n+1/2)h\nu} \\ &= \sum_{n=0}^{\infty} e^{-\beta h\nu/2} \cdot e^{-\beta n h\nu} \\ &= e^{-\beta h\nu/2} \sum_{n=0}^{\infty} (e^{-\beta h\nu})^n \end{aligned}$$

Now, note that $\beta h\nu$ is a positive number, so $e^{-\beta h\nu} < 1$. We can now use the following expression from the theory of infinite series:

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots = \frac{1}{1-x} \quad \text{if } x < 1$$

Substituting $e^{-\beta h\nu}$ for x , we can write

$$Q = \frac{e^{-\beta h\nu/2}}{1 - e^{-\beta h\nu}}$$

4.2.3 Internal Energy

$$\begin{aligned} U &= -\frac{\partial \ln Q}{\partial \beta} \\ &= -\frac{\partial}{\partial \beta} \ln \left(\frac{e^{-\beta h\nu/2}}{1 - e^{-\beta h\nu}} \right) \\ &= -\frac{\partial}{\partial \beta} \ln(e^{-\beta h\nu/2}) + \frac{\partial}{\partial \beta} \ln(1 - e^{-\beta h\nu}) \\ &= \frac{h\nu}{2} + \frac{h\nu e^{-\beta h\nu}}{1 - e^{-\beta h\nu}} \\ &= \frac{h\nu}{2} + \frac{h\nu}{e^{\beta h\nu} - 1} \end{aligned}$$

There are two things to note in this expression:

1. When $T \rightarrow 0$, $\beta \rightarrow \infty$ and $e^{\beta h\nu} - 1 \rightarrow \infty$. Which means

$$T \rightarrow 0 \implies U \rightarrow \frac{h\nu}{2}$$

Therefore, even at an absolute temperature of zero, the harmonic oscillator retains some amount of internal energy. This is known as the **zero-point energy**, and it is a consequence of quantum mechanics.

2. When $T \rightarrow \infty$, $\beta \rightarrow 0$. From the mathematics of infinite series it can be shown:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty \text{ if } x < 1$$

If, however, $x \rightarrow 0$, i.e., $x \ll 1$, the terms with power two or more become negligibly small. So

$$e^x \rightarrow 1 + x \text{ if } x \rightarrow 0$$

Therefore, when $T \rightarrow \infty$, $e^{\beta h\nu} \rightarrow 1 + \beta h\nu$ and

$$U \rightarrow \frac{h\nu}{2} + \frac{h\nu}{1 + \beta h\nu - 1} = h\nu + \frac{1}{\beta} = k_B T$$

as $h\nu \ll k_B T$ at high temperature. So for 1D harmonic oscillator,

$$\boxed{T \rightarrow \infty \implies U \rightarrow k_B T}$$

This is the classical limit where quantum effects (such as Planck's constant) become negligible.

4.2.4 Heat Capacity

We cannot experimentally measure the internal energy of a molecular system, but we can measure the heat capacity C . Therefore, going forward, we will compute the heat capacity contribution from all the energy modes of a molecule (vibration, rotation, and translation).

$$C = \left(\frac{\partial U}{\partial T} \right) = \left(\frac{\partial U}{\partial \beta} \right) \left(\frac{\partial \beta}{\partial T} \right) = -\frac{1}{k_B T^2} \left(\frac{\partial U}{\partial \beta} \right)$$

We see that

$$\left(\frac{\partial U}{\partial \beta} \right) = \frac{\partial}{\partial \beta} \left(\frac{h\nu}{e^{\beta h\nu} - 1} \right) = -\frac{(h\nu)^2 e^{\beta h\nu}}{(e^{\beta h\nu} - 1)^2}$$

Therefore,

$$C = \frac{(h\nu)^2}{k_B T^2} \frac{e^{\beta h\nu}}{(e^{\beta h\nu} - 1)^2} = k_B \frac{(\beta h\nu)^2 e^{\beta h\nu}}{(e^{\beta h\nu} - 1)^2}$$

To measure the heat capacity experimentally, it would be easier if we could express it in terms of T instead of β .

$$C(T) = k_B \frac{(h\nu/k_B T)^2 e^{h\nu/k_B T}}{(e^{h\nu/k_B T} - 1)^2} = k_B \left(\frac{\Theta_{\text{vib}}}{T} \right)^2 \frac{e^{\Theta_{\text{vib}}/T}}{(e^{\Theta_{\text{vib}}/T} - 1)^2}$$

where $\Theta_{\text{vib}} = h\nu/k_B$.

The term Θ_{vib} is an effective vibrational temperature of the system. At $T > \Theta_{\text{vib}}$ we can use the classical limit, as the energy difference between the levels becomes so low that the discrete energy levels effectively become a continuum. The Θ_{vib} term therefore defines the boundary of the classical and quantum regimes. For example, the ν for the C-O bond stretching in carbon monoxide is $6.42 \times 10^{13} \text{ s}^{-1}$. Using $h = 6.626 \times 10^{-34} \text{ Js}$ and $k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$, $\Theta_{\text{vib}} = 3082 \text{ K}$. Therefore at room temperature $T < \Theta_{\text{vib}}$ and the C-O bond stretching will show significant quantum mechanical behaviour.

Interactive Tutorial

4.2.5 Heat Capacity of a Solid Crystal

Before proceeding with the molecular partition function, let's examine an example that utilizes the statistical mechanics of the simple harmonic oscillator to understand the properties of solid materials. Specifically, we will examine the heat capacity of homogeneous, isotropic solid crystals. The **Dulong-Petit Law** states that the molar heat capacity of solid elements at high temperatures is approximately constant and equal to $3R$, where R is the universal gas constant. While useful for understanding solids with simple structures at high temperatures, the law fails at low temperatures, where the heat capacity goes to zero. Albert **Einstein** used the quantum mechanical simple harmonic oscillator model to explain this effect.

Note that the problem of crystal vibration is different from the 1D harmonic oscillator model in two ways:

1. The atoms in crystal lattice in all three directions x, y, z , instead of just 1D
2. There is more than one atom in a crystal lattice.

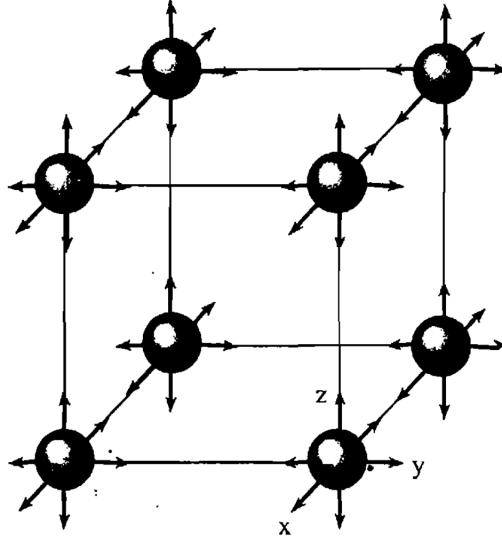


Fig. 3.5 In a very simple model of the vibrational motion of atoms in a crystal we consider the atoms to form an array of SHOs, each with natural frequency ν , that can vibrate in all three directions.

Figure 16: Figure Courtesy: Wark, J. S., and A. M. Glazer. Statistical Mechanics: A Survival Guide. Oxford University Press, 2001.

For simplicity, we assume that the crystal is isotropic, i.e., the vibrational frequency in each of the x, y , or z direction is identical: $\nu_x = \nu_y = \nu_z = \nu$. We also assume that the energy states of each atom are independent from those of the other atoms and that the atoms are distinguishable. These are reasonable assumptions for a solid atomic crystal, such as a diamond or metal. We also assume that the vibrations along each Cartesian coordinate; therefore, the 3D single particle partition function is

$$q_{3D} = q_x q_y q_z = q^3$$

as all directions are equivalent. The q is the single particle partition function in 1D:

$$q = \frac{e^{-\beta h\nu/2}}{1 - e^{-\beta h\nu}}$$

The 3D N -particle partition function is

$$Q_N = q_{3D}^N = q^{3N} = \left(\frac{e^{-\beta h\nu/2}}{1 - e^{-\beta h\nu}} \right)^{3N}$$

Before we plug in the expression for q into this formula, let us see if we can utilize the U and C_V we already calculated for the 1D system to simplify our calculations for the 3D N -particle system:

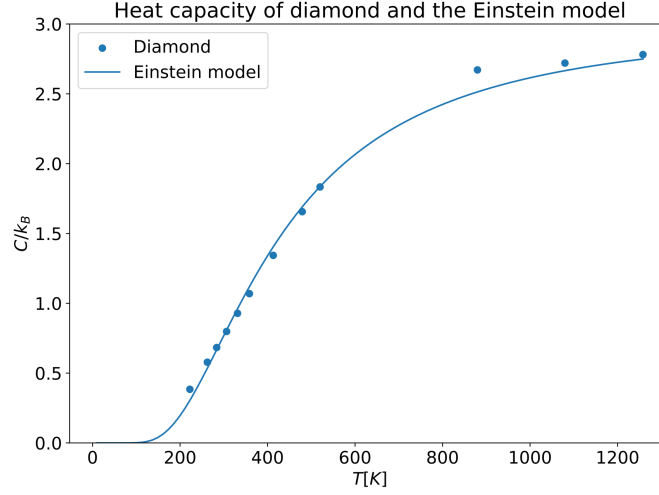
$$U_{N,3D} = -\frac{\partial \ln Q_N}{\partial \beta} = -\frac{\partial \ln q^{3N}}{\partial \beta} = -3N \frac{\partial \ln q}{\partial \beta} = 3NU_{1,1D}$$

$$C_{N,3D} = \frac{\partial U_{N,3D}}{\partial T} = 3N \left(\frac{\partial U_{N,3D}}{\partial T} \right) = 3NC_{1,1D}$$

We observe that the 1-particle 1D heat capacity can be directly used to obtain the 3D N -particle heat capacity. To make the heat capacity independent of system size, we often talk about the per-particle heat capacity: $C = C_{N,3D}/N$. From the expression we already derived for the 1D simple harmonic oscillator:

$$C(T) = 3k_B \left(\frac{\Theta_{\text{vib}}}{T} \right)^2 \frac{e^{\Theta_{\text{vib}}/T}}{(e^{\Theta_{\text{vib}}/T} - 1)^2} \quad \text{where} \quad \Theta_{\text{vib}} = \frac{h\nu}{k_B}$$

This is the Einstein model of a solid's heat capacity. In the plot below, we see that it agrees very well with the experimentally measured heat capacities at various temperatures for diamond. In comparison, the Dulong-Petit law would have predicted $C = 3k_B$ for all temperatures.



4.3 Rotational Motion: Rigid Rotor

Apart from vibration, another internal degree of freedom of a molecule is the rotational motion, which contributes rotational energy to the system. To understand this energy source, we will utilize the rigid rotor model, which is analogous to a diatomic molecule with a fixed bond length (bond length variations are accounted for by the vibrational partition function).

4.3.1 Classical Mechanics of the Rigid Rotor Model.

$$\text{Kinetic Energy of Translation} = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2$$

$$\text{Kinetic Energy of Rotation} = \frac{1}{2}I\omega^2 = \frac{1}{2}I\left(\frac{d\theta}{dt}\right)^2$$

Here x is the linear displacement, v is the linear velocity, θ is the angular displacement, ω is the angular velocity, and I is the moment of inertia which is the equivalent of mass in rotational energy. The moment of inertia can be computed as

$$I = \sum_i m_i r_i^2$$

where m_i is the mass of the i -th constituent atom and r_i is its distance from the center of mass. In the rigid rotor model in the picture below: $I = M_1 R_1^2 + M_2 R_2^2$. If we want to express I in terms of the bond length $R = R_1 + R_2$: $I = \mu R^2$, where $\mu = \frac{M_1 M_2}{M_1 + M_2}$.

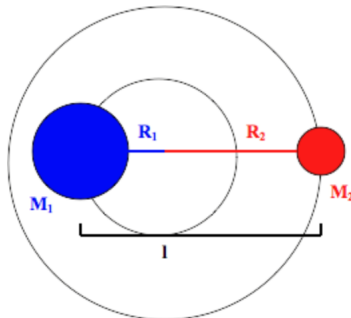


Figure 1: Rigid Rotor Model of a Diatomic Molecule

4.3.2 Energy Levels

From quantum mechanics, we can derive that the energy levels of a rigid rotor are

$$E_J = J(J+1) \frac{\hbar^2}{2I}; \quad J = 0, 1, 2, \dots, \infty$$

Here, J is the integer quantum number which ensures that the energy levels are discrete; $\hbar = \frac{h}{2\pi}$, and I is the moment of inertia.

Moreover, each energy level has a degeneracy of

$$g_J = 2J + 1$$

4.3.3 Partition Function

The partition function

$$Q = \sum_{J=0}^{\infty} g_J e^{-\beta E_J} = \sum_{J=0}^{\infty} (2J+1) e^{-\beta J(J+1) \hbar^2 / 2I} = \sum_{J=0}^{\infty} (2J+1) e^{-J(J+1) \Theta_{\text{rot}} / T}$$

where $\Theta_{\text{rot}} = \frac{\hbar^2}{2Ik_B}$ is the effective rotational temperature, similar to the Θ_{vib} quantity we introduced in the vibrational partition function. At $T > \Theta_{\text{rot}}$, the system will behave classically as the energy levels will be so close to each other (with respect to the thermal energy $k_B T$) that they will appear almost continuous. Contrarily, at $T < \Theta_{\text{rot}}$, quantum effects and the discrete nature of the energy levels will be significant.

For example, the moment of inertia (I) of the carbon monoxide (CO) is 1.46×10^{-46} kg m². Therefore, $\Theta_{\text{rot}} = 2.76$ K, which is significantly lower than the room temperature. Therefore, at room temperature, the rotational energy levels will appear continuous. In this situation it is advisable to replace the summation with an integration:

$$Q = \sum_{J=0}^{\infty} (2J+1) e^{-J(J+1) \Theta_{\text{rot}} / T} = \int_0^{\infty} (2J+1) e^{-J(J+1) \Theta_{\text{rot}} / T} dJ$$

Let us consider $x^2 = J(J+1) \implies 2x dx = (2J+1) dJ$. Therefore,

$$Q = \int_0^{\infty} 2x e^{-x^2 \Theta_{\text{rot}} / T} dx$$

We now use the known Gaussian integral $\int_0^{\infty} x e^{-ax^2} dx = \frac{1}{2a}$

$$Q = \frac{2}{2\Theta_{\text{rot}}/T} = \frac{T}{\Theta_{\text{rot}}} = \frac{2Ik_B T}{\hbar^2} = \frac{2I}{\beta \hbar^2}$$

Now, similar to the multiparticle vibrational partition function, the rotation can take place around each of the three Cartesian coordinates, and there are N distinguishable particles in the system. So the 3D partition function for the N -molecule system is:

$$Q_{N,3D} = Q_{1,1D}^{3N} = \left(\frac{2I}{\beta \hbar^2} \right)^{3N}$$

4.3.4 Internal Energy and Heat Capacity

$$U = -\frac{\partial \ln Q_{N,3D}}{\partial \beta} = -3N \frac{\partial \ln Q_{1,1D}}{\partial \beta} = -3N \frac{\partial}{\partial \beta} \ln \left(\frac{2I}{\beta \hbar^2} \right) = -3N \frac{\partial}{\partial \beta} \left[\ln \left(\frac{2I}{\hbar^2} \right) - \ln \beta \right] = \frac{3N}{\beta} = 3N k_B T$$

$$C = \left(\frac{\partial U}{\partial T} \right) = 3N k_B$$

Note that these results are only true for the high T limit; however, for all practical purposes, these expressions are helpful as the crossover point between classical and quantum regimes is very close to absolute zero temperature.

4.4 Translational Motion: Monoatomic Ideal Gas

Another source of energy we will discuss is translational motion, i.e., the movement of an atom or, generally, a molecule in three-dimensional space, without considering its internal degrees of freedom. As the translation of gas molecule allows them to interchange their locations, we need to treat them as indistinguishable particles.

4.4.1 Energy Levels

The quantum mechanical energy levels of a monoatomic ideal gas can be obtained using the particle in a box model. This model assumes that a gas particle is trapped between two infinitely high walls that it cannot escape. In between the walls, the particle moves freely.

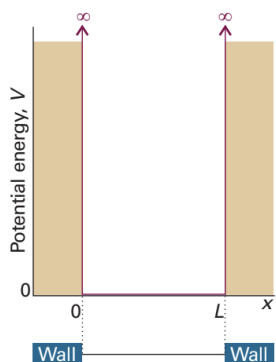


Fig. 9.1 A particle in a one-dimensional region with impenetrable walls. Its potential energy is zero between $x=0$ and $x=L$, and rises abruptly to infinity as soon as it touches the walls.

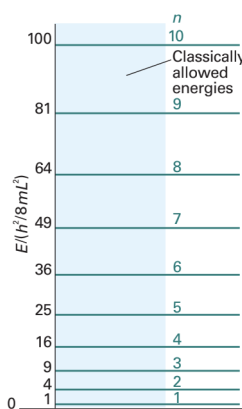


Fig. 9.2 The allowed energy levels for a particle in a box. Note that the energy levels increase as n^2 , and that their separation increases as the quantum number increases.

Figure 17: Figure Courtesy: Atkins, Peter William, Julio De Paula, and James Keeler. Atkins' Physical Chemistry. Oxford University Press, 2023.

The energy levels of a particle in a 1-dimensional box are given by:

$$E_n = \frac{n^2 h^2}{8mL^2}; \quad n = 1, 2, 3, \dots, \infty$$

Note that n starts from 1 and not 0. m is the mass of the particle and L is the length of the 1D box.

4.4.2 Partition Function

The 1D, 1 particle partition function for the particle in a box model is

$$Q = \sum_{n=1}^{\infty} e^{-\beta E_n} = \sum_{n=1}^{\infty} e^{-n^2 h^2 / 8mL^2 k_B T} = \sum_{n=1}^{\infty} e^{-n^2 \Theta_{\text{trans}} / T}$$

Here $\Theta_{\text{trans}} = \frac{h^2}{8mL^2 k_B}$ is the characteristic temperature for energetic transitions for the translational motion. Let's compute the value of this temperature for a typical atomic gas, Argon. Let us say the gas is kept in a 1D box of length 1m. The mass of a single Ar atom is 6.63×10^{-26} kg.

$$\Theta_{\text{trans}} = \frac{h^2}{8mL^2 k_B} = \frac{(6.626 \times 10^{-34})^2}{8(6.63 \times 10^{-26}) \cdot 1^2 \cdot 1.38 \times 10^{-23}} \approx 6 \times 10^{-20} \text{ K}$$

As quantum effects and the discretization of energy are only significant at $T < \Theta_{\text{trans}}$, at any finite temperature, the translational energy levels are continuous.

At this point, you may wonder why we bother with quantum mechanics if we're only going to say that energy levels are continuous. The answer to that lies in classical mechanics, where the energy of a particle often depends on

its position and velocity, which can change constantly and is challenging to track. Whereas quantum mechanical energies are only dependent on the inherent properties of the atom, such as mass or moment of inertia, which can be readily obtained, they do not vary with time. It is therefore much easier to calculate the quantum mechanical energy levels and then perform an approximation to the classical level.

As we already established that at any temperature of practical interest translational energy levels are continuous, we can replace the summation with an integration:

$$Q = \int_0^{\infty} e^{-n^2 \Theta_{\text{trans}}/T} dn$$

Here we need to use another Gaussian integral expression:

$$\int_0^{\infty} e^{-ax^2} dx = \frac{1}{2} \left(\frac{\pi}{a} \right)^{1/2}$$

which gives

$$Q = \int_0^{\infty} e^{-n^2 \Theta_{\text{trans}}/T} dn = \frac{1}{2} \left(\frac{\pi T}{\Theta_{\text{trans}}} \right)^{1/2} = \frac{1}{2} \left(\frac{8\pi m L^2 k_B T}{h^2} \right)^{1/2} = \left(\frac{2\pi m L^2 k_B T}{h^2} \right)^{1/2}$$

4.4.3 Multiparticle translational partition function in 3D

The next step is to generalize this partition function to three dimensions and multiple particles. Let us say that the gas particle is in a box which is aligned with the Cartesian axes and with length, breadth, and height of L_x , L_y , and L_z . Therefore

$$Q_{3D} = Q_x Q_y Q_z = \left(\frac{2\pi m L_x^2 k_B T}{h^2} \right)^{1/2} \left(\frac{2\pi m L_y^2 k_B T}{h^2} \right)^{1/2} \left(\frac{2\pi m L_z^2 k_B T}{h^2} \right)^{1/2} = \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} L_x L_y L_z = \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} V$$

where $V = L_x L_y L_z$ is the volume of the container.

We also define the thermal De-Broglie wavelength $\Lambda = \left(\frac{h^2}{2\pi m k_B T} \right)^{1/2}$, to avoid having to write this factor again and again. (Do not worry about the name, as this will be apparent in the QM class next quarter.) So,

$$Q_{3D} = \frac{V}{\Lambda^3}$$

To generalize it further to N particle system, we need to use the expression for the indistinguishable particle.

$$Q_{N,3D} = \frac{Q_{3D}^N}{N!} = \frac{V^N}{\Lambda^{3N} N!}$$

Example: Calculate the translational partition function for an O_2 molecule in a 1 L vessel at 25°C .

$$\begin{aligned} \Lambda &= \frac{h}{(2\pi m k T)^{1/2}} = \frac{6.626 \times 10^{-34} \text{ J}\cdot\text{s}}{(2 \times \pi \times 32 \text{ amu} \times 1.67 \times 10^{-27} \text{ kg/amu} \times 1.38 \times 10^{-23} \text{ J/K} \times 298 \text{ K})^{1/2}} \\ &= 1.78 \times 10^{-11} \text{ m} = 17.8 \text{ pm} \\ q^T &= \frac{V}{\Lambda^3} = \frac{1 \times 10^{-3} \text{ m}^3}{(1.78 \times 10^{-11})^3 \text{ m}^3} = 1.77 \times 10^{29} \end{aligned}$$

4.4.4 Internal Energy and Heat Capacity

Internal energy:

$$\begin{aligned}
 U &= -\frac{\partial \ln Q}{\partial \beta} \\
 &= -\frac{\partial}{\partial \beta} \ln \left(\frac{V^N}{\Lambda^{3N} N!} \right) \\
 &= -\frac{\partial \ln V^N}{\partial \beta} - \frac{\partial \ln(1/\Lambda^{3N})}{\partial \beta} - \frac{\partial \ln(1/N!)}{\partial \beta} \\
 &= \frac{\partial \ln \Lambda^{3N}}{\partial \beta} \quad (\text{Only } \Lambda \text{ depends on } \beta) \\
 &= 3N \frac{\partial \ln \Lambda}{\partial \beta} \\
 &= 3N \frac{\partial}{\partial \beta} \ln \left(\frac{h^2}{2\pi m k_B T} \right)^{1/2} \\
 &= 3N \frac{\partial}{\partial \beta} \ln \left(\frac{h^2 \beta}{2\pi m} \right)^{1/2} \\
 &= 3N \frac{\partial}{\partial \beta} \ln \left(\frac{h^2}{2\pi m} \right)^{1/2} + 3N \frac{\partial}{\partial \beta} \ln \beta^{1/2} \quad (\text{Only 2nd term depends on } \beta) \\
 &= \frac{3N}{2\beta} = \frac{3}{2} N k_B T
 \end{aligned}$$

You might remember studying this expression for the internal energy of an ideal gas in thermodynamics class. Here, we reached the same expression from molecular-level descriptions. This is how statistical mechanics forms a bridge between microscopic properties and macroscopic properties.

Constant volume heat capacity:

$$C_v = \left(\frac{\partial U}{\partial T} \right)_v = \frac{3}{2} \left(\frac{\partial N k_B T}{\partial T} \right)_v = \frac{3}{2} N k_B$$

4.4.5 Pressure

Because there is an explicit volume dependence of the partition function, we can compute the pressure as well. We know that

$$P = -\left(\frac{\partial U}{\partial V} \right)_T$$

But here, the internal energy does not have explicit volume dependence. So we can work with the Helmholtz free energy instead. Recall

$$A = U - TS$$

$$dU = TdS - PdV \quad (N \text{ is constant, so we ignore } \mu dN \text{ term})$$

$$dA = dU - TdS - SdT = TdS - PdV - TdS - SdT = -PdV - SdT$$

Therefore,

$$P = -\left(\frac{\partial A}{\partial V} \right)_T$$

From the 3D partition function:

$$A = -k_B T \ln Q = -k_B T \ln \left(\frac{V^N}{\Lambda^{3N} N!} \right) = -k_B T \ln V^N + k_B T \ln \Lambda^{3N} + k_B T \ln N!$$

As T is constant and the N has no V dependence, the pressure

$$P = -\left(\frac{\partial A}{\partial V} \right)_T = -\left(\frac{\partial (-k_B T \ln V^N)}{\partial V} \right)_T = N k_B T \left(\frac{\partial \ln V}{\partial V} \right)_T = \frac{N k_B T}{V}$$

Therefore,

$$PV = Nk_B T \text{ or } nRT$$

We can write the second expression in terms of molarity instead of the number of particles as $R = k_B \cdot N_A$ where N_A is Avogadro's number and R is the ideal gas constant.

Note that you have derived this expression $PV = nRT$ using empirical observations such as Boyal's Law and Charles' Law. Here, we derive them from the microscopic energy levels of each individual atom. So statistical mechanics again connects the microscopic world with the macroscopic world.

4.5 Putting It All Together: The Molecular Partition Function

The only thing left right now is to calculate the electronic partition function and we will find the molecular partition function. However, we will omit the electronic partition function from this course. One reason is that it requires a significant mastery of quantum mechanics, which we will attain in your future courses. But a more scientific reason for skipping the electronic partition function is that the electronic energy levels are extremely far apart. The electronic energy levels are at the opposite end of the spectrum relative to the translation. Therefore, an enormous amount of energy is required to access the higher energy levels; such a high amount of energy is not available from thermal effects. Unless we can use a very powerful and different type of energy source, such as lightwave, we cannot excite a molecule to its higher electronic energy levels.

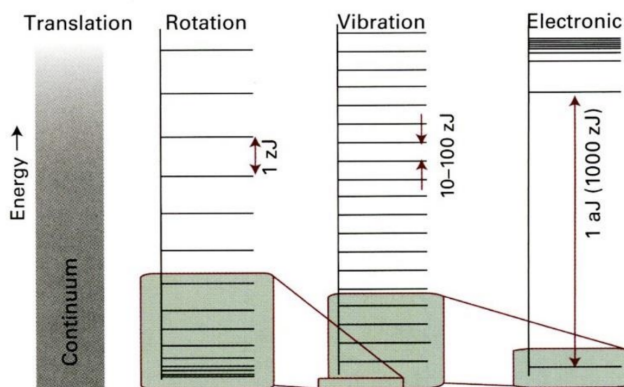


Figure B.3 The energy level separations typical of four types of system. (1 zJ = 10^{-21} J; in molar terms, 1 zJ is equivalent to about 0.6 kJ mol^{-1} .)

Figure 18: Figure Courtesy: Atkins, Peter William, Julio De Paula, and James Keeler. Atkins' Physical Chemistry. Oxford University Press, 2023.

For the sake of closure, we will keep Q_{elec} component in our expression for the molecular partition function, to show that the electronic energy levels do exist; however, we are not computing them here. Therefore, the molecular partition function is given by

$$Q_{\text{mol}} = Q_{\text{trans}} Q_{\text{rot}} Q_{\text{vib}} Q_{\text{elec}} = \left(\frac{V^N}{\Lambda^{3N} N!} \right) \left(\frac{2I}{\beta \hbar^2} \right)^{3N} \left(\frac{e^{-\beta h \nu / 2}}{1 - e^{-\beta h \nu}} \right)^{3N} \cdot Q_{\text{elec}}$$

As we need to take derivative of the $\ln Q$ to get the energy and heat capacity, these properties are additive:

$$U_{\text{mol}} = U_{\text{trans}} + U_{\text{rot}} + U_{\text{vib}} + U_{\text{elec}}$$

and

$$C_{\text{mol}} = C_{\text{trans}} + C_{\text{rot}} + C_{\text{vib}}$$

We ignore the C_{elec} term as increasing temperature generally cannot increase the electronic energy of a system, due to the largely spaced energy levels. Although the total heat capacity is the sum of the translational, rotational, and vibrational heat capacities, one of them can be more prominent over the others depending on the temperature.

At a very low temperature, there is not enough energy to induce transitions in the rotational and vibrational energy levels; therefore, increasing the temperature actually contributes only to the translational internal energy. This means that at low temperatures, if we provide the system with some energy, that energy contributes entirely to the translational motion of the system. Still, the molecules do not start to rotate or vibrate.

$$C_v = C_{\text{trans}} = \frac{3}{2}Nk_B T \quad \text{at low temperature}$$

At an intermediate temperature, the rotational states start to get populated. Still, the temperature is too low to get a finite population at the higher energy levels of the vibrational partition function. In this regime,

$$C_v = C_{\text{trans}} + C_{\text{rot}} = \frac{3}{2}Nk_B T + Nk_B T = \frac{5}{2}Nk_B T \quad \text{at intermediate temperature}$$

When we increase the temperature even higher, vibrational higher energy states also start getting populated, and the vibrational partition function starts contributing to the heat capacity.

$$C_v = C_{\text{trans}} + C_{\text{rot}} + C_{\text{vib}} = \frac{3}{2}Nk_B T + Nk_B T + Nk_B T = \frac{7}{2}Nk_B T \quad \text{at high temperature}$$

This leads to a unique step-like increase in the heat capacity of a molecular gas with the increase in temperature. This step-like behaviour can only be explained using quantum mechanics, which allows the possibility of discrete energy states. The boundaries between the different regimes of the heat capacity are determined by the characteristic temperatures of rotation and vibration: Θ_{rot} and Θ_{vib} .

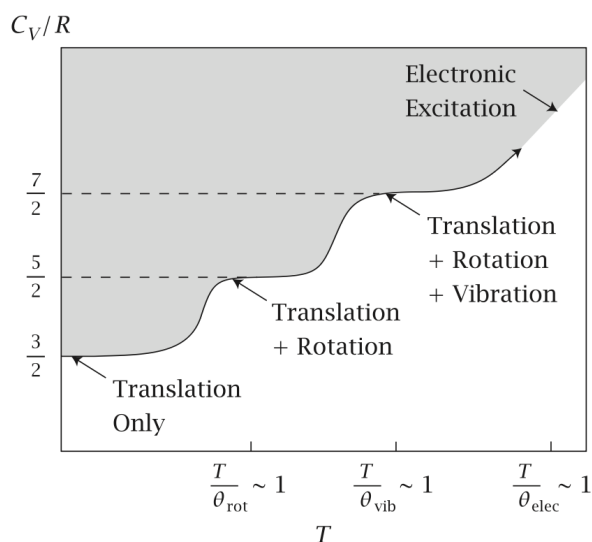


Figure 19: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.

[Interactive Tutorial](#)

4.6 Summary

- The molecular partition can be expressed as a product of the translational, rotational, vibrational, and electronic partition functions of a molecule if those degrees of freedom are energetically independent from each other:

$$Q_{\text{mol}} = Q_{\text{trans}} Q_{\text{rot}} Q_{\text{vib}} Q_{\text{elec}}$$

- The vibrational contribution to the molecular partition function can be computed by assuming the vibrational modes of a molecule as simple harmonic oscillators. According to quantum mechanics, the energy levels of a one-dimensional simple harmonic oscillator are given by:

$$E_n = \left(n + \frac{1}{2}\right)h\nu \quad \text{where } n = 0, 1, 2, \dots, \infty$$

Here ν is the frequency of oscillation and h is Planck's Constant with a value of 6.626×10^{-34} Joule-second.

- The partition function for a 1D simple harmonic oscillator is

$$Q = \frac{e^{-\beta h\nu/2}}{1 - e^{-\beta h\nu}}$$

- For the 1D simple harmonic oscillator, the internal energy $U \rightarrow h\nu/2$ at $T \rightarrow 0$. This is called zero-point energy. At $T \rightarrow \infty$; $U \rightarrow k_B T$.
- For a simple harmonic oscillator, the heat capacity is given by

$$C(T) = k_B \left(\frac{\Theta_{\text{vib}}}{T}\right)^2 \frac{e^{\Theta_{\text{vib}}/T}}{(e^{\Theta_{\text{vib}}/T} - 1)^2}$$

where $\Theta_{\text{vib}} = h\nu/k_B$. The Θ_{vib} term therefore defines the boundary of the classical and quantum regimes. At $T < \Theta_{\text{vib}}$, the spacing between energy levels is larger than the thermal energy $k_B T$. In this regime quantum effects dominate. At $T > \Theta_{\text{vib}}$ we can use the classical limit, as the energy difference between the levels becomes so low that the discrete energy levels effectively become a continuum.

- The energy levels of a rigid rotor, which can be used to represent the rotational degrees of freedom of a molecule, are given by:

$$E_J = J(J+1)\frac{\hbar^2}{2I}; \quad J = 0, 1, 2, \dots, \infty$$

$\hbar = \frac{h}{2\pi}$, and I is the moment of inertia. For a diatomic molecule: $I = \mu R^2$ where R is the bond length and $\mu = m_1 m_2 / (m_1 + m_2)$, where m_1 and m_2 are the masses of the two atoms. Each energy level also has a degeneracy of $g_J = 2J + 1$.

- The rotational partition function of a rigid rotor is

$$Q = \sum_{J=0}^{\infty} (2J+1) e^{-J(J+1)\Theta_{\text{rot}}/T} \approx \frac{2I}{\beta \hbar^2}$$

where $\Theta_{\text{rot}} = \frac{\hbar^2}{2Ik_B}$ is the effective rotational temperature. For most molecules, Θ_{rot} is much less than room temperature, so the classical limit can be applied. The rotational contribution to heat capacity is k_B for each rotational mode of the system.

- The translational component of the partition function can be represented using the ideal gas model, whose energy levels are obtained from the particle in a box model of quantum mechanics:

$$E_n = \frac{n^2 \hbar^2}{8mL^2}; \quad n = 1, 2, 3, \dots, \infty$$

- The partition function of an ideal gas atom in a 3D volume V is:

$$Q_{3D} = \frac{V}{\Lambda^3} \quad \left[\Lambda = \sqrt{\hbar^2 / 2\pi m k_B T} \right]$$

For N indistinguishable particles.

$$Q_{N,3D} = \frac{Q_{3D}^N}{N!} = \frac{V^N}{\Lambda^{3N} N!}$$

The translational energy levels are virtually continuous at room temperature, exhibiting only classical behavior.

- The internal energy and heat capacity of the ideal gas are: $U = \frac{3}{2} N k_B T$ and $C_v = \frac{3}{2} N k_B$
- The electronic energy levels are so far apart in energy for most molecules that electronic excitations are not achievable with thermal energy.

5 Concept of Ensembles

We have seen how the partition function can be used to calculate properties of molecules. Now we will go back to one of the basic concepts of statistical mechanics: the **ensembles**. This will help us generalize the principles of statistical mechanics into more complex and realistic problems, where we cannot predict the properties just from thermodynamics.

5.1 Idea of ensemble averaging

Let us revisit the simple two-level system with two energy states $E_1 = 0$ and $E_2 = \epsilon_0$. We want to compute the average energy of the system. If you recall the **simulation** approach, we can generate a random number to decide when the system will transit from the ground state to the excited state. Then repeat the simulation several hundred or thousands of times, and compute the average energy from all the observations. If out of N such trials, the system remains in the excited state for N_2 trials, the average energy is $N_2\epsilon_0/N$.

However, if we know something about the system, such as whether it is in a constant temperature condition, we can predict from **theory** all possible states the system can attain and their respective probabilities. We then perform an average over these different possible states. In this case, there are only two possible configurations: either in the energy 0 state or in the energy ϵ_0 state. Under constant temperature conditions, their relative probabilities will be proportional to their Boltzmann probabilities. Therefore,

$$\text{Average Energy} = p(1)E_1 + p(2)E_2 = 1.0 + e^{-\beta\epsilon_0}.\epsilon_0$$

Averaging over all possible configurations of the system (without waiting for the system to reach that conformation and counting) is known as **ensemble averaging**. The collection of all possible configurations of a system under a given condition is called an **ensemble**.

This might sound obvious, but it has important consequences. The above example is about a **constant temperature ensemble**, where we got the expected result. But what would happen if I did the measurement under constant energy conditions? We can ask, for example, what is the probability of being in either of the states if the system is in constant energy $\epsilon_0/2$. (Note, we did not ask what the average energy is because we already set that constant). We see that in this condition, the only solution is $p(1) = p(2) = 0.5$, and not the Boltzmann probability. This is a **constant energy ensemble**. Therefore, to measure properties under different conditions, such as constant energy, temperature, or pressure, we need to create the appropriate ensemble and then construct the partition function for that specific ensemble.

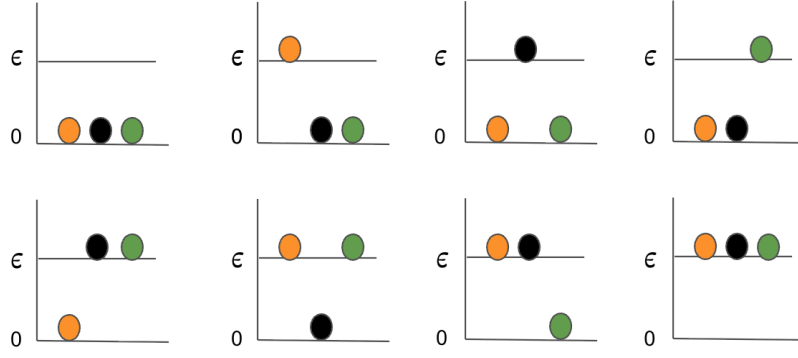
5.2 Microcanonical, Canonical, and Grand Canonical Ensembles

In this course, we will discuss primarily three types of ensembles:

- **Microcanonical Ensemble:** Constant energy ensemble, and is applicable to **isolated systems** where there is no exchange of heat or matter with the surroundings. We often call this the NVE ensemble if we keep the volume (V) constant, in addition to the number of particles (N) and energy (E).
- **Canonical Ensemble:** Constant temperature ensemble, and is applicable to **closed systems** where there is exchange of heat but no exchange of matter with the surroundings. This is the ensemble we have discussed so far in this course. We often call this the NVT ensemble if we keep the volume (V) constant, in addition to the number of particles (N) and energy (E). If we keep the pressure (P) constant instead of volume (most chemical reactions and processes), we call it an NPT ensemble or isothermal-isobaric ensemble.
- **Grand Canonical Ensemble:** In this ensemble, the number of particles is allowed to vary while keeping the chemical potential μ constant. This is applicable to **open systems** where there is an exchange of both heat and matter with the surroundings. We also call it μVT or μPT ensemble, depending on whether we kept the volume or pressure constant. This is the most realistic of all the ensembles, but also mathematically the most complex to tackle.

5.2.1 Example of all ensembles on a model system

Let us now examine the model system of 3 distinguishable particles in the two-level system derived from these three ensembles. All possible configurations for the system are shown below:



Let us name these configurations based on the energy levels: $(0,0,0)$, $(\epsilon,0,0)$, $(0,\epsilon,0)$, $(0,0,\epsilon)$, $(\epsilon,\epsilon,0)$, $(0,\epsilon,\epsilon)$, $(\epsilon,0,\epsilon)$, and $(\epsilon,\epsilon,\epsilon)$.

Microcanonical Ensemble: This is a fixed energy ensemble. But these configurations have different total energies. So, all of these configurations cannot be included in a single microcanonical ensemble with a specific energy. Let us say that I want to define a microcanonical ensemble with constant energy ϵ . Only the following three configurations: $(\epsilon,0,0)$, $(0,\epsilon,0)$, and $(0,0,\epsilon)$ can be part of this ensemble. Since there is nothing to differentiate between each configuration, they will be equally likely, with probability $1/3$. The partition function for this ensemble is just the number of microstates associated with the macrostate with energy ϵ .

Canonical Ensemble: Here, we do not fix the energy but the temperature. So all configurations are possible, although with different probabilities determined from Boltzmann probabilities. The canonical partition function is

$$Q(T) = 1 + 3e^{-\epsilon/k_B T} + 3e^{-2\epsilon/k_B T} + e^{-3\epsilon/k_B T}$$

We discussed this type of partition function throughout the course.

Grand Canonical Ensemble: In this case, the number of particles is also not fixed. So there can be any number of particles starting from 1 to ∞ . And each will have a partition function associated with it.

$$\begin{aligned} Q_1(T) &= 1 + e^{-\epsilon/k_B T} \\ Q_2(T) &= 1 + 2e^{-\epsilon/k_B T} + e^{-2\epsilon/k_B T} \\ Q(T) &= 1 + 3e^{-\epsilon/k_B T} + 3e^{-2\epsilon/k_B T} + e^{-3\epsilon/k_B T} \\ &\dots \end{aligned}$$

However, the likelihood of having different numbers of particles is not equal. Therefore, if we could write a grand partition function it should take the form:

$$Z = \mathcal{P}(0)Q_0 + \mathcal{P}(1)Q_1 + \mathcal{P}(2)Q_2 + \mathcal{P}(3)Q_3 + \dots + \mathcal{P}(\infty)Q_\infty$$

where $\mathcal{P}(N)$ is the probability of observing N particles in the system.

As you might have guessed by now, this grand canonical partition function not only depends on T but also on the other intensive variable, chemical potential μ , that is related to N in the expression for entropy:

$$dS = \frac{1}{T}dU + \frac{P}{T}dV - \frac{\mu}{T}dN$$

In the canonical partition function and in Boltzmann probability, the energy is multiplied by $\frac{1}{k_B} \cdot \frac{1}{T}$ before being exponentiated in the probability expression: $e^{-E/k_B T}$. There should be a similar expression for $\mathcal{P}(N)$. By analogy, N should be multiplied by $\frac{1}{k_B} \cdot \frac{-\mu}{T}$ before being exponentiated, i.e., $\mathcal{P}(N) \propto e^{\mu/k_B T}$. (For a more formal proof, check Glazer's book, pages 108-109.)

$$\begin{aligned}
Z(\mu, T) &= \mathcal{P}(0)Q_0 + \mathcal{P}(1)Q_1 + \mathcal{P}(2)Q_2 + \dots + \mathcal{P}(\infty)Q_\infty \\
&= e^{\mu \cdot 0/k_B T} Q_0 + e^{\mu \cdot 1/k_B T} Q_1 + e^{\mu \cdot 2/k_B T} Q_2 + \dots + e^{\mu \cdot \infty/k_B T} Q_\infty \\
&= \sum_{N=0}^{\infty} e^{\mu N/k_B T} Q_N \\
&= \sum_{N=0}^{\infty} e^{\mu N/k_B T} \sum_{i=1}^M e^{-E/k_B T} \quad (\text{M is the number of energy levels}) \\
&= \sum_{N=0}^{\infty} \sum_{i=1}^M e^{-(E - \mu N)/k_B T}
\end{aligned}$$

For simplicity, the $Z(\mu, T)$ is also written as

$$Z(\mu, T) = \sum_{N=0}^{\infty} \zeta^N Q_N(T)$$

where $\zeta = e^{\mu/k_B T} = e^{\beta\mu}$

We use the grand canonical partition function when we are interested in calculating the average number of particles in a given system or a state.

$$\langle N \rangle = \frac{\sum_{N=0}^{\infty} N \zeta^N Q_N(T)}{Z(\mu, T)}$$

Note, that

$$\frac{\partial Z}{\partial \zeta} = \sum_{N=0}^{\infty} \frac{\partial}{\partial \zeta} \zeta^N Q_N = \sum_{N=0}^{\infty} N \zeta^{N-1} Q_N = \frac{1}{\zeta} \sum_{N=0}^{\infty} N \zeta^N Q_N$$

Therefore,

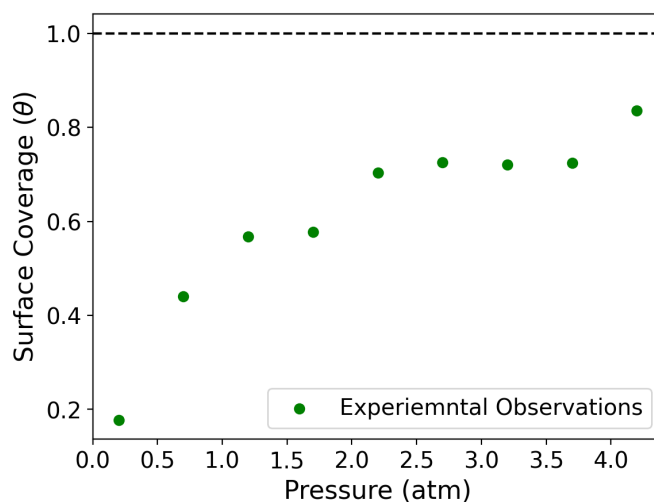
$$\langle N \rangle = \frac{\zeta}{Z(\mu, T)} \frac{\partial Z}{\partial \zeta} = \zeta \frac{\partial}{\partial \zeta} \ln Z$$

5.3 Application of Grand Canonical Partition Function: Langmuir Adsorption Isotherm

There are many practical scenarios where the Grand Canonical Partition Function will be relevant, as many chemical, biological, and material processes involve the change of the number of particles in a system. One prominent example is crystal growth, where molecules move from the solution phase to the crystal phase during crystallization. But here we will discuss a simpler example: the adsorption of gas on a solid surface. The amount of gas molecules adsorbed on a solid surface is measured by the surface coverage, which is defined by the following fraction:

$$\theta = \frac{\text{Number of adsorption site occupied by incoming gas molecules}}{\text{Total number of adsorption sites on the surface}}$$

It has been found experimentally that the fraction of the surface coverage θ increases with the pressure P of the gas on top of the surface, but eventually the value saturates. To explain this behaviour from the microscopic level, we will make use of the grand canonical ensemble.



We will only focus on the physisorption (no chemical reaction between the surface and the adsorbed molecule) in a single layer (no new molecule can be adsorbed on top of an already adsorbed molecule). We will also assume the adsorbed molecules do not interact with neighbouring adsorbed molecules.

Below is a simplified model of adsorption that follows this assumption. There are a fixed number of adsorption sites, and only one gas molecule can be adsorbed on a single site.

Consider an adsorbent surface having N sites each of which can adsorb one gas molecule. Suppose that it is in contact with an ideal gas with the chemical potential μ . An adsorbed molecule has energy $-\varepsilon_0$ compared to one in a free state. We determine the coverage ratio θ .

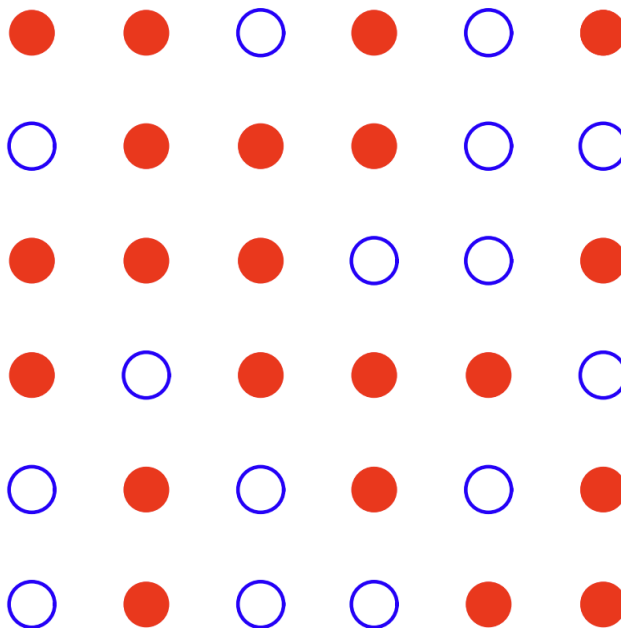


Fig. Langmuir adsorption. The number of total sites is N . N_1 sites are occupied by atoms (red, one atom in each site). $(N - N_1)$ sites are not occupied (blue circles). The number of particles is N_1 . The energy of the system is $(-N_1\varepsilon_0)$

The N_1 particle canonical partition function of the system is

$$Q_{N_1} = g_{N_1} e^{-\beta E_{N_1}}$$

as there is only one possible energy state $E_{N_1} = -N_1 \epsilon_0$. The degeneracy is the number of possible ways we can rearrange N_1 occupied sites out of N total sites. This problem is identical to finding the number of ways one can have N_1 heads in N coin flips. So the number of microstates, i.e., rearrangements possible is

$$g_{N_1} = \frac{N!}{N_1!(N - N_1)!}$$

So

$$Q_{N_1} = g_{N_1} e^{-\beta E_{N_1}} = \frac{N!}{N_1!(N - N_1)!} e^{\beta N_1 \epsilon_0}$$

The grand canonical partition function:

$$\begin{aligned} Z &= \sum_{N_1=0}^N \zeta^{N_1} Q_{N_1} \quad (\text{Sum goes to } N \text{ as that is the maximum number of occupied sites}) \\ &= \sum_{N_1=0}^N \zeta^{N_1} \frac{N!}{N_1!(N - N_1)!} e^{\beta N_1 \epsilon_0} \\ &= \sum_{N_1=0}^N \frac{N!}{N_1!(N - N_1)!} (\zeta e^{\beta \epsilon_0})^{N_1} \\ &= (1 + \zeta e^{\beta \epsilon_0})^N \quad \left[\text{Using } \sum_{k=0}^n \frac{n!}{k!(n-k)!} x^k = (1+x)^n \right] \end{aligned}$$

The average number of occupied sites

$$\begin{aligned} \langle N_1 \rangle &= \zeta \frac{\partial}{\partial \zeta} \ln Z \\ &= \zeta \frac{\partial}{\partial \zeta} \ln(1 + \zeta e^{\beta \epsilon_0})^N \\ &= N \zeta \frac{\partial}{\partial \zeta} \ln(1 + \zeta e^{\beta \epsilon_0}) \\ &= \frac{N \zeta e^{\beta \epsilon_0}}{1 + \zeta e^{\beta \epsilon_0}} \end{aligned}$$

Therefore, the fraction of surface coverage

$$\theta = \frac{\langle N_1 \rangle}{N} = \frac{\zeta e^{\beta \epsilon_0}}{1 + \zeta e^{\beta \epsilon_0}}$$

Now we have derived the expression for the surface coverage. But this expression shows how θ depends on $\zeta = e^{\beta \mu}$, but not on the pressure P , which is measured experimentally. Therefore, we need to derive an expression that relates ζ with P , so that we can substitute ζ with a function of P in the expression of θ . We note from our recapitulation of thermodynamics

$$dA = -SdT - PdV + \mu dN$$

Therefore,

$$\begin{aligned}
\mu &= \left(\frac{\partial A}{\partial N} \right)_{T,V} \\
&= -k_B T \frac{\partial}{\partial N} \ln Q_N \\
&= -k_B T \frac{\partial}{\partial N} \ln \left(\frac{V^N}{\Lambda^{3N} N!} \right) \\
&= -k_B T \frac{\partial}{\partial N} \left[N \ln \left(\frac{V}{\Lambda^3} \right) - \ln N! \right] \\
&= -k_B T \frac{\partial}{\partial N} \left[N \ln \left(\frac{V}{\Lambda^3} \right) - N \ln N + N \right] \\
&= -k_B T \left[\ln \left(\frac{V}{\Lambda^3} \right) - \ln N - N/N + 1 \right] \\
&= -k_B T \ln \left(\frac{V}{\Lambda^3 N} \right)
\end{aligned}$$

Note that we substituted the ideal gas partition function because we assume the gas from which the adsorption is taking place is a monoatomic ideal gas. Besides, we can only measure pressure for the gaseous state. Also note that we use the same μ for both the gas and the adsorbed state as we are working in a constant chemical potential ensemble or μVT ensemble. Now for an ideal gas, $PV = Nk_B T \implies \frac{V}{N} = \frac{k_B T}{P}$. Therefore

$$\begin{aligned}
\mu &= -k_B T \ln \left(\frac{k_B T}{\Lambda^3 P} \right) = k_B T \ln \left(\frac{\Lambda^3 P}{k_B T} \right) \\
\zeta &= e^{\mu/k_B T} = \frac{\Lambda^3 P}{k_B T} = \beta \Lambda^3 P
\end{aligned}$$

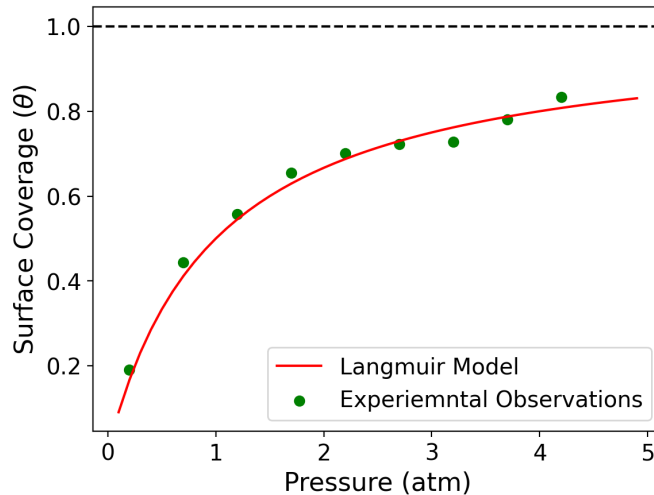
Substituting this into the expression for surface coverage θ

$$\theta = \frac{\zeta e^{\beta \epsilon_0}}{1 + \zeta e^{\beta \epsilon_0}} = \frac{\beta \Lambda^3 P e^{\beta \epsilon_0}}{1 + \beta \Lambda^3 P e^{\beta \epsilon_0}} = \frac{P}{\frac{1}{\beta \Lambda^3 e^{\beta \epsilon_0}} + P}$$

As $\beta \Lambda^3 e^{\beta \epsilon_0}$ only depends on β , i.e., T and not on P , we can define a temperature dependent constant $P_0(T) = \frac{1}{\beta \Lambda^3 e^{\beta \epsilon_0}}$. Therefore:

$$\theta = \frac{P}{P + P_0(T)}$$

This result is the Langmuir adsorption isotherm, which agrees very well with the experiment as long as our assumptions are valid. Note the solid line is the expression directly derived from statistical mechanics and not a fit to the data points.



Here, we derived it from the statistical mechanical standpoint, i.e., starting from microscopic properties (per-atom adsorption energy) to the macroscopic result (surface coverage as a function of pressure) that can be measured in experiment.

5.4 Chemical Equilibrium

The chemical potential μ can be used to derive an expression of the chemical equilibrium constant in terms of the partition function for each individual component. Let us consider the simple reactive equilibrium:



The equilibrium constant

$$K_{eq} = \frac{[B]_{eq}}{[A]_{eq}} = \frac{N_B/V}{N_A/V} = \frac{N_B}{N_A}$$

Here $[]_{eq}$ indicates concentration at equilibrium, N_A and N_B are the number of particles of A and B , and the reaction takes place at constant temperature T and volume V .

Now the molecules of each individual component, i.e., all A and all B molecules, are indistinguishable. So the partition function for each component will be $Q = q^N/N!$, where q is the single particle partition function.

$$\mu = \left(\frac{\partial A}{\partial N} \right)_{V,T} = -k_B T \frac{\partial}{\partial N} \ln \frac{q^N}{N!} = -k_B T \frac{\partial}{\partial N} (N \ln q - N \ln N + N) = -k_B T (\ln q - \ln N) = -k_B T \ln \left(\frac{q}{N} \right)$$

So

$$\mu_A = -k_B T \ln \left(\frac{q_A}{N_A} \right) \quad \text{and} \quad \mu_B = -k_B T \ln \left(\frac{q_B}{N_B} \right)$$

For a constant T,V system, we compute the change of Helmholtz Free Energy:

$$dA = -SdT - PdV + \mu_A dN_A + \mu_B dN_B = \mu_A dN_A + \mu_B dN_B \quad (\text{as } T \text{ and } V \text{ are constant})$$

For a constant T, P system, we compute the change of Gibbs Free Energy:

$$dG = -SdT - VdP + \mu_A dN_A + \mu_B dN_B = \mu_A dN_A + \mu_B dN_B \quad (\text{as } T \text{ and } P \text{ are constant})$$

Now, from the stoichiometry of the reaction, the number of A molecules consumed is equal to the number of B molecules produced. So $dN_A = -dN_B$. Besides, at equilibrium, the free energy is minimized, so $dA = 0$ (or $dG = 0$ at constant pressure)

$$dA = 0 \implies \mu_A dN_A + \mu_B dN_B = 0 \implies \mu_A dN_A - \mu_B dN_A = 0 \implies \mu_A = \mu_B$$

Therefore,

$$-k_B T \ln \left(\frac{q_A}{N_A} \right) = -k_B T \ln \left(\frac{q_B}{N_B} \right) \implies \frac{q_A}{N_A} = \frac{q_B}{N_B} \implies \frac{q_B}{q_A} = \frac{N_B}{N_A} = K_{eq}$$

So the equilibrium constant is

$$K_{eq} = \frac{q_B}{q_A}$$

Note that the ratio is on the single particle partition function and not the bulk N particle partition function.

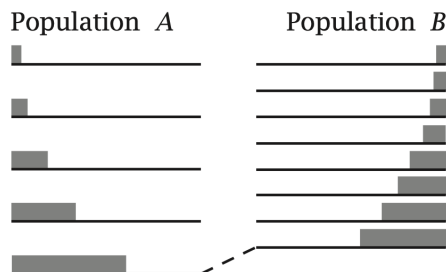


Figure 13.1 Hypothetical energy ladders for the two-state equilibrium $A \rightarrow B$. Two factors contribute to the equilibrium populations of A and B : low energies and high densities of states. A is shown here to have the lowest-energy ground state. B has the higher density of states.

Figure 21: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.

In isolation, the partition functions for reactant or product start counting the energy levels from zero (relative to the zero-point energy in quantum treatment)

$$q_A = e^{-\beta \cdot 0} + e^{-\beta E_1^A} + e^{-\beta E_2^A} + \dots = 1 + e^{-\beta E_1^A} + e^{-\beta E_2^A} + \dots$$

$$q_B = e^{-\beta \cdot 0} + e^{-\beta E_1^B} + e^{-\beta E_2^B} + \dots = 1 + e^{-\beta E_1^B} + e^{-\beta E_2^B} + \dots$$

But in a reactive system, especially in equilibrium, we cannot use these expressions directly. That is because the energy levels of both A and B can be accessed simultaneously, and their zero-point energies are not the same. The difference in zero-point energies arises from the difference in bond energies or other forms of non-covalent interaction energies between the reactant and product. To account for this discrepancy, we must explicitly include the zero-point energies (E_0^A for A and E_0^B for B), as we can no longer set them to the same level.

In the reaction mixture:

$$q'_A = e^{-\beta E_0^A} + e^{-\beta(E_1^A + E_0^A)} + e^{-\beta(E_2^A + E_0^A)} + \dots = e^{-\beta E_0^A} q_A$$

$$q'_B = e^{-\beta E_0^B} + e^{-\beta(E_1^B + E_0^B)} + e^{-\beta(E_2^B + E_0^B)} + \dots = e^{-\beta E_0^B} q_B$$

Therefore, in the reaction mixture:

$$K_{eq} = \frac{q'_B}{q'_A} = \frac{e^{-\beta E_0^B} q_B}{e^{-\beta E_0^A} q_A} = \frac{q_B}{q_A} e^{-\beta(E_0^B - E_0^A)} = \frac{q_B}{q_A} e^{-\beta \Delta E}$$

Where ΔE is the difference of ground state energies between the reactants and products. This expression is very general across the number of constituents in the reaction mixture and stoichiometry:

$$A + B \rightleftharpoons C + D \quad K_{eq} = \frac{q_C q_D}{q_A q_B} e^{-\beta(E_0^C + E_0^D - E_0^A - E_0^B)}$$

$$\nu_A A \rightleftharpoons \nu_B B \quad K_{eq} = \frac{q_B^{\nu_B}}{q_A^{\nu_A}} e^{-\beta(\nu_B E_0^B - \nu_A E_0^A)}$$

EXAMPLE 13.1 Stoichiometry. To construct one car C, suppose you need four tires T and six windows W. The 'equilibrium' for the 'reaction' that transforms tires and windows into cars can be written as



In this case, $a = 4$, $b = 6$, and $c = 1$. Making one car is represented by $d\xi = 1$. So, one reaction transformation gives $c d\xi = +1$ new car and it uses up $-a d\xi = -4$ tires and $-b d\xi = -6$ windows.

Exercise: Prove the equilibrium constant expression for the reaction: $\nu_A A \rightleftharpoons \nu_B B$. Hint: a similar problem has been solved on pages 238-239 of the Dill and Bromberg book.

5.4.1 Relationship between Free Energy and Equilibrium Constant

Inside the reaction mixture:

$$G_A = -k_B T \ln q'_A \quad \text{and} \quad G_B = -k_B T \ln q'_B$$

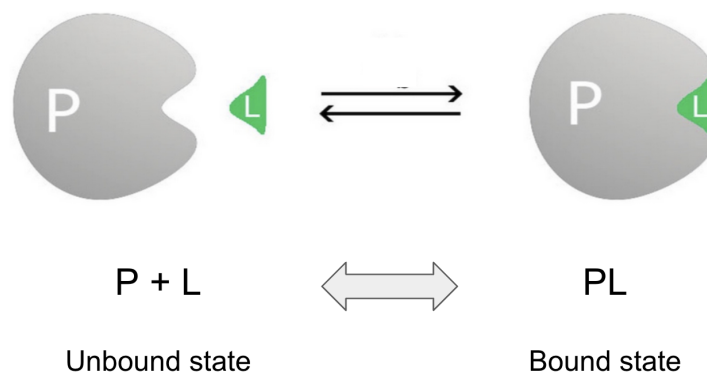
We consider the Gibbs free energy, as most chemical reactions are conducted at constant T and P. Therefore,

$$q'_A = e^{-G_A/k_B T} \quad \text{and} \quad q'_B = e^{-G_B/k_B T}$$

$$K_{eq} = \frac{q'_B}{q'_A} = e^{-(G_B - G_A)/k_B T} = e^{-\Delta G_{rxn}/k_B T} \quad \text{or} \quad \Delta G_{rxn} = -k_B T \ln K_{eq}$$

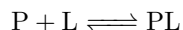
Here ΔG_{rxn} is the free energy difference between the products and reactants of a reaction. Note that these expressions do not contain the partition functions anymore. So ΔG_{rxn} can be computed from the K_{eq} and vice versa without having to compute the partition function, which requires the knowledge of all energy levels of the system.

5.4.2 Practical Application: Ligand Binding Equilibrium and Computational Drug Discovery



One of the most important applications is in the field of computational drug discovery. To design one new drug molecule, we need to test millions of potential candidate molecules. Experimentally testing the efficacy of all these molecules will be extremely expensive, harm the environment by producing a large amount of toxic waste, and may be outright dangerous if animal or human testing is performed. One way to address this problem is to computationally screen the large set of molecules and select the most promising ones for chemical or physiological experiments.

These molecules (*ligands*) (L) are screened primarily based on how strongly they bind to the disease-causing target protein (P). It can be assumed that there exists an equilibrium between the unbound state involving free proteins and free ligands and the bound state comprising a protein ligand complex (PL). The equilibrium can be represented as:



Although this is a non-reactive process, we can apply the concepts of chemical equilibrium. The equilibrium constant of the binding (K_b) and dissociation (K_d) can be expressed as

$$K_b = \frac{[PL]}{[P][L]} \quad ; \quad K_d = \frac{[P][L]}{[PL]}$$

Note that $K_b = 1/K_d$, and this is true for the equilibrium constants of all forward and backward reactions. The units of K_b and K_d are M^{-1} and M respectively: M is mols/liter.

A ligand is a suitable drug candidate when it binds strongly to the protein, i.e., in equilibrium, the concentration of the protein-ligand complex is significantly higher than free proteins and ligands: $[PL] \gg [P], [L]$. Therefore, for a strong binder, $K_b \gg 1$ and $K_d \ll 1$.

The free energy difference of ligand binding (ΔG_b) and unbinding (ΔG_u) are given by:

$$\Delta G_b = -RT \ln K_b \quad \text{and} \quad \Delta G_u = -RT \ln K_d$$

Note, we are using R instead of k_B because we are computing molar free energies. You may have realized by now that for strong binders, ΔG_b is a large negative number and ΔG_u is a large positive number. Also note that:

$$\Delta G_b = -RT \ln K_b = -RT \ln(1/K_d) = RT \ln K_d$$

(We need to divide the K_d by $1M$ to ensure the argument of the logarithm is unitless.)

The $\Delta G_b = RT \ln K_d$ expression is frequently used in the field of computational drug design, as K_d has the unit of molar concentration, and its logarithm directly yields the binding free energy ΔG_b . The K_d is therefore referred to as the binding affinity.

The K_d can also be used as an estimate to measure the effective drug concentration, as it has the unit of molar concentration. For example a drug X has $K_d = 10^{-9}M = 1nM$ and another drug Y has $K_d = 10^{-6}M = 1\mu M$. Therefore, one needs approximately nM concentration of drug X to have the desired (healing) effect as opposed to μM concentration of drug Y . The drug X is effective at a lower concentration, therefore it is more potent.

The binding free energies of these two drugs are:

$$\Delta G_b^X = RT \ln K_d^X = 0.6 \ln 10^{-9} = -12.43 \text{ kcal/mol} \quad [RT \approx 0.6 \text{ kcal/mol at } T = 300K]$$

$$\Delta G_b^Y = RT \ln K_d^Y = 0.6 \ln 10^{-6} = -8.29 \text{ kcal/mol}$$

Therefore drug X has a more negative binding free energy and is a stronger binder.

This effect can also be understood in terms of higher stability from lower free energy. The binding free energy

$$\Delta G_b = G_{\text{bound}} - G_{\text{unbound}} = G_{PL} - (G_P + G_L)$$

$$\Delta G_b \ll 0 \implies G_{PL} \ll (G_P + G_L)$$

which means the free energy of the bound state is significantly lower than that of the unbound state; i.e., the bound state is very stable. So $\Delta G_b \ll 0$ implies a strong binder.

5.5 Summary

- In statistical mechanics, we average over many copies of a system to obtain analytical expressions of quantities of interest. We primarily use three different types of ensembles:

1. **Microcanonical Ensemble:** Constant energy (NVE) ensemble applicable to isolated systems.

2. **Canonical Ensemble:** Constant temperature (NVT or NPT) ensemble applicable to closed systems.

Partition function:
$$Q_N(T) = \sum_i e^{-\beta E_i}$$

3. **Grand Canonical Ensemble:** Constant chemical potential (μ VT or μ PT) ensemble where the number

of particles N can change; applicable to open systems. Partition function:

$$Z(\mu, T) = \sum_{N=0}^{\infty} \zeta^N Q_N(T)$$

where $\zeta = e^{\mu/k_B T} = e^{\beta\mu}$. In this ensemble the average number of particles in the system can be computed as

$$\langle N \rangle = \zeta \frac{\partial}{\partial \zeta} \ln Z$$

- Chemical potential $\mu = \left(\frac{\partial A}{\partial N} \right)_{V,T}$. For indistinguishable molecules $\mu = -k_B T \ln \left(\frac{q}{N} \right)$ where q is single particle canonical partition function.

- For a chemical reaction: $aA + bB \rightleftharpoons cC + dD$, the equilibrium constant

$$K_{eq} = \frac{(q'_C)^c (q'_D)^d}{(q'_A)^a (q'_B)^b} = \frac{(q_C)^c (q_D)^d}{(q_A)^a (q_B)^b} e^{-((cE_C + dE_D) - (aE_A + bE_B))/k_B T} = \frac{(q_C)^c (q_D)^d}{(q_A)^a (q_B)^b} e^{-\Delta E_{\text{rxn}}/k_B T}$$

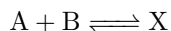
The partition functions with prime ($'$) are evaluated in the reaction mixture, and the ones without are evaluated in isolation.

- The difference of the Gibbs free energies of the products and the reactants is known as ΔG_{rxn} and is related to the equilibrium constant:

$$\Delta G_{\text{rxn}} = -k_B T \ln K_{eq}$$

6 Chemical Kinetics

Now we will switch gears into a completely different aspect of the molecular system: chemical kinetics. So far, we have been concerned with the equilibrium states of a system, specifically what the average energy, heat capacity, and number of particles would be once equilibrium is achieved. In this module, we will examine the preceding step and how quickly equilibrium is achieved. This part is also equally important. Let's say you are a chemist who wants to design a reaction to synthesize molecule X . You designed a reaction between A and B to form X :



You computed the partition function of the reactant product system.

$$Q = e^{-\beta E_A} + e^{-\beta E_B} + e^{-\beta E_X}$$

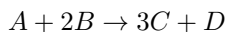
and determined that $p_X \gg (p_A + p_B)$. Therefore, the forward reaction is thermodynamically favourable, and if we mix A and B , after the reaction we will obtain significantly more of the product X than the reactants A and B . Therefore, this reaction should have a very high yield.

But what if the reaction is very slow? If we mix A and B , it takes several days before equilibrium is reached, where we can obtain a significant amount of X . This reaction will not be commercially viable to make the compound X . That is why, as chemists, we also have to care about the rates of chemical reactions.

6.1 Rates of Chemical Reaction

We are monitoring a reaction from reactant (R) to product (P): $R \rightarrow P$. At some instant, the molar concentration of R and P are $[R]$ and $[P]$ respectively, and the volume of the system is constant. The instantaneous rate of consumption of the reactant at a given time is $-d[R]/dt$. This rate is a positive quantity. The rate of formation of the product is $d[P]/dt$ (note the difference in sign). This rate is also positive.

But how are the rates of consumption of reactants and the formation of products related to each other? Consider a chemical reaction:



. Let us assume we mix n_A moles of A and n_B moles of B at time zero in a vessel of constant volume 1L:

$$\text{at } t = 0; \quad [A] = n_A, \quad [B] = n_B, \quad [C] = 0, \quad [D] = 0$$

This is because at $t = 0$ no product has formed. After a short time Δt , the amount of A that is consumed is Δx moles. Due to the stoichiometry of the reaction, this will lead to a consumption of $2\Delta x$ moles of B and the formation of $3\Delta x$ amount of C and Δx amount of D . So

$$\text{at } t = \Delta t; \quad [A] = n_A - \Delta x, \quad [B] = n_B - 2\Delta x, \quad [C] = 3\Delta x, \quad [D] = \Delta x$$

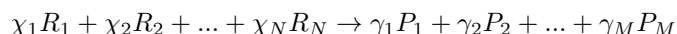
The rate of change of concentration of different reactants and products:

$$\begin{aligned} \frac{d[A]}{dt} &= -\frac{\Delta x}{\Delta t} \\ \frac{d[B]}{dt} &= -2\frac{\Delta x}{\Delta t} \\ \frac{d[C]}{dt} &= 3\frac{\Delta x}{\Delta t} \\ \frac{d[D]}{dt} &= \frac{\Delta x}{\Delta t} \end{aligned}$$

Therefore

$$-\frac{d[A]}{dt} = -\frac{1}{2} \frac{d[B]}{dt} = \frac{1}{3} \frac{d[C]}{dt} = \frac{d[D]}{dt} = \frac{\Delta x}{\Delta t}$$

When we refer to the **Rate of a Chemical Reaction**, we can refer to any of these derivatives of the concentrations. To generalize, for a reaction with the following stoichiometry:

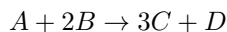


The rate of the reaction is defined as:

$$\text{Rate} = -\frac{1}{\chi_1} \frac{d[R_1]}{dt} = -\frac{1}{\chi_2} \frac{d[R_2]}{dt} = \dots = -\frac{1}{\chi_N} \frac{d[R_N]}{dt} = \frac{1}{\gamma_1} \frac{d[P_1]}{dt} = \frac{1}{\gamma_2} \frac{d[P_2]}{dt} = \dots = \frac{1}{\gamma_M} \frac{d[P_M]}{dt}$$

6.1.1 Rate constants and Reaction Orders

The rate of reaction is often found to be proportional to the concentrations of the reactants raised to a power. For example, the rate of the reaction



may be proportional to the molar concentrations of two reactants, A and B, so we write

$$\text{Rate} = k[A][B]$$

with each concentration raised to the first power. The coefficient k is called the **Rate Constant** for the reaction. The rate constant is independent of the concentrations but depends on the temperature. Note that the powers of the concentrations may not be related to the stoichiometry of the reaction. The power of the concentration is called the **Order of a Reaction** and is often determined by the molecular mechanism of the reaction. For example:

$$\text{Rate} = k \quad (\text{zeroth order kinetics})$$

$$\text{Rate} = k[A] \quad (\text{1st order kinetics})$$

$$\text{Rate} = k[A]^2 \quad (\text{2nd order kinetics})$$

$$\text{Rate} = k[A][B]^2 \quad (\text{overall 3rd order kinetics, 1st order w.r.t A and 2nd order w.r.t B})$$

Unit: Note that the unit of the rate is fixed: concentration/time, such as Ms^{-1} where M indicates moles/litre. Therefore, depending on the order of the reaction, the unit of the rate constant would change. For a zeroth-order reaction, the unit of k is Ms^{-1} , for first order reaction it is just s^{-1} , for a second order reaction it is $\text{M}^{-1}\text{s}^{-1}$, and so on.

6.1.2 Integrated Rate Laws

As the rate of a reaction can be expressed as the time derivative of the reactant concentration, it is possible to write differential equations whose solutions demonstrate how reactant or product concentration changes with time during the reaction.

Zeroth Order Kinetics

$$\begin{aligned} -\frac{d[A]}{dt} &= k \\ \implies d[A] &= -kdt \\ \implies \int_{[A]_0}^{[A]} d[A] &= -k \int_0^t dt \\ \implies [A] - [A]_0 &= -kt \\ \implies [A] &= [A]_0 - kt \end{aligned}$$

First Order Kinetics

$$\begin{aligned} -\frac{d[A]}{dt} &= k[A] \\ \implies \frac{d[A]}{[A]} &= -kdt \\ \implies \int_{[A]_0}^{[A]} \frac{d[A]}{[A]} &= -k \int_0^t dt \\ \implies \ln[A] - \ln[A]_0 &= -kt \\ \implies \frac{[A]}{[A]_0} &= e^{-kt} \\ \implies [A] &= [A]_0 e^{-kt} \end{aligned}$$

Second Order Kinetics

$$\begin{aligned}
 -\frac{d[A]}{dt} &= k[A]^2 \\
 \Rightarrow \frac{d[A]}{[A]^2} &= -k dt \\
 \Rightarrow \int_{[A]_0}^{[A]} \frac{d[A]}{[A]^2} &= -k \int_0^t dt \\
 \Rightarrow -\frac{1}{[A]} + \frac{1}{[A]_0} &= -kt \\
 \Rightarrow \frac{1}{[A]} &= \frac{1}{[A]_0} + kt \\
 \Rightarrow [A] &= \frac{[A]_0}{1 + [A]_0 kt}
 \end{aligned}$$

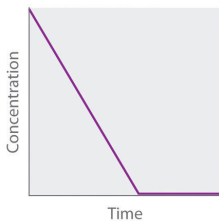
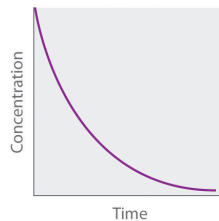
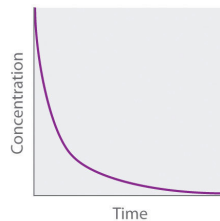
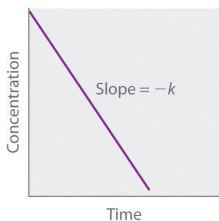
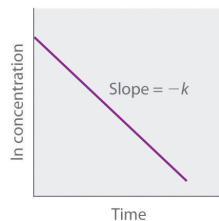
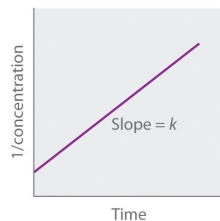
	Zeroth Order	First Order	Second Order																								
Differential rate law	Rate = $-\frac{\Delta[A]}{\Delta t} = k$	Rate = $-\frac{\Delta[A]}{\Delta t} = k[A]$	Rate = $-\frac{\Delta[A]}{\Delta t} = k[A]^2$																								
Concentration vs. time																											
Integrated rate law	$[A] = [A]_0 - kt$	$[A] = [A]_0 e^{-kt}$ or $\ln[A] = \ln[A]_0 - kt$	$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$																								
Straight-line plot to determine rate constant																											
Relative rate vs. concentration	<table><tr><th>[A], M</th><th>Rate, M/s</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>1</td></tr><tr><td>3</td><td>1</td></tr></table>	[A], M	Rate, M/s	1	1	2	1	3	1	<table><tr><th>[A], M</th><th>Rate, M/s</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table>	[A], M	Rate, M/s	1	1	2	2	3	3	<table><tr><th>[A], M</th><th>Rate, M/s</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>9</td></tr></table>	[A], M	Rate, M/s	1	1	2	4	3	9
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3	9																										
Half-life	$t_{1/2} = \frac{[A]_0}{2k}$	$t_{1/2} = \frac{0.693}{k}$	$t_{1/2} = \frac{1}{k[A]_0}$																								
Units of k, rate constant	M/s	1/s	M ⁻¹ s ⁻¹																								

Figure 22: Courtesy: LibreTexts Chemistry

6.1.3 Rate Determining Step

In a multistep reaction, **the slowest step determines the overall rate of the process**. This slowest step often involves the crossing of the **highest activation barrier** and is called the **rate determining step**.

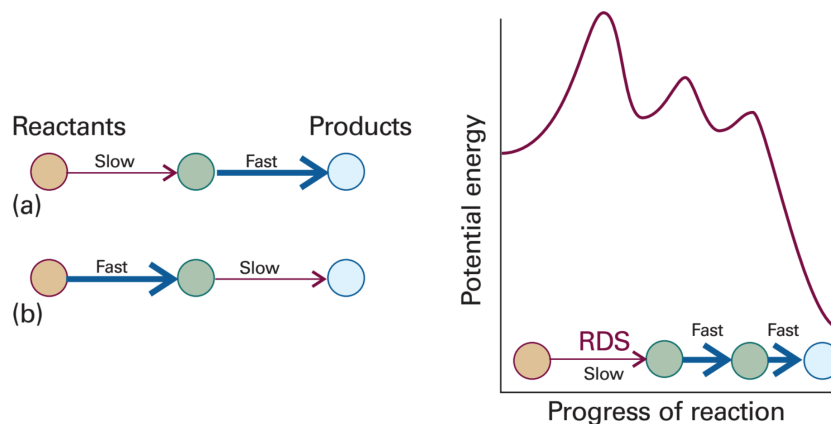
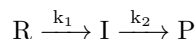


Figure 23: Figure Courtesy: Atkins, Peter William, Julio De Paula, and James Keeler. Atkins' Physical Chemistry. Oxford University Press, 2023.

Let us illustrate this concept with a simple consecutive reaction where all reactions are first order:



The R only gets depleted with rate constant k_1 ,

$$\frac{d[R]}{dt} = -k_1[R]$$

The intermediate, I gets formed at rate constant k_1 at the same time gets depleted at rate constant k_2

$$\frac{d[I]}{dt} = k_1[R] - k_2[I]$$

while the product P only gets formed at rate constant k_2

$$\frac{d[P]}{dt} = k_2[I]$$

- If $k_2 \gg k_1$, i.e., the second step is much faster than the first one, every time any intermediate molecule I is formed it is converted to the product. So the second step of the reaction literally waits for the first step of the reaction. So the concentration of the I molecule hardly changes as the rate of depletion is equal to the rate of formation.

$$\frac{d[I]}{dt} = k_1[R] - k_2[I] \approx 0 \implies k_2[I] \approx k_1[R]$$

So

$$\text{Overall Rate: } \frac{d[P]}{dt} = k_2[I] \approx k_1[R]$$

The overall reaction behaves like a first order reaction with respect to the reactant, and the rate of the reaction is equivalent to the first (slowest) step of the reaction.

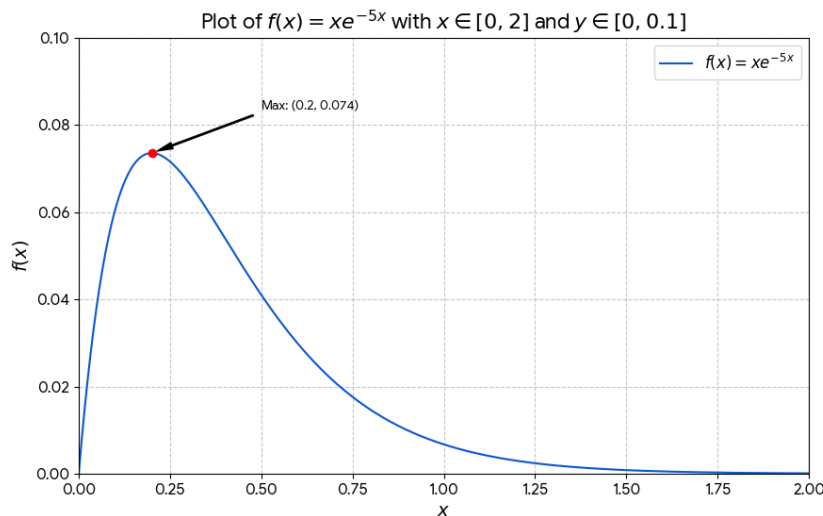
- If $k_1 \gg k_2$, i.e., the first step is faster than the second one, any reactant in the system will be quickly converted into the intermediate, while due to the slower conversion of intermediate to product, there will be accumulation of intermediate in the system. Now using the integrated rate law for the first order rate constant:

$$[R] = [R]_0 e^{-k_1 t}$$

So

$$\frac{d[I]}{dt} = k_1[R] - k_2[I] = k_1[R]_0 e^{-k_1 t} - k_2[I]$$

The first term $k_1[R]_0 e^{-k_1 t}$ has two different types of k_1 dependence. At low k_1 , the linear term dominates and at high k_1 the exponential term dominates. So the value of the first term first increases and decreases to zero with increasing values of k_1 . See the plot of the function $f(x) = xe^{-5x}$ below.



So when $k_1 \gg k_2$, $k_1[R]_0 e^{-k_1 t} \approx 0$. Therefore,

$$\frac{d[P]}{dt} = -\frac{d[I]}{dt} = k_2[I]$$

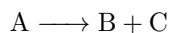
So the overall reaction is just a first-order reaction with respect to I , with a rate constant k_2 , which is the rate constant of the slowest (second) step.

Interactive Tutorial

6.1.4 Stoichiometry and Reaction Order: A Hint at the Mechanism

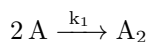
In the previous example, we saw how the mechanism of a reaction can change its rate law or maybe the order. We previously mentioned that the stoichiometry of a reaction and its order are unrelated to each other. One exception to this is an elementary reaction step, which is part of a larger reaction mechanism. For a single reactive step, the stoichiometry and order are often the same, and this property can be used to decipher reaction mechanisms.

Let us understand this with a hypothetical example of an unimolecular dissociation reaction:

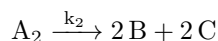


If this is a single step elementary reaction, it will be a first order reaction with respect to A: rate = $k[A]$. But let's say the mechanism is more complex and involves two steps.

First, two A molecules collide to form an A_2 molecule.



Next, the A_2 dissociates into B and C.



Let's say the first step is the rate-determining step, i.e., $k_1 \ll k_2$. The reaction will therefore be second order with respect to A: rate = $k[A]^2$.

Once we identify the rate law through experimental measurements, we attempt to construct mechanistic models that align well with the observed reaction order. This is one of the key ways we discover reaction mechanisms. However, it is necessary to obtain other parallel evidence in support of the mechanism before we can be absolutely certain.

6.2 Potential Energy Surface and Temperature Dependence of Reaction Rates

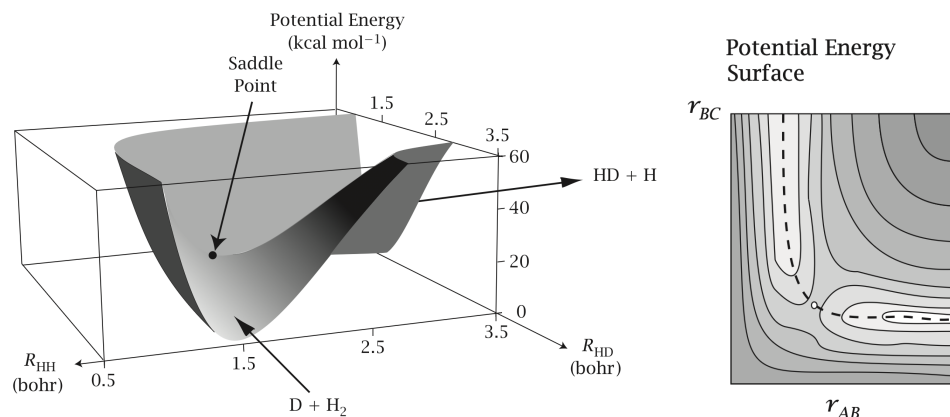
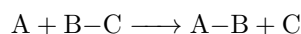


Figure 24: Figure Courtesy: Ken Dill and Sarina Bromberg. *Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience*. Garland Science, 2010.

The next question we ask is how the rate constant k depends on temperature. To address that, we need to revert to the microscopic description of the system and connect that to the macroscopic rate constants through statistical mechanics. For simplicity, we look at a reaction of the form:



where A approaches the molecule B-C, breaks its bond and forms a bond between A-B, and the atom C leaves the AB molecule. One example of such a reaction is the $D + H_2 \longrightarrow HD + H$ reaction shown in the figure above. To simplify the problem, we assume the entire reaction occurs on a single straight line. Therefore, there are only two important degrees of freedom: (i) distance between A and B (r_{AB}) and (ii) distance between B and C (r_{BC}). If we compute the potential energies of the systems for a grid of all possible values of r_{AB} and r_{BC} , we obtain a potential energy surface shown in the above figure. In the reactant state, r_{AB} is high and r_{BC} is low, as the B and C atoms are covalently bonded. In this regime, the potential energy surface has a minimum that corresponds to r_{BC} = the B-C bond distance. Similarly, in the product state, the potential energy function has a minimum at r_{AB} = the A-B bond distance, as the C atom has drifted apart. The reaction rate and mechanism are determined by what happens in the middle of these two extremes.

In the potential energy function, there exists a low-energy channel along which the reaction proceeds. This is called the minimum energy path (MEP). Along this path, there is a point called the saddle point, where the potential energy has a maximum along the MEP and a minimum along its orthogonal direction. The state of the system at this saddle point on the potential energy surface is called a **Transition State (TS)**. If we plot the potential energy as a function of the progress of the reaction along the MEP (dotted line), we get an energy profile of the reaction similar to the figure below.

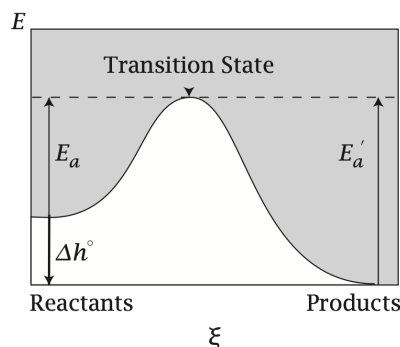
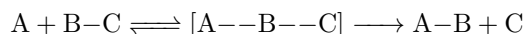


Figure 19.3 An activation energy diagram relates three quantities: the activation energy E_a for the forward reaction, the activation energy E'_a for the reverse reaction, and the enthalpy Δh° for the equilibrium. ξ is the reaction coordinate. A more microscopic interpretation of the reaction coordinate is given on page 363.

Figure 25: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.

The reaction therefore, can be represented as:



where $[A-B-C]$ is the transition state, where the $A-B$ bond is neither fully formed nor the $B-C$ bond fully broken. Being the maximum of the potential energy surface, the transition state is transient and unstable (such as balancing a ball on top of a wooden edge). Note my use of the $\rightleftharpoons$ and $\longrightarrow$ signs in the above reaction scheme. This indicates that the reactant state is in equilibrium with the transition state. From the transition state, the system can go back toward the reactant or go forward toward the product. Technically, an equilibrium also exists between the TS and the products; however, when we are discussing the rate constant of the forward reaction, we can effectively ignore that process.

To simplify the notations for the calculations that follow, we will express the reaction as:



The rate of this reaction is

$$\frac{d[P]}{dt} = \kappa^\dagger [TS]^n$$

As the transformation from the TS to P is a straightforward conversion without the involvement of any other species, the rate depends only on the concentration of the TS, and the reaction is a first-order reaction. But the rate depends on the concentration of the transition state, which is near impossible to measure due to the short-lived nature of the TS. So we invoke the equilibrium constant of the first step:

$$K_{eq} = \frac{[TS]}{[R]} \implies [TS] = K_{eq}[R]$$

Therefore

$$\frac{d[P]}{dt} = \kappa^\dagger [TS] = \kappa^\dagger K_{eq}[R]$$

Comparing with a general first-order rate equation,

$$\text{Rate} = \frac{d[P]}{dt} = k[R] \implies \text{Rate constant } k = \kappa^\dagger K_{eq}$$

The K_{eq} is an equilibrium constant and can be written as the ratio of partition functions and the TS and R.

$$K_{eq} = \frac{q_{TS}}{q_R} e^{-\beta(E_{TS}-E_R)} = \frac{q_{TS}}{q_R} e^{-E_a/k_B T}$$

$$\text{Rate constant } k = \kappa^\dagger K_{eq} = \kappa^\dagger \frac{q_{TS}}{q_R} e^{-E_a/k_B T}$$

The term E_a is the difference between the energies of the transition state and the reactant. This energy difference is called the **Activation Energy** or the activation barrier, as this is the energetic barrier the system needs to overcome to go from the reactant to the product state.

Note that the pre-exponential factor is a ratio of two canonical partition functions, the values of which depend only on one external intensive variable, the temperature. If we do not need to gain microscopic insight about the system, we can simplify the prefactor as just some function of temperature:

$$\text{Rate constant } k = A(T) \cdot e^{-E_a/k_B T}$$

This is known as the Arrhenius equation and was originally discovered empirically. We also see that

$$\ln k = \ln A - \frac{E_a}{k_B T}$$

By plotting the natural log of the rate constant of a reaction at various temperatures and creating a linear fit with $1/T$, one can experimentally estimate the Arrhenius constant A and the activation energy E_a . Note that in the plot below it is assumed that the Arrhenius prefactor A is independent of the temperature. This assumption is not always true, particularly in dense solutions and low temperatures.

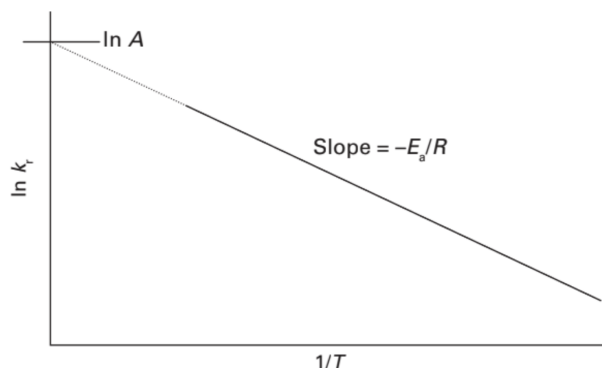


Figure 17D.1 An Arrhenius plot, a plot of $\ln k$, against $1/T$, is a straight line when the reaction follows the behaviour described by the Arrhenius equation (eqn 17D.1). The slope is $-E_a/R$ and the intercept at $1/T = 0$ is $\ln A$.

Figure 26: Figure Courtesy: Atkins, Peter William, Julio De Paula, and James Keeler. Atkins' Physical Chemistry. Oxford University Press, 2023.

6.2.1 The Eyring-Polanyi Equation and Experimental Measurement of Enthalpy and Entropy of Activation

Now we will consider the temperature dependence of the pre-exponential factor or the Arrhenius factor $A(T)$. Note that there is no single general expression for the temperature dependence of $A(T)$, and it will change depending on the type of reaction and reaction conditions. For simple one-step first-order reactions in the gas phase or dilute solution, one can often express the rate using the Eyring-Polanyi Equation:

$$\text{Rate constant } k = \frac{k_B T}{h} e^{-\Delta G^\ddagger/k_B T}$$

Where $\Delta G^\ddagger$ is the relative Gibbs Free Energy with respect to the reactants. We call it the Gibbs Free Energy of activation. (Note: we are using Gibbs free energy instead of Helmholtz free energy, as most reactions happen at constant temperature and pressure. Controlling the total volume of a reaction is near impossible.) We also know from thermodynamics: $\Delta G = \Delta H - T\Delta S$. Substituting we get:

$$\text{Rate constant } k = \frac{k_B T}{h} e^{-\Delta H^\ddagger/k_B T} e^{\Delta S^\ddagger/k_B}$$

This is the final expression for the Eyring-Polanyi Equation. Note that now we are taking the exponential of the **activation free energy** and not the activation energy.

Experimentally, from the temperature dependence of a chemical reaction's rate constant, k , one can determine the thermodynamic activation parameters ($\Delta H^\ddagger$, $\Delta S^\ddagger$, and $\Delta G^\ddagger$) based on Transition State Theory (TST). For facilitating a linear fit, the linearized form of the Eyring equation is often used.

$$\ln\left(\frac{k}{T}\right) = -\frac{\Delta H^\ddagger}{R} \left(\frac{1}{T}\right) + \left[\ln\left(\frac{k_B}{h}\right) + \frac{\Delta S^\ddagger}{R}\right]$$

Note, we used R instead of k_B when we measure the molar $\Delta H^\ddagger$, $\Delta S^\ddagger$, and $\Delta G^\ddagger$.

- **Procedure:** The reaction is monitored over time at a minimum of four distinct, accurately controlled absolute temperatures (T).
- **Data Collection:** Monitor the change in concentration of a reactant or product using a suitable analytical technique (e.g., titration, spectrophotometry).
- **Rate Constant (k):** For each temperature, the concentration-time data is fitted to the appropriate integrated rate law (e.g., first-order: $\ln[A]_t = -kt + \ln[A]_0$) to determine the rate constant, k .

The **Eyring Plot** is generated by plotting $\ln(k/T)$ (y-axis) versus $1/T$ (x-axis).

Linear regression of the Eyring Plot, represented as $y = mx + c$, yields the following relationships:

Enthalpy of Activation ($\Delta H^\ddagger$):

$$\begin{aligned}\text{Slope}(m) &= -\frac{\Delta H^\ddagger}{R} \\ \implies \Delta H^\ddagger &= -R \times \text{Slope}\end{aligned}$$

Entropy of Activation ($\Delta S^\ddagger$):

$$\begin{aligned}\text{Intercept}(c) &= \ln\left(\frac{k_B}{h}\right) + \frac{\Delta S^\ddagger}{R} \\ \implies \Delta S^\ddagger &= R \times \left[\text{Intercept} - \ln\left(\frac{k_B}{h}\right)\right]\end{aligned}$$

Gibbs Energy of Activation ($\Delta G^\ddagger$): Calculated at a specific temperature, T :

$$\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$$

These parameters provide insight into the energetic barrier ($\Delta H^\ddagger$) and the change in molecular order ($\Delta S^\ddagger$) involved in forming the transition state.

6.2.2 Optional Appendix: Derivation of the Eyring-Polanyi Equation

Now we will try to determine the temperature dependence of the pre-exponential factor or the Arrhenius factor $A(T)$. Note that there is no single general expression for the temperature dependence of $A(T)$, and it will change depending on the type of reaction and reaction conditions. However, some general considerations can be made anyways. The transition state theory (TST) says the rate constant

$$k = \kappa \nu \frac{q'_{TS}}{q'_R} \quad (6.1)$$

Here I use the ($'$) symbol as the partition functions are evaluated inside the reaction mixture. Let us look at the other two terms:

- **Characteristic frequency (ν):** The TST assumes that the transition state has one weak bond that breaks to form the product. The rate of product formation will be proportional to the vibrational frequency of that bond. For an intuitive understanding, consider a pendulum with a string and a bob. The string is very worn out and can tear any time. The more oscillations we observe per unit time the more is the chance that the string gets torn. Very similar concept is applied here.

- **Transmission coefficient (κ):** Going back to the worn out pendulum, not all oscillations will lead to breakage but only some will. Transmission coefficient is the proportion of those vibrations that leads to product formation. So if 1 out of 3 oscillations of the bond leads to bond breakage, $\kappa = 1/3$.

Note that TST rate expression in Eq. 6.1 is very general and can be applied to any molecular systems with covalent bond breaking (e.g., chemical reactions), non-covalent bond breaking (e.g., dissociation of a protein-ligand complex), and conformational transitions and rearrangements (e.g., protein folding, etc.). There are some exceptions, particularly where quantum mechanical tunneling is involved but that is beyond the scope of this course.

But the big disadvantage of the TST equation is that none of the terms in the equation can be evaluated or measured in a straightforward manner for molecular systems. Therefore we need to make approximations. We will describe the Eyring-Polanyi equation which is one of the simplest (and most crude) approximations to the TST equation.

The Eyring-Polanyi approach introduces two very drastic assumptions.

- The reactive vibrations are completely uncoupled from the rest of motions in the transition state structure.
- The transmission coefficient is 1, i.e., all vibrations of the bond in the TS leads to formation of product. (Many books and papers now avoid this assumption leading to slightly different expression for the Eyring-Polanyi rate constant).

The second approximation is fairly easy to comprehend so we focus on the first assumption below. Although the transition state structure is unstable (at a maximum of energy) along the reaction coordinate, it is stable (at a minimum) of energy in all other directions that are normal to the reaction coordinate (see figure below).

(b) Three-Dimensional Transition State

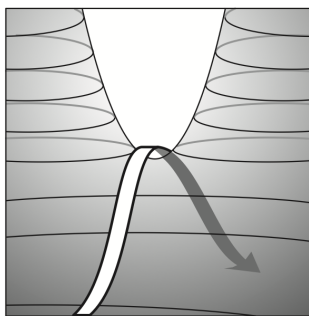


Figure 27: Figure Courtesy: Ken Dill and Sarina Bromberg. Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience. Garland Science, 2010.

For the stable degrees of freedom, the system is presumed to act like any other equilibrium system. Those degrees of freedom are described by equilibrium partition functions. Provided that the unstable degrees of freedom are independent of the stable degrees of freedom, q_{TS} can be factorized into two components:

$$q'_{TS} = \overline{q'_{TS}} q_{\xi}$$

the $\overline{q_{TS}}$ is for the stable part and q_{ξ} is along the reaction coordinate, i.e., MEP. This is the central result of the first assumption. This approximation can be applied only when the reaction coordinate is completely decoupled from the other degrees of freedom (such as vibration, rotation, translation, etc. of other atoms and bonds in the transition state). This is **in general not true** as energy transfer from one degree of freedom to another is prevalent in most condensed phase reactions (i.e. reactions in solution or dense media) or biomolecular processes. But in gas phase unimolecular or bimolecular reactions involving very small molecules with < 10 atoms in each molecule, this assumption is somewhat valid. Despite these severe limitations we use this approximation as it simplifies the math a lot and help us come to some easy to understand conclusions.

We approximate the q_{ξ} as the vibrational partition function of the bond that is breaking (or forming) in the reactive event. Because the bond along ξ is only partly formed (or broken) in the transition-state structure, we

treat it as a weak vibration. The frequency ν of vibration depends on the spring constant k_ξ and the reduced mass $\mu = m_A m_B / (m_A + m_B)$

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k_\xi}{\mu}}$$

As the bond along ξ is weak, k_{xi} is very small and so is ν . Therefore:

$$q_\xi = \frac{1}{1 - e^{-\beta h \nu}} \approx \frac{k_B T}{h \nu}$$

Here we used $e^x = 1 - x$ for small x .

Note that we ignored the zero point energy contribution $e^{-\beta h \nu / 2}$ in the numerator, as the zero-point energies are negligible for very low frequency vibrations.

So we get

$$\text{Rate constant } k = \kappa \nu \frac{q'_{TS}}{q'_R} = \kappa \nu q_\xi \frac{\overline{q'_{TS}}}{q'_R} = \kappa \nu \frac{k_B T}{h \nu} \frac{\overline{q'_{TS}}}{q'_R} = \kappa \frac{k_B T}{h} \frac{\overline{q'_{TS}}}{q'_R}$$

If we now ignore the energetic contribution of that very slowly vibrating bond, we can write:

$$K_{eq} = \frac{\overline{q'_{TS}}}{q'_R}$$

If the difference of Gibbs free energy between the TS and the reactant is $G_{TS} - G_R = \Delta G^\ddagger$:

$$\Delta G^\ddagger = -k_B T \ln K_{eq} \implies K_{eq} = \frac{\overline{q'_{TS}}}{q'_R} = e^{-\Delta G^\ddagger / k_B T}$$

Substituting we get:

$$k = \kappa \frac{k_B T}{h} \frac{\overline{q'_{TS}}}{q'_R} = \kappa \frac{k_B T}{h} e^{-\Delta G^\ddagger / k_B T}$$

Invoking the first assumption of $\kappa = 1$

$$k = \frac{k_B T}{h} e^{-\Delta G^\ddagger / k_B T}$$

We also know from thermodynamics: $\Delta G = \Delta H - T \Delta S$. Substituting we get:

$$\text{Rate constant } k = \frac{k_B T}{h} e^{-\Delta G^\ddagger / k_B T} = \frac{k_B T}{h} e^{-\Delta H^\ddagger / k_B T} e^{\Delta S^\ddagger / k_B}$$

This is the final expression for the Eyring-Polanyi Equation. Note that now we are taking the exponential of the **activation free energy** and not the activation energy.

As we discussed earlier, the Eyring-Polanyi equation is not generally applicable, particularly for condensed phase processes. However, from Eq. 6.1 one can derive expressions for rate constants using different approximations that are more appropriate for the situations one is interested in. For example, it is possible to account for the effect of solvent from the effective friction along the reaction. The rate constant we obtain in such a situation we obtain the Kramer's rate theory. In more complex cases, if we include the effect of memory in friction, we get the Grote-Hynes Theory of rate. These more sophisticated approaches are beyond the scope of this class.

6.2.3 Catalysis

A **catalyst** is a substance that accelerates a reaction but undergoes no net chemical change. The catalyst lowers the activation energy of the reaction by providing an alternative path that avoids the slow, rate-determining step of the uncatalysed reaction. Catalysts can be very effective; for instance, the activation energy for the decomposition of hydrogen peroxide in solution is 76 kJ mol⁻¹, and the reaction is slow at room temperature. When a little iodide ion is added, the activation energy falls to 57 kJ mol⁻¹ and the rate constant increases by a factor of 2000.

Enzymes, which are biological catalysts, are very specific and can have a dramatic effect on the reactions they control. For example, the enzyme catalase reduces the activation energy for the decomposition of hydrogen peroxide to 8 kJ mol⁻¹, corresponding to an acceleration of the reaction by a factor of 10¹⁵ at 298 K. Enzymes often have

a binding site that complements the transition state of a reaction. As the process of binding stabilizes a structure through electrostatic, hydrophobic, and sometime, covalent interactions, this reduces the energy of the transition state and reduces the activation barrier. Note that if an enzyme perfectly complements the substrate (i.e. reactant) it will actually increase the activation barrier, slowing down the reaction. Some enzymes change conformation upon substrate binding to ensure that the binding pocket can bind stringly with the substrate and also can complement the transition state.

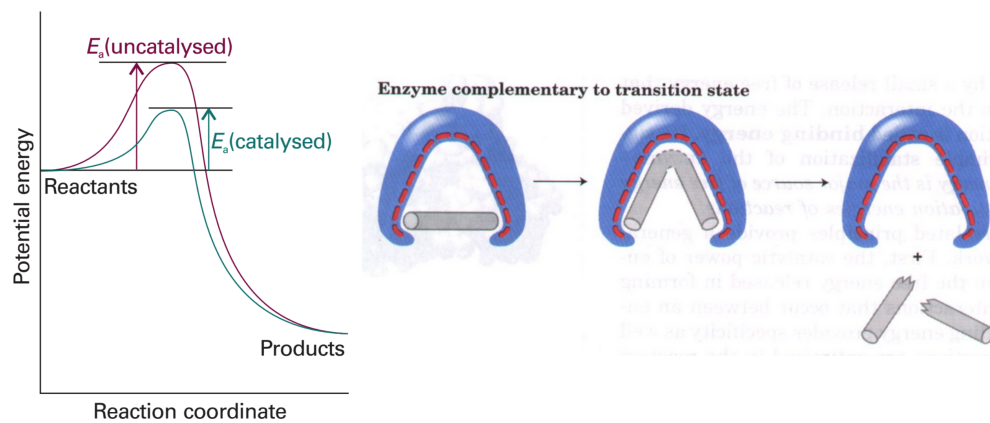


Figure 28: Figure Courtesy: Atkins, Peter William, Julio De Paula, and James Keeler. Atkins' Physical Chemistry. Oxford University Press, 2023.

6.2.4 Thermodynamic and Kinetic Control

Often in chemical reactions, we encounter a situation where more than one alternative product is possible from the same reactants. If kinetics were not relevant, the product with the lowest energy is the one that should form in larger quantities. If, however, the activation barrier to reach the lower energy product is higher than that of reaching the higher energy product, a conflicting scenario may arise.

At shorter timescales, the higher energy product will show a higher yield, as that product is formed faster than the lower energy product. However, if we wait long enough, we will see a higher yield for the lower energy product as we reach the thermodynamic equilibrium. The higher energy product with a lower activation barrier is called the **Kinetic Product**, and such a reaction is called **Kinetically Controlled** reaction, as faster kinetics leads to a higher yield of this product. The lower energy product with a higher activation barrier is called the **Thermodynamic Product**, and such a reaction is called **Thermodynamically Controlled** reaction, as thermodynamic stability is responsible for the higher yield of this product.

The **kinetic product** is also prevalent in **low-temperature conditions** where the system does not have enough energy to cross the higher energy barrier to reach the thermodynamic product. If we increase the temperature, we make it possible to cross the higher energy barrier; therefore, at **high temperature conditions**, we often get a higher yield for the **thermodynamic product**. A significant aspect of research in synthetic chemistry involves modifying conditions to optimize the balance between thermodynamic and kinetic control of the reaction, thereby achieving a high yield of the desired product.

Interactive Tutorial

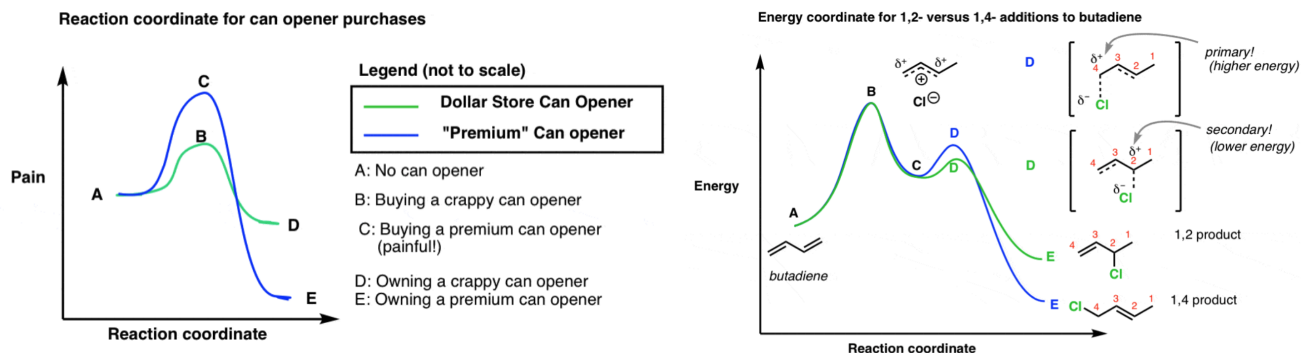
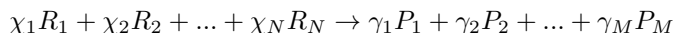


Figure 29: Figure Courtesy: James Ashenhurst

6.3 Summary

- For a reaction with the following stoichiometry:



The rate of the reaction is defined as:

$$\text{Rate} = -\frac{1}{\chi_1} \frac{d[R_1]}{dt} = -\frac{1}{\chi_2} \frac{d[R_2]}{dt} = \dots = -\frac{1}{\chi_N} \frac{d[R_N]}{dt} = \frac{1}{\gamma_1} \frac{d[P_1]}{dt} = \frac{1}{\gamma_2} \frac{d[P_2]}{dt} = \dots = \frac{1}{\gamma_M} \frac{d[P_M]}{dt}$$

- Rate law of a reaction, such as $A + 2B \rightarrow C + D$ is expressed as:

$$\text{Rate} = k[A]^n[B]^m$$

Where $[]$ indicates concentration. The order of the reaction is n with respect to A and m with respect to B . The k is the rate constant.

The unit of the Rate is always (concentration/time). However, the unit of the rate constant will vary depending on the reaction order.

The order of a reaction may or may not be related to the stoichiometry of the reaction.

- The differential forms of the rate laws for specific reaction orders are:

$$\text{Zeroth Order: } -\frac{d[A]}{dt} = k; \quad \text{1st Order: } -\frac{d[A]}{dt} = k[A]; \quad \text{2nd Order: } -\frac{d[A]}{dt} = k[A]^2$$

- The integrated rate laws express the change of concentration of the reactant $[A]$ over time t .

$$\text{Zeroth Order: } [A] = [A]_0 - kt; \quad \text{1st Order: } [A] = [A]_0 e^{-kt}; \quad \text{2nd Order: } \frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

- In a multistep reaction, the slowest step determines the overall rate of the process and is referred to as the rate-determining step.
- Chemical reactions often proceed through several elementary reactive steps. The rate constant can be deduced from the stoichiometry of the slowest elementary reactive step (rate-determining step). This property can be used to determine the mechanism of a reaction from experimentally measured reaction orders.
- In the potential energy surface (PES) of a chemical reaction, the reaction proceeds through the minimum energy path (MEP). The potential energy has a maximum along the MEP, and the chemical species at this potential energy maximum is called a Transition State (TS). The potential energy difference between the reactant and the TS is called the activation energy, and it dictates how the rate of the reaction changes with temperature:

$$\text{Rate constant} \propto \frac{q_{TS}}{q_R} e^{-E_a/k_B T} \implies \text{Rate constant} = A(T) e^{-E_a/k_B T}$$

where q denotes the partition function.

- The Eyring-Polanyi equation provides an expression for the rate constant in terms of the Gibbs free energy of activation ($\Delta G^\ddagger$) or the Enthalpy ($\Delta H^\ddagger$) and entropy ($\Delta S^\ddagger$) of activation:

$$\text{Rate constant } k = \frac{k_B T}{h} e^{-\Delta G^\ddagger / k_B T} = \frac{k_B T}{h} e^{-\Delta H^\ddagger / k_B T} e^{\Delta S^\ddagger / k_B}$$

Experimentally, from the temperature dependence of a chemical reaction's rate constant, k , one can determine the thermodynamic activation parameters ($\Delta H^\ddagger$, $\Delta S^\ddagger$, and $\Delta G^\ddagger$) based on Transition State Theory (TST). For facilitating a linear fit, the linearized form of the Eyring equation is often used.

$$\ln \left(\frac{k}{T} \right) = -\frac{\Delta H^\ddagger}{R} \left(\frac{1}{T} \right) + \left[\ln \left(\frac{k_B}{h} \right) + \frac{\Delta S^\ddagger}{R} \right]$$

- A catalyst accelerates a chemical reaction by reducing the activation energy, aka the barrier height of a reaction. Enzymes catalyse biological reactions and often complement the structure of the transition state to stabilize the TS and reduce the activation barrier.
- If there is more than one competing reactive pathway starting from the same reactant, the product involving a lower activation energy barrier will show a higher yield at low temperature and shorter timescales. This product is referred to as the kinetic product, and such a reaction is classified as a kinetically controlled reaction. Contrarily, at higher temperatures and longer timescales, the yield of the most thermodynamically stable product (lowest free energy) will be higher. This product is referred to as the thermodynamic product, and such a reaction is classified as a thermodynamically controlled reaction.